MASTERFUNCTIONALSKILLS

★ OFQUAL REGULATED STANDARDS

MATHEMATICS • ENGLISH • DIGITAL SKILLS

All Levels (Ofqual Regulated)

Functional Skills Revision Cards

Complete All-In-One Visual Revision Deck

- ★ Mapped to UK DfE & Ofqual Subject Criteria
- ★ Pearson Edexcel & City & Guilds Aligned
- ★ Visual Diagrams & Golden Rules on Every Card
- ★ Quick Exam Challenges & Step-by-Step Model Steps
- ★ A6 Postcard Cardstock Size (148 × 105 mm)

A6 Postcard Cardstock (148 × 105 mm)

Turn over for Deck Contents & Checklist ➔

masterfunctionalskills.co.uk

1. Order of Operations (BIDMAS)

NUMBER SYSTEM

B - I - D/M - A/S

Operations must be completed in order: 1. Brackets (), 2. Indices (², ³, √), 3. Division & Multiplication (left-to-right), 4. Addition & Subtraction (left-to-right).

B | I | D | M | A | S

Example: 5 + 3 × (4 - 2)²

1. (4-2)=2 → 2. 2²=4 → 3. 3×4=12 → 4. 5+12 = 17

★ Golden Rule: Always solve brackets and powers first before multiplying or adding!

⚠️ Watch Out: Don't just calculate left-to-right! 5 + 3 × 2 is 11, NOT 16.

DfE L2.N1 • Level 2

Turn over for Practice Challenge ➔

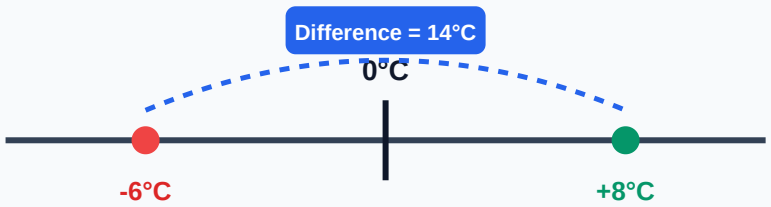
A6 Pocket Card

2. Negative Numbers in Context

NUMBER SYSTEM

Rules of Signs & Difference

Difference between positive & negative = Add their absolute values (|a| + |b|). Multiplying/Dividing: Same signs = POSITIVE (+). Different signs = NEGATIVE (-).



(-6) to (+8) = 6 steps to zero + 8 steps = 14

★ Golden Rule: Subtracting a negative is the same as ADDING! 5 - (-3) = 5 + 3 = 8.

⚠️ Watch Out: -5°C is COLDER than -2°C. The larger the numeral after the minus sign, the lower the value.

DfE L2.N2 • Level 2

Turn over for Practice Challenge ➔

A6 Pocket Card

3. Fractions, Decimals & % Equivalence

NUMBER SYSTEM

Conversion Benchmarks

To compare values, convert them all to percentages or decimals: Fraction to %: (Numerator ÷ Denominator) × 100. Decimal to %: Multiply by 100.

Fraction	Decimal	Percentage
1/8	0.125	12.5%
3/8	0.375	37.5%
5/8	0.625	62.5%

★ Golden Rule: Eighths are common exam traps: 1/8 = 12.5% (0.125), not 18%!

⚠️ Watch Out: 0.4 is 40%, NOT 4%! 0.04 is 4%. Count your decimal places carefully.

DfE L2.N3 • Level 2

Turn over for Practice Challenge ➔

A6 Pocket Card

Challenge: Order of Operations (BIDMAS)

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N1

Evaluate: 8 + 4 × (7 - 4)² ÷ 6

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

1. Brackets: (7 - 4) = 3. 2. Index: 3² = 9. 3. Multiply: 4 × 9 = 36. 4. Divide: 36 ÷ 6 = 6. 5. Add: 8 + 6 = 14.

📋 DECK CONTENTS & REVISION CHECKLIST

266 Revision Cards

Tick [✓] Read when you understand the rule, and [★] Solved when you answer the challenge! Pearson & C&G Aligned

📐 MATHEMATICS SUITE (88 Cards)

Deck 1: Maths Level 2 • 50 Cards (#1–#50)
Number (1–18) • Measures & Space (19–36) • Data (37–50)

Deck 2: Maths Level 1 • 38 Cards (#1–#38)
Number (1–14) • Shape & Space (15–28) • Handling Data (29–38)

✍️ ENGLISH SUITE (98 Cards)

Deck 3: English Level 2 • 52 Cards (#1–#52)
Reading (1–18) • Writing (19–33) • SPaG (34–48) • SLC (49–52)

Deck 4: English Level 1 • 46 Cards (#1–#46)
Reading (1–15) • Writing (16–28) • SPaG (29–42) • SLC (43–46)

💻 DIGITAL SKILLS (50 Cards)

Devices (1–10) • Creating & Spreadsheets (11–24) • Communicating (25–32) • Transacting (33–40) • Security (41–50)

🌟 ENTRY LEVEL 3 (30 Cards)

Entry 3 Maths Foundations (1–14) • Entry 3 English Reading, Writing, SPaG & SLC (15–24) • Entry 3 Digital Skills (25–30)

Total Cards in Collection: 266 Dual-Coded Postcards

Standards: DfE Statutory Subject Content • Pearson Edexcel & City & Guilds

💡 **Tutor Golden Tip (Abdul):** Do not flip the card immediately! Attempt the Quick Challenge on the back using a dry-wipe pen or scratchpad first. Verify with the step-by-step model steps only after completing your working.

Challenge: Fractions, Decimals & % Equivalence

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N3

Put in order from smallest to largest: 3/8, 35%, 0.36, 1/3.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Convert all to decimals: 1/3 = 0.333..., 35% = 0.35, 0.36 = 0.36, 3/8 = 0.375. Order: 1/3, 35%, 0.36, 3/8.

Challenge: Negative Numbers in Context

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N2

A vaccine freezer is at -18°C. A power failure causes it to rise to -4°C. By how many degrees did the temperature rise?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Rise = -4 - (-18) = -4 + 18 = 14°C.

4. Fraction of an Amount (Non-Calc)

NUMBER SYSTEM

Divide by Bottom, Times by Top

To find a fraction (a/b) of any amount: 1. Divide total by denominator (b) to find 1 part. 2. Multiply result by numerator (a) to get required parts.

Find 3/5 of £180

Step 1: Divide by 5

£180 ÷ 5 = £36 (1 part)

Step 2: Multiply by 3

£36 × 3 = £108

⋆ Golden Rule: Divide by the denominator, multiply by the numerator!

⚠️ Watch Out: If a question asks for the REMAINING amount, remember to subtract your answer from the original total.

5. Adding & Subtracting Mixed Fractions

NUMBER SYSTEM

Improper & Common Denominator

1. Convert mixed numbers to improper fractions (Whole × Denom + Num). 2. Find lowest common denominator (LCM). 3. Add or subtract numerators only. 4. Simplify.

2 1/4 + 1 2/3

1. Improper: 9/4 + 5/3
2. Common Denom (12): 27/12 + 20/12
3. Add numerators: 47/12 = 3 11/12

⋆ Golden Rule: NEVER add denominators together! 1/4 + 2/3 is NEVER 3/7.

⚠️ Watch Out: When subtracting, always convert to top-heavy improper fractions first to prevent negative fractional parts.

6. Multiplying & Dividing Fractions

NUMBER SYSTEM

Multiply Straight / Keep-Change-Flip

Multiplication: Multiply tops, multiply bottoms (cross-cancel if possible). Division: KEEP the first fraction, CHANGE ÷ to ×, FLIP the second fraction (reciprocal).

MULTIPLY

2/3 × 5/7

Top: 2 × 5 = 10

Bottom: 3 × 7 = 21 → 10/21

DIVIDE (K-C-F)

3/4 ÷ 2/5

Keep 3/4 • Change × • Flip 5/2

3/4 × 5/2 = 15/8 = 1 7/8

⋆ Golden Rule: Dividing by a fraction is the same as multiplying by its reciprocal!

⚠️ Watch Out: Never flip the FIRST fraction — only flip the SECOND fraction after changing the sign.

7. Percentage Increase & Decrease (Multiplier)

NUMBER SYSTEM

Single Multiplier Method

Increase: Add % to 100%, divide by 100. (e.g. +17.5% = × 1.175). Decrease: Subtract % from 100%, divide by 100. (e.g. -15% = × 0.85).

INCREASE (+20%)

100% + 20% = 120%

Multiplier = × 1.20

DECREASE (-15%)

100% - 15% = 85%

Multiplier = × 0.85

⋆ Golden Rule: Using a single multiplier is faster and eliminates rounding errors on calculator papers!

⚠️ Watch Out: Be careful with single digit percentages: +5% is × 1.05, NOT × 1.5 (which is +50%)!

Challenge: Adding & Subtracting Mixed Fractions

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N4

Work out 3 1/3 - 1 3/4. Give your answer as a mixed number in simplest form.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

1. Convert: 10/3 - 7/4. 2. Common denom 12: 40/12 - 21/12 = 19/12. 3. Mixed: 1 7/12.

Challenge: Fraction of an Amount (Non-Calc)

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N4

A clinic orders 240 boxes of gloves. 5/8 of them are size Medium. How many boxes are NOT size Medium?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Non-Medium fraction = 1 - 5/8 = 3/8. 240 ÷ 8 = 30. 30 × 3 = 90 boxes.

Challenge: Percentage Increase & Decrease (Multiplier)

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N6

A laptop originally costs £640. In a sale, it is reduced by 18%. Use a single multiplier to calculate the sale price.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Multiplier = 1 - 0.18 = 0.82. Sale price = £640 × 0.82 = £524.80.

Challenge: Multiplying & Dividing Fractions

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N5

Work out 2/5 ÷ 4/15. Give your answer in its simplest form.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Keep 2/5, change to ×, flip to 15/4: (2 × 15) ÷ (5 × 4) = 30 ÷ 20 = 3/2 = 1 1/2.

8. Percentage Change Formula

NUMBER SYSTEM

(Change ÷ Original) × 100

Percentage Change = (Actual Difference ÷ Original Starting Value) × 100. Works for percentage profit, loss, discount, and inflation.

% Change = (Change ÷ Original) × 100

Always divide by the STARTING value, never the new value!

Car bought for £8,000, sold for £6,800

Change = £1,200 → (1,200 ÷ 8,000) × 100 = 15% Loss

⋆ Golden Rule: Denominator is ALWAYS the original starting value before the change occurred!

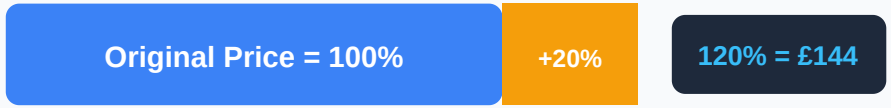
⚠ Watch Out: Don't divide by the new price! Dividing 1,200 by 6,800 gives 17.6%, which is incorrect.

9. Reverse Percentages (Original Value)

NUMBER SYSTEM

Divide by Multiplier

To find the original price before VAT or sale: 1. Identify the new percentage (e.g. Price + 20% VAT = 120% = 1.20). 2. Divide final price by multiplier (Price ÷ 1.20).



DO NOT CALCULATE 20% OF £144!

Original = £144 ÷ 1.20 = £120

⋆ Golden Rule: To go BACKWARDS to the original, DIVIDE by the multiplier!

⚠ Watch Out: Never calculate the percentage of the sale/final price and subtract! 20% of £144 is £28.80 (£115.20 = WRONG).

10. Simple vs Compound Interest

NUMBER SYSTEM

Fixed Amount vs Interest on Interest

Simple: Same fixed interest paid every year. Compound: Total = Principal × (Multiplier)^Years. Interest is earned on accumulated interest.

SIMPLE INTEREST

£1,000 at 5% for 3 yrs

£50 + £50 + £50 = £150

Total = £1,150

COMPOUND INTEREST

£1,000 × $(1.05)^3$

£1,000 × 1.157625

Total = £1,157.63

⋆ Golden Rule: Compound formula: Total = Principal × $(1 + r/100)^n$

⚠ Watch Out: Check whether the question asks for the TOTAL ACCUMULATED AMOUNT or just the INTEREST EARNED!

11. Sharing in a 3-Part Ratio

NUMBER SYSTEM

Add Parts, Divide, Multiply

1. Add all parts together to find total parts (e.g. 2 + 3 + 5 = 10 parts). 2. Divide total amount by total parts to find value of 1 part. 3. Multiply 1 part by each share.

Share £480 in ratio 2 : 3 : 5

1. Total: 10 parts

2. £480÷10 = £48

3. ×2, ×3, ×5

Shares: £96 : £144 : £240 (Sum = £480 ✓)

⋆ Golden Rule: Check your work: Add the final shares together — they must equal the starting total!

⚠ Watch Out: Don't divide the total by just one number! Always sum ALL parts first.

Challenge: Reverse Percentages (Original Value)

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N8

A coat costs £136 in a 15% off sale. What was its original price before the sale?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Sale price = 100% - 15% = 85% = 0.85. Original = £136 ÷ 0.85 = £160.

Challenge: Percentage Change Formula

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N7

A gym membership increases from £35 per month to £42 per month. Calculate the percentage increase.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Change = £42 - £35 = £7. % Increase = (7 ÷ 35) × 100 = 1/5 × 100 = 20%.

Challenge: Sharing in a 3-Part Ratio

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N10

Three partners invest in a business in the ratio 3 : 4 : 5. Profits of £72,000 are shared in this ratio. What is the largest share?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Total parts = 3 + 4 + 5 = 12. Value of 1 part = £72,000 ÷ 12 = £6,000. Largest share (5 parts) = 5 × £6,000 = £30,000.

Challenge: Simple vs Compound Interest

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N9

£4,000 is invested in a bond paying 4% compound interest annually. What is the total value after 3 years?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Multiplier = 1.04. Total = £4,000 × $(1.04)^3$ = £4,000 × 1.124864 = £4,499.46.

12. Reverse Ratios (Difference / One Share)

NUMBER SYSTEM

Find 1 Part from the Difference

When told 'Person A gets £X more than Person B': 1. Find difference in ratio parts. 2. Divide given money by difference in parts = 1 part. 3. Multiply by required share.

Ratio 5 : 3 • Liam gets £40 more than Maya

1. Difference in parts: 5 - 3 = 2 parts
2. Value of 1 part: £40 ÷ 2 = £20
3. Total money: (5 + 3) = 8 parts × £20 = £160

⋆ Golden Rule: Match the given amount to its specific parts, NEVER the total parts!

⚠ Watch Out: Don't divide £40 by 8! The £40 represents the DIFFERENCE (2 parts), not the total.

13. Direct & Inverse Proportion

NUMBER SYSTEM

Direct (Both Up) vs Inverse (One Up, One Down)

Direct: As X doubles, Y doubles (Scale factor or Unitary method). Inverse: As workers increase, time decreases! Workers × Time = Constant Total Work.

DIRECT PROPORTION

4 books cost £18

1 book = 18 ÷ 4 = £4.50

7 books = 7 × 4.5 = £31.50

INVERSE PROPORTION

3 painters take 8 hours

Total = 3 × 8 = 24 hrs

6 painters = 24 ÷ 6 = 4 hrs

⋆ Golden Rule: More workers take LESS time! Multiply workers by hours to get total person-hours.

⚠ Watch Out: Don't scale inverse proportion by multiplying both! If 3 workers take 8 hrs, 6 workers do NOT take 16 hrs!

14. Best Buy & Unit Pricing

NUMBER SYSTEM

Cost per Unit (Price ÷ Quantity)

To compare value for money, find the cost for 1 unit (100g, 1ml, or 1 item): Cost per 100g = (Price in pence ÷ Grams) × 100. The lowest cost per unit is the best value.

Pack A: 500g for £2.40

240p ÷ 5 =

48p per 100g

Pack B: 800g for £3.60

360p ÷ 8 =

45p per 100g ⋆

⋆ Golden Rule: Convert pounds to pence first to avoid difficult decimal calculations!

⚠ Watch Out: Don't just pick the cheaper price tag — a smaller pack might cost more per 100g.

15. Rounding to Significant Figures

NUMBER SYSTEM

Count from First Non-Zero Digit

1st sig fig is the first digit that is NOT zero reading left-to-right. Look at next digit: 5 or more rounds UP, 4 or less rounds DOWN. Fill trailing spaces with placeholder zeroes.

Round 0.004782 to 2 sig figs:

Leading zeros do NOT count! First non-zero = 4 (1st), 7 (2nd).

Next digit is 8 (rounds 7 UP to 8) → 0.0048

Round 48,290 to 1 sig fig: 48,290 → 50,000 (keep placeholder zeroes!)

⋆ Golden Rule: Leading zeros in decimals NEVER count as significant figures!

⚠ Watch Out: Don't turn 48,290 into 5! You must keep the placeholder zeros (50,000).

Challenge: Direct & Inverse Proportion

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N12

4 technicians take 6 hours to test a server system. How long would 3 technicians take working at the same rate?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Total work = 4 × 6 = 24 person-hours. For 3 technicians: 24 ÷ 3 = 8 hours.

Challenge: Reverse Ratios (Difference / One Share)

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N11

In a bakery, loaves to rolls are baked in the ratio 4 : 7. There are 45 more rolls than loaves. How many loaves were baked?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Difference in parts = 7 - 4 = 3 parts. 1 part = 45 ÷ 3 = 15 items. Loaves (4 parts) = 4 × 15 = 60 loaves.

Challenge: Rounding to Significant Figures

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N14

Round: (a) 63,740 to 2 significant figures. (b) 0.05084 to 2 significant figures.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

(a) 63,740 → 64,000 (7 rounds 3 up). (b) 0.05084 → 0.051 (8 rounds 0 up; first non-zero is 5).

Challenge: Best Buy & Unit Pricing

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N13

Cereal Brand X: 450g costs £2.25. Brand Y: 600g costs £2.88. Which is better value for money?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Brand X: 225p ÷ 4.5 = 50p per 100g. Brand Y: 288p ÷ 6 = 48p per 100g. Brand Y is better value by 2p per 100g.

16. Estimation & Approximation

NUMBER SYSTEM

Round Every Number to 1 Sig Fig

To estimate calculations and verify checks in Section A: Round EACH number to 1 significant figure BEFORE performing any multiplication or division.

Estimate: (48.7 × 19.3) ÷ 0.49

1. Round to 1 sig fig: 48.7 → 50, 19.3 → 20, 0.49 → 0.5
2. Calculate numerator: 50 × 20 = 1,000
3. Divide: 1,000 ÷ 0.5 = 2,000 (Dividing by 0.5 doubles!)

⋆ Golden Rule: Dividing by 0.5 is the exact same as multiplying by 2!

⚠ Watch Out: Do NOT calculate the exact answer first and round at the end! Examiners award zero marks if unrounded values are used.

17. Powers, Roots & Indices

NUMBER SYSTEM

Square, Cube & Roots

$x^2 = x \times x$. $x^3 = x \times x \times x$. Square root ($\sqrt{x}$) is inverse of squaring. Cube root ($\sqrt[3]{x}$) is the number multiplied by itself 3 times to give x.

SQUARE ROOTS

$8^2 = 64 \rightarrow \sqrt{64} = 8$

$11^2 = 121 \rightarrow \sqrt{121} = 11$

$\sqrt{144} = 12$

CUBE ROOTS

$2^3 = 2 \times 2 \times 2 = 8 \rightarrow \sqrt[3]{8} = 2$

$3^3 = 3 \times 3 \times 3 = 27 \rightarrow \sqrt[3]{27} = 3$

$5^3 = 125 \rightarrow \sqrt[3]{125} = 5$

⋆ Golden Rule: 5^3 is NOT 5 × 3 = 15! 5^3 is 5 × 5 × 5 = 125.

⚠ Watch Out: Square root of a number between 0 and 1 gets BIGGER! √0.25 = 0.5.

18. Formula Substitution

NUMBER SYSTEM

Replace Letters with Numbers

1. Replace each variable with given numbers (use brackets around negatives!). 2. Follow BIDMAS: Multiply adjacent terms before adding. 3. Check units.

Wage Formula: W = 12h + 1.5(oh)

If regular hours h = 35 and overtime hours oh = 6:

W = (12 × 35) + 1.5 × (12 × 6)

W = £420 + £108 = £528 Total Wage

⋆ Golden Rule: Letters side-by-side mean MULTIPLY: 4ab means 4 × a × b!

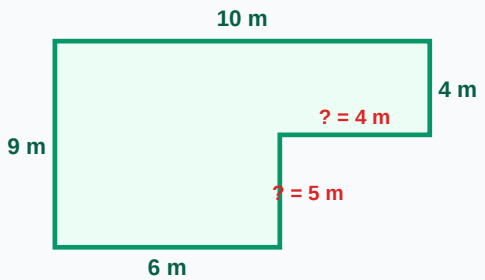
⚠ Watch Out: Squaring a negative number gives a positive: $(-4)^2 = +16$, NOT -16.

19. Perimeter of Composite Shapes

SHAPE & SPACE

Find Missing Sides, Then Add All

1. Find every missing side: Horizontal missing side = Total width - given width. Vertical missing side = Total height - given height. 2. Sum ALL outer sides.



Missing Sides:

Width = 10 - 6 = 4 m

Height = 9 - 4 = 5 m

⋆ Golden Rule: Tick off each side as you add it to avoid counting sides twice or skipping one!

⚠ Watch Out: Never add internal dotted lines when finding perimeter — perimeter is ONLY the outside border!

Challenge: Powers, Roots & Indices

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N16

Evaluate: 4^3 - √169 + 2^4.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

4^3 = 64. √169 = 13. 2^4 = 16. Calculation: 64 - 13 + 16 = 51 + 16 = 67.

Challenge: Estimation & Approximation

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N15

Use approximation to 1 significant figure to estimate: (312 × 4.89) ÷ 0.19.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Round to 1 sf: 300 × 5 ÷ 0.2. 300 × 5 = 1,500. 1,500 ÷ 0.2 = 1,500 × 5 = 7,500.

Challenge: Perimeter of Composite Shapes

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS18

An L-shaped lawn has outer dimensions 12 m by 8 m, with an inset cut of 5 m wide and 3 m deep. What is the perimeter?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Missing horizontal = 12 - 5 = 7 m. Missing vertical = 8 - 3 = 5 m. Perimeter = 12 + 8 + 7 + 3 + 5 + 5 = 40 m.

Challenge: Formula Substitution

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N17

Kinematics: $v^2 = u^2 + 2as$. Find v when u = 3, a = 4, and s = 5.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

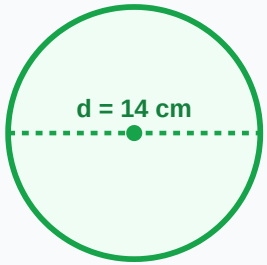
$v^2 = 3^2 + (2 \times 4 \times 5) = 9 + 40 = 49$. v = √49 = 7.

20. Circumference of a Circle

SHAPE & SPACE

Circumference = 3.142 × Diameter

Circumference is the distance around the outside edge of a circle. Diameter (d) goes straight across through center. Radius (r) is from center to edge (d = 2r).



CIRCUMFERENCE

C = 3.142 × Diameter

C = 3.142 × 14

= 43.99 cm

★ Golden Rule: Distance around outside: Circumference = 3.142 × Diameter!

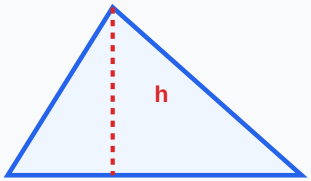
⚠ Watch Out: If given radius (r), you MUST double it before multiplying by 3.142 (Diameter = 2 × r)!

21. Area of Triangle & Parallelogram

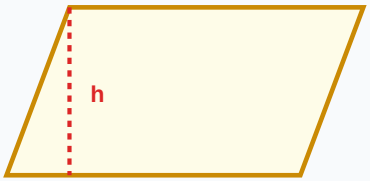
SHAPE & SPACE

Triangle: 1/2 × b × h | Parallelogram: b × h

Both use PERPENDICULAR vertical height at 90° to the base. Triangle is halved because two identical triangles make a parallelogram/rectangle.



Area = (b × h) ÷ 2



Area = b × h (NO ÷ 2!)

★ Golden Rule: Always use the perpendicular vertical height, NEVER the slanted side!

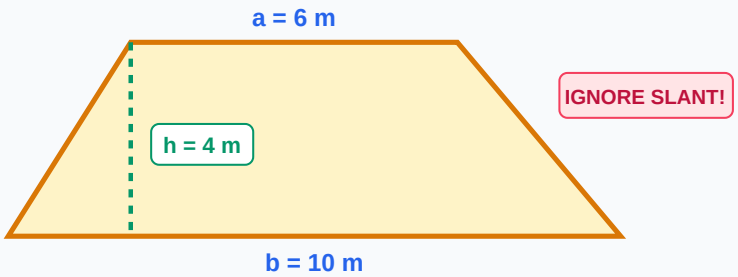
⚠ Watch Out: Don't divide by 2 for a parallelogram — only halve for a triangle!

22. Area of a Trapezium

SHAPE & SPACE

Area = 1/2(a + b)h

1. Add parallel top and bottom sides (a + b). 2. Divide by 2 to get average parallel width. 3. Multiply by perpendicular vertical height (h).



★ Golden Rule: Half the sum of parallel sides times vertical height!

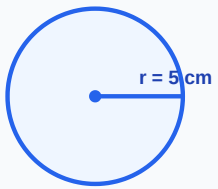
⚠ Watch Out: Examiners include slanted side lengths as a trap. Cross out the slant immediately.

23. Area of a Circle & Semicircle

SHAPE & SPACE

Circle: π r² = 3.142 × r² | Semicircle: (3.142 × r²) ÷ 2

The ² symbol means squared: r² means radius multiplied by itself (r × r). Calculate r² (r × r) first, then multiply by 3.142 (Area = 3.142 × r²). If given diameter, halve it first to get radius. Semicircle area is half the full circle: (3.142 × r²) ÷ 2.



A = 3.142 × 5² = 78.55 cm²



Semicircle Area = (π r²) ÷ 2

(r² means r × r = 5 × 5)

★ Golden Rule: Circle Area = 3.142 × r² (where r² means Radius × Radius)!

⚠ Watch Out: Do NOT multiply diameter by itself! Always halve diameter first to find radius.

Challenge: Area of Triangle & Parallelogram

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS20

A triangular roof truss has a base of 8.4 m, perpendicular height of 3.5 m, and sloping sides of 5.5 m. Find its area.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Discard sloping side 5.5 m. Area = 1/2 × base × height = 0.5 × 8.4 × 3.5 = 14.7 m².

Challenge: Circumference of a Circle

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS19

A circular bicycle wheel has a radius of 35 cm. How far does the bicycle travel in 1 full rotation of the wheel? (Use π = 3.142)

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Diameter = 35 × 2 = 70 cm. Circumference = 3.142 × 70 = 219.94 cm (or 2.20 m).

Challenge: Area of a Circle & Semicircle

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS21

A semicircular patio has a straight diameter of 8 metres. Calculate the area of the patio. (Use π = 3.142)

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Radius = 8 ÷ 2 = 4 m. Full circle = 3.142 × 4² = 3.142 × 4 × 4 = 50.272 m². Semicircle = 50.272 ÷ 2 = 25.14 m².

Challenge: Area of a Trapezium

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS20

A pedestrian walkway has parallel edges of 7 m and 11 m, vertical width of 4.5 m, and slant edge of 5.2 m. Calculate its area.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

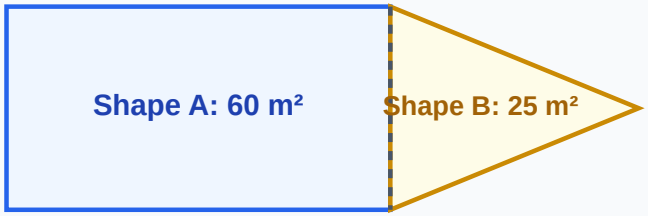
Ignore slant 5.2 m! Area = ((7 + 11) ÷ 2) × 4.5 = 9 × 4.5 = 40.5 m².

24. Area of Composite Shapes

SHAPE & SPACE

Split into Simpler Shapes, Then Add

1. Split compound floor plan into non-overlapping rectangles and triangles with dashed lines. 2. Calculate area of each individual section. 3. Sum all areas.



Total Area = 60 + 25 = 85 m²

⋆ Golden Rule: Draw clear dotted split lines and label areas 'A' and 'B'!

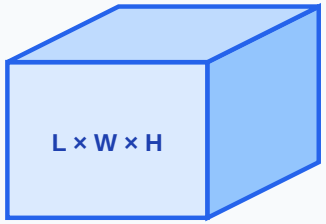
⚠ Watch Out: Check if a section is CUT OUT (e.g. pond in a lawn). If cut out, SUBTRACT its area.

25. Volume of Cuboids & Prisms

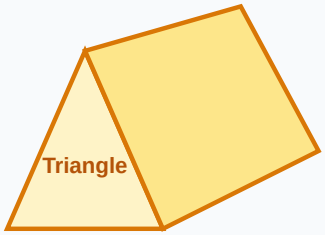
SHAPE & SPACE

Volume = Cross-Sectional Area × Length

A prism has the same 2D shape all the way through: 1. Find area of the front face (cross-section). 2. Multiply by the 3D length/depth. Units are cubic (cm³, m³).



Cuboid



Triangular Prism: Area × Length

⋆ Golden Rule: Always multiply the front face area by the full length/depth!

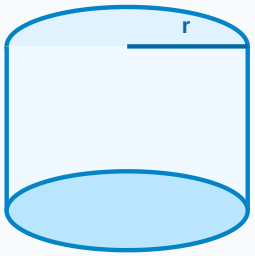
⚠ Watch Out: Don't confuse Volume (3D space in m³) with Surface Area (2D skin in m²).

26. Volume of a Cylinder

SHAPE & SPACE

Volume = π r² h = 3.142 × r² × h

A cylinder is a circular prism: 1. Circular base area = π r² = 3.142 × r² (where r² means r × r). 2. Multiply by vertical height (h): Volume = 3.142 × r² × h. Units are cubic (cm³, m³), where ³ means cubed (x × x × x). 1,000 cm³ = 1 Litre.



CYLINDER FORMULA

V = π r² h = 3.142 × r² × h

(r² means radius × radius)

1,000 cm³ = 1 Litre

⋆ Golden Rule: Cylinder Volume = 3.142 × r² × h (where r² means Radius × Radius)!

⚠ Watch Out: If given diameter, divide by 2 first to get radius! (r = d ÷ 2).

27. Surface Area of a Cuboid

SHAPE & SPACE

SA = 2(lw + lh + wh)

A cuboid has 6 rectangular faces in 3 identical opposite pairs: 1. Front & Back (l × h) × 2. 2. Top & Bottom (l × w) × 2. 3. Left & Right (w × h) × 2. Sum all 6 faces.

Box dimensions: 5 m long, 4 m wide, 3 m high

Top/Bot: 2×(5×4)=40 | Front/Back: 2×(5×3)=30 | Sides: 2×(4×3)=24

Total Surface Area = 40 + 30 + 24 = 94 m²

⋆ Golden Rule: Calculate 3 faces, add them up, then multiply by 2!

⚠ Watch Out: If painting a room, remember to EXCLUDE the floor (or ceiling and windows) if instructed!

Challenge: Volume of Cuboids & Prisms

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS23

A triangular prism has a cross-section triangle with base 6 cm and height 4 cm. Its length is 15 cm. What is its volume?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Triangle area = 1/2 × 6 × 4 = 12 cm². Volume = Cross-section × Length = 12 × 15 = 180 cm³.

Challenge: Area of Composite Shapes

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS22

A warehouse floor is a 15 m by 10 m rectangle with a semicircular loading bay of diameter 6 m extending outwards on one side. What is the total area? (π = 3.14)

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Rectangle = 15 × 10 = 150 m². Semicircle radius = 3 m. Semicircle area = (3.14 × 3²) ÷ 2 = 14.13 m². Total area = 150 + 14.13 = 164.13 m².

Challenge: Surface Area of a Cuboid

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS25

A storage crate with an open top (no lid) is 2 m long, 1.5 m wide, and 1 m high. What is its outside surface area?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Open top = 1 bottom + 2 front/backs + 2 sides. Bottom = 2 × 1.5 = 3 m². Front/Back = 2 × (2 × 1) = 4 m². Ends = 2 × (1.5 × 1) = 3 m². Total SA = 3 + 4 + 3 = 10 m².

Challenge: Volume of a Cylinder

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS24

A hot water tank is a cylinder with diameter 60 cm and height 120 cm. Calculate its capacity in litres (π = 3.14).

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

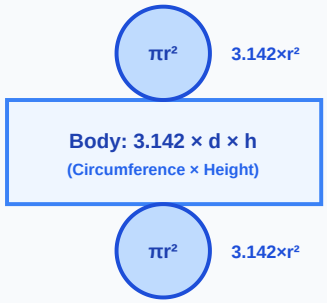
Radius = 60 ÷ 2 = 30 cm. Volume = 3.14 × 30² × 120 = 3.14 × (30 × 30) × 120 = 3.14 × 900 × 120 = 339,120 cm³. Litres = 339,120 ÷ 1,000 = 339.12 L.

28. Surface Area of a Cylinder

SHAPE & SPACE

Total SA = 2πr² + πdh = (2 × 3.142 × r²) + (3.142 × d × h)

A closed cylinder net has 3 parts: Two circular ends (2 × 3.142 × r², where r² means r × r) plus one unwrapped rectangular body (3.142 × d × h).



CLOSED CYLINDER NET

2 Ends = 2 × 3.142 × r²

+ Body = 3.142 × d × h

(r² means radius × radius)

★ Golden Rule: Closed Cylinder: 2 circular ends (2 × 3.142 × r²) + curved body (3.142 × d × h)!

⚠ Watch Out: Check if the pipe or can is OPEN at one or both ends — only add circular ends that actually exist.

29. Metric Unit Conversions

SHAPE & SPACE

Length, Mass & Capacity Ladders

Big unit to small unit = MULTIPLY (×). Small unit to big unit = DIVIDE (÷). Length: km (×1000) m (×100) cm (×10) mm. Mass: tonne (×1000) kg (×1000) g. Volume: L (×1000) mL.

METRIC CONVERSION LADDERS

1 km = 1,000 m • 1 m = 100 cm • 1 cm = 10 mm

1 tonne = 1,000 kg • 1 kg = 1,000 g

1 Litre = 1,000 ml • 1 Litre = 1,000 cm³

★ Golden Rule: Kilo = 1,000 • Centi = 1/100 • Milli = 1/1,000.

⚠ Watch Out: Don't divide 3.5 m by 100 to get cm! When going from metres to cm, you get MORE units (× 100 = 350 cm).

30. Converting Square & Cubic Units

SHAPE & SPACE

1 m² = 10,000 cm² | 1 m³ = 1,000,000 cm³

To convert square units, apply the linear conversion TWICE (100² = 10,000). To convert cubic units, apply conversion THREE TIMES (100³ = 1,000,000).

SQUARE UNITS (m²)

1 m = 100 cm

1 m² = 100 × 100 cm

1 m² = 10,000 cm²

CUBIC UNITS (m³)

1 m = 100 cm

1 m³ = 100 × 100 × 100

1 m³ = 1,000,000 cm³

★ Golden Rule: Square the conversion factor for area; cube it for volume!

⚠ Watch Out: 1 m² is NOT 100 cm²! That is the most failed unit conversion in Level 2 exams.

31. Metric to Imperial Approximations

SHAPE & SPACE

Core UK Benchmark Ratios

Memorise standard conversion factors: 5 miles ≈ 8 km (1 mile ≈ 1.6 km). 1 inch ≈ 2.5 cm. 1 kg ≈ 2.2 lbs. 1 gallon ≈ 4.5 litres. 1 foot = 12 inches ≈ 30 cm.

KEY IMPERIAL BENCHMARKS

Distance: 5 miles ≈ 8 km (or 1 mile ≈ 1.6 km)

Weight: 1 kg ≈ 2.2 pounds (lbs)

Liquid: 1 gallon ≈ 4.5 litres • 1 pint ≈ 568 ml

Length: 1 inch ≈ 2.54 cm • 1 foot ≈ 30 cm

★ Golden Rule: 5 miles = 8 km. To convert km to miles, multiply by 5 then divide by 8.

⚠ Watch Out: Kilograms are HEAVIER than pounds! 10 kg is about 22 lbs, NOT 4.5 lbs.

Challenge: Metric Unit Conversions

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS26

A patient requires 450 mg of medication per dose, 3 times a day. How many grams of medication is taken in 4 days?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Per day = 450 × 3 = 1,350 mg. In 4 days = 1,350 × 4 = 5,400 mg. Grams = 5,400 ÷ 1,000 = 5.4 g.

Challenge: Surface Area of a Cylinder

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS25

A closed tin can has radius 4 cm and height 10 cm. Find total surface area to 1 decimal place (π = 3.142).

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

2 Ends = 2 × 3.142 × 4² = 2 × 3.142 × 16 = 100.54 cm². Curved body = 3.142 × 8 × 10 = 251.36 cm². Total SA = 100.54 + 251.36 = 351.9 cm².

Challenge: Metric to Imperial Approximations

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS28

A UK highway speed limit is 70 mph. A foreign driver's speedometer is in km/h. What is 70 mph in km/h? (5 miles ≈ 8 km)

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

70 ÷ 5 = 14. 14 × 8 = 112 km/h.

Challenge: Converting Square & Cubic Units

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS27

Convert 3.5 m² into cm².

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

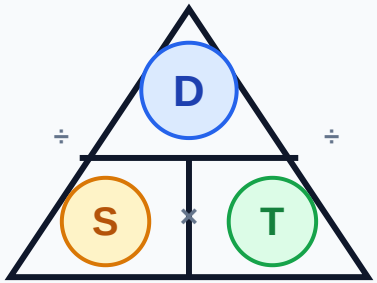
1 m² = 10,000 cm². 3.5 × 10,000 = 35,000 cm².

32. Speed, Distance & Time

SHAPE & SPACE

D = S × T | Decimal Time Warning

Distance = Speed × Time. Speed = Distance ÷ Time. Time = Distance ÷ Speed. Decimal hours must be converted: Multiply decimal minutes by 60 (0.4 hrs = 24 mins).



S-D-T FORMULAS

- Distance = Speed × Time
- Speed = Distance ÷ Time
- Time = Distance ÷ Speed

⚠️ 45m = 0.75h (NEVER 0.45!)

★ Golden Rule: Minutes to decimal: Divide by 60. Decimal to minutes: Multiply by 60.

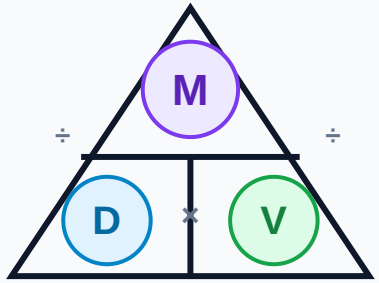
⚠️ Watch Out: 2 hours 45 minutes on a calculator is 2.75, NEVER 2.45!

33. Density, Mass & Volume

SHAPE & SPACE

M = D × V (MDV Triangle)

Mass = Density × Volume. Density = Mass ÷ Volume (written in g/cm³ or kg/m³). Volume = Mass ÷ Density. Density measures how tightly matter is packed.



M-D-V FORMULAS

- Mass = Density × Volume
- Density = Mass ÷ Volume
- Volume = Mass ÷ Density

⚖️ g with cm³ | kg with m³

★ Golden Rule: Remember: 'Department of Motor Vehicles' (D-M-V) or Mass on top!

⚠️ Watch Out: Check units: If density is in g/cm³, volume must be in cm³ and mass must be in grams.

34. Scale Drawings & Maps

SHAPE & SPACE

Scale Ratios (1 : n)

Scale 1 : 50 means 1 cm on paper represents 50 cm in real life (0.5 m). Scale 1 : 25,000 means 1 cm = 250 m. Map to Real: MULTIPLY. Real to Map: DIVIDE.

ARCHITECTURAL: 1:50

1 cm on plan = 50 cm real

50 cm = 0.5 metres

6 cm on plan = 3.0 m

MAP SCALE: 1:25,000

1 cm = 25,000 cm

25,000 cm ÷ 100 = 250 m

4 cm = 1,000 m (1 km)

★ Golden Rule: Both sides of a scale ratio start in the SAME unit (centimetres)!

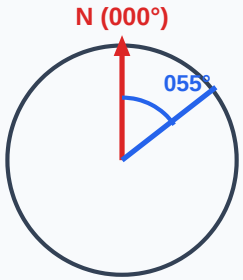
⚠️ Watch Out: Always convert cm to metres (÷ 100) or km (÷ 100,000) for real-world answers.

35. Bearings (3-Figure Bearings)

SHAPE & SPACE

North • Clockwise • 3 Figures

1. Measure angle clockwise starting from North line (000°). 2. Always write using THREE digits (e.g. 045°, 090°, 260°). Back bearing = Add or subtract 180°.



★ Golden Rule: Always 3 digits! 55° must be written as 055°.

⚠️ Watch Out: Check which location the bearing is measured FROM: 'Bearing of B from A' means draw North at A!

Challenge: Density, Mass & Volume

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS30

A solid brass statue has a volume of 450 cm³ and a density of 8.5 g/cm³. What is the mass of the statue in kilograms?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Mass = Density × Volume = 8.5 × 450 = 3,825 grams. In kilograms = 3,825 ÷ 1,000 = 3.825 kg.

Challenge: Speed, Distance & Time

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS29

A train travels 140 miles at an average speed of 56 mph. How long does the journey take in hours and minutes?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Time = Distance ÷ Speed = 140 ÷ 56 = 2.5 hours. 0.5 hours × 60 = 30 minutes. Journey time = 2 hours 30 minutes.

Challenge: Bearings (3-Figure Bearings)

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS32

The bearing of town B from town A is 072°. What is the back-bearing of town A from town B?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Since 072° < 180°, add 180°: Back bearing = 072° + 180° = 252°.

Challenge: Scale Drawings & Maps

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS31

On an architectural blueprint with scale 1 : 200, a sports hall wall measures 8.5 cm. What is the actual length of the wall in metres?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Real length = 8.5 × 200 = 1,700 cm. In metres = 1,700 ÷ 100 = 17 metres.

36. Plans & Elevations

SHAPE & SPACE

Plan View • Front • Side Elevation

Plan View = Looking straight down from the bird's eye view. Front Elevation = Looking straight from front. Side Elevation = Looking straight from side.



⋆ Golden Rule: Draw each 2D projection as if you are standing directly in line with that face!

⚠️ Watch Out: Plan view NEVER shows height — it only shows width and depth.

37. Averages: Mean, Median, Mode & Range

HANDLING DATA

The 4 Core Statistics

Mean = Sum of values ÷ Count. Median = Middle value after ordering low to high. Mode = Most frequent value. Range = Highest - Lowest (spread, NOT an average).

MEAN = Total ÷ Count

Add all up and share equally

MEDIAN = Middle Value

MUST put in order first!

MODE = Most Common

Number that appears most often

RANGE = High - Low

Spread of data (NOT an average!)

⋆ Golden Rule: Range is a measure of consistency/spread, NEVER an average!

⚠️ Watch Out: For the median with an even count of numbers, add the middle two and divide by 2.

38. Mean from a Frequency Table

HANDLING DATA

Mean = Σfx ÷ Σf

1. Multiply value (x) by frequency (f) to make an fx column. 2. Sum the fx column (Total score Σfx). 3. Sum the frequency column (Total count Σf). 4. Divide: Σfx ÷ Σf.

Score (x)	Freq (f)	x × f (fx)
1	3	1 × 3 = 3
2	5	2 × 5 = 10
13 ÷ 8 = 1.625	Total f = 8	Total fx = 13

⋆ Golden Rule: Divide by total frequency (Σf), NEVER by the number of rows in the table!

⚠️ Watch Out: Don't just average the 'x' column — that ignores how many times each value occurred.

39. Estimated Mean (Grouped Data)

HANDLING DATA

Midpoints × Frequency ÷ Total Frequency

1. Find midpoint of each group: (Lower bound + Upper bound) ÷ 2. 2. Multiply midpoint by frequency (mp × f). 3. Sum (mp × f). 4. Divide by total frequency.

Grouped Data: 10 < t ≤ 20 (f = 4)

1. Midpoint = (10 + 20) ÷ 2 = 15

2. mp × f = 15 × 4 = 60

Estimated Mean = Total (mp × f) ÷ Total Frequency

⋆ Golden Rule: It is an 'ESTIMATE' because the exact individual raw values inside the group are unknown!

⚠️ Watch Out: Be careful finding midpoints: for 0 to 10, midpoint is 5. For 20 to 50, midpoint is 35 (not 30).

Challenge: Averages: Mean, Median, Mode & Range

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L2.HD34

Find the median and range of this dataset: 14, 8, 22, 19, 14, 27, 11.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Ordered: 8, 11, 14, 14, 19, 22, 27 (7 numbers). Middle is 4th number = 14 (Median). Range = 27 - 8 = 19.

Challenge: Plans & Elevations

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS33

A 3D stepped staircase block is 3 cubes wide, 2 cubes deep, and 3 cubes high. Which 2D elevation shows the step pattern?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

The side elevation shows the step profile (staircase outline).

Challenge: Estimated Mean (Grouped Data)

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L2.HD36

Times: 0–10 mins (f=2, mp=5), 10–20 mins (f=6, mp=15), 20–30 mins (f=2, mp=25). Calculate estimated mean.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Total (mp × f) = (5×2) + (15×6) + (25×2) = 10 + 90 + 50 = 150. Total f = 2 + 6 + 2 = 10. Estimated Mean = 150 ÷ 10 = 15 mins.

Challenge: Mean from a Frequency Table

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L2.HD35

Cars per household: 0 cars (f=4), 1 car (f=10), 2 cars (f=6). Calculate the mean number of cars per household.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

fx: (0×4) + (1×10) + (2×6) = 0 + 10 + 12 = 22. Total frequency Σf = 4 + 10 + 6 = 20. Mean = 22 ÷ 20 = 1.1 cars.

40. Modal Class & Median Group

HANDLING DATA

Identifying Class Intervals

Modal Class = The group interval with the HIGHEST frequency number. Median Group = The interval that contains the middle position ((Total f + 1) ÷ 2).

MODAL CLASS

Look down frequency column

HIGHEST Frequency

MEDIAN GROUP

Find position: (n + 1) ÷ 2

Cumulative Count

⋆ **Golden Rule: Always write the ENTIRE interval inequality (e.g. 20 < t ≤ 30), not just one number!**

⚠️ **Watch Out: The modal class is the interval itself, NOT the frequency count.**

41. Comparing Two Data Sets

HANDLING DATA

Average + Spread in Context

To earn full marks on Pearson comparison questions, you MUST state: 1. One Average comparison in context ('Branch A has a higher mean sales by £50'). 2. One Range/Spread comparison ('Branch B is more consistent because its range is smaller').

PEARSON GOLDEN 2-PART COMPARISON

1. AVERAGE: "Group A on average is higher because mean is..."
2. SPREAD: "Group B is more consistent because range is smaller..."

⚠️ Must mention numbers AND the real-world context!

⋆ **Golden Rule: Smaller range = more consistent / reliable results!**

⚠️ **Watch Out: Always write comparison words (higher, lower, more consistent) in context!**

42. Probability of Single Events

HANDLING DATA

P(Event) = Favourable ÷ Total

Probability is always between 0 (Impossible) and 1 (Certain). Can be written as a fraction, decimal, or percentage. P(Not event) = 1 - P(Event).

P = Favourable Outcomes ÷ Total



0 (Impossible) 0.5 (Even chance) 1 (Certain)

⋆ **Golden Rule: Probabilities can NEVER be greater than 1 or negative!**

⚠️ **Watch Out: Don't write probability as a ratio (e.g. 1 : 4 is WRONG). Write as a fraction (1/5), decimal (0.2), or % (20%).**

43. Combined Independent Events

HANDLING DATA

P(A and B) = P(A) × P(B)

Independent events do not affect each other (e.g. flipping a coin and rolling a die). 'AND' rule = MULTIPLY probabilities. 'OR' rule = ADD probabilities.

P(A and B) = P(A) × P(B)

Flipping Heads AND rolling a 6:

P(Heads) = 1/2 • P(Rolling a 6) = 1/6

Combined: 1/2 × 1/6 = 1/12

⋆ **Golden Rule: 'AND' means MULTIPLY; 'OR' means ADD!**

⚠️ **Watch Out: Don't add when you need both to happen: 1/2 + 1/6 = 4/6 (which is much too high for both happening together).**

Challenge: Comparing Two Data Sets

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L2.HD38

Shift 1: Mean = 45 units, Range = 12. Shift 2: Mean = 41 units, Range = 5. Compare the two shifts.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

1. Shift 1 produced more units on average (mean of 45 vs 41). 2. Shift 2 was more consistent because its range was smaller (range of 5 vs 12).

Challenge: Modal Class & Median Group

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L2.HD37

Salaries: £15k–£20k (f=5), £20k–£25k (f=12), £25k–£30k (f=3). State: (a) Modal class, (b) Median group.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

(a) Modal class = £20k–£25k (highest frequency 12). (b) Total f = 20. Middle is 10.5th person. 5 in 1st group, next 12 in 2nd group. Median group = £20k–£25k.

Challenge: Combined Independent Events

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L2.HD40

The probability that train A is delayed is 0.2. The probability that train B is delayed is 0.15. What is the probability that BOTH trains are delayed?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

P(Both delayed) = 0.2 × 0.15 = 0.03 (or 3%).

Challenge: Probability of Single Events

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L2.HD39

A bag contains 5 red, 7 blue, and 8 green tokens. What is the probability of picking a token that is NOT red? Give as a simplified fraction.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

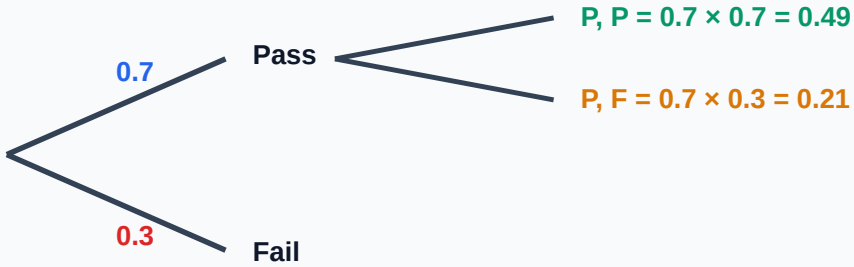
Total tokens = 5 + 7 + 8 = 20. Non-red tokens = 7 + 8 = 15. Probability = 15/20 = 3/4.

44. Probability Tree Diagrams

HANDLING DATA

Multiply Across, Add Down

1. Branches from any single node must add up to 1. 2. MULTIPLY along branches to find outcome probability. 3. ADD individual outcome probabilities together if multiple paths satisfy the question.



⋆ Golden Rule: Every pair of branch splits MUST add up to 1 (100%)!

⚠️ Watch Out: Check whether picking is WITH replacement or WITHOUT replacement (fractions change on 2nd pick!).

45. Two-Way Tables & Probability

HANDLING DATA

Row Totals + Column Totals

1. Every row must add up to its row total. 2. Every column must add up to its column total. 3. Grand total bottom-right must match both row totals and column totals.

Group	Bus	Walk	Total
Men	18	12	30
Women	22	28	50
Total	40	40	80

⋆ Golden Rule: Find the cell with only ONE missing number in its row or column first!

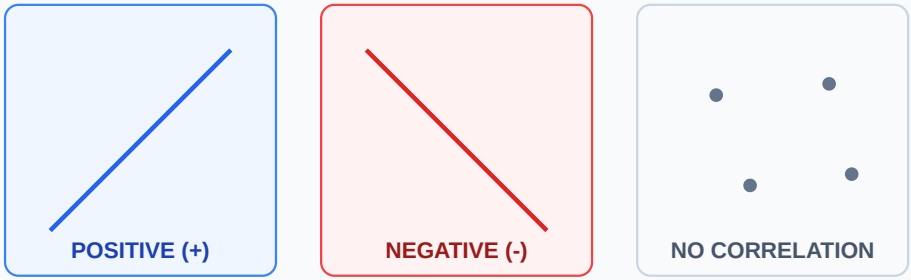
⚠️ Watch Out: For conditional probability ('Given that the person is a woman...'), divide by the WOMEN TOTAL (50), not the grand total (80)!

46. Scatter Diagrams & Correlation

HANDLING DATA

Positive • Negative • Zero Correlation

Positive: As X goes up, Y goes up (slopes uphill). Negative: As X goes up, Y goes down (slopes downhill). Zero: Scattered random cloud with no trend.



⋆ Golden Rule: Draw a line of best fit with an equal balance of points above and below the line!

⚠️ Watch Out: Correlation does NOT prove causation! (e.g. ice cream sales and sunburn both correlate with hot weather).

47. Interpolation vs Extrapolation

HANDLING DATA

Inside Range (Reliable) vs Beyond (Unreliable)

Interpolation: Estimating a value INSIDE the given data range (Reliable). Extrapolation: Extending line of best fit BEYOND the data points (Unreliable / Risky).

INTERPOLATION

Within tested range

e.g. data for 10–50 yrs

RELIABLE PREDICTION

EXTRAPOLATION

Outside data range

Predicting at 90 yrs

UNRELIABLE / RISKY

⋆ Golden Rule: Predictions outside the original data range are UNRELIABLE because trends may flatten or reverse!

⚠️ Watch Out: Examiners will ask 'Why might this prediction be unreliable?' — answer: 'It is extrapolated beyond the data range.'

Challenge: Two-Way Tables & Probability

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L2.HD42

In a staff survey of 80 people (30 Men: 18 Bus, 12 Walk; 50 Women: 22 Bus, 28 Walk), what is the probability that a random person is a man who walks to work?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Men who walk = 12. Grand total = 80. Probability = 12/80 = 3/20 (or 0.15 / 15%).

Challenge: Probability Tree Diagrams

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L2.HD41

A student has an 80% (0.8) chance of passing Theory, and a 70% (0.7) chance of passing Practical. What is the probability of passing AT LEAST ONE test?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

P(Fails both) = (1 - 0.8) × (1 - 0.7) = 0.2 × 0.3 = 0.06. P(At least one pass) = 1 - 0.06 = 0.94 (or 94%).

Challenge: Interpolation vs Extrapolation

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L2.HD44

A company measures sales between 2018 and 2023. They predict sales for 2035 using their line of best fit. Is this reliable?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

No, it is unreliable because 2035 is an extrapolation far beyond the recorded data range, and market conditions may change.

Challenge: Scatter Diagrams & Correlation

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L2.HD43

Describe the relationship between temperature and heating fuel consumption: As temperature rises, heating fuel usage drops.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

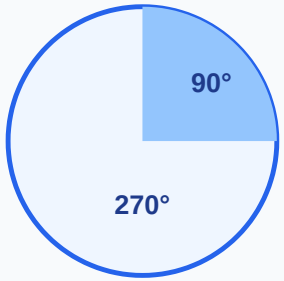
Negative correlation (as temperature increases, heating fuel used decreases).

48. Pie Charts: Angles & Frequencies

HANDLING DATA

Angle = (Value ÷ Total) × 360°

Total degrees in any circle = 360°. Degrees per item = 360° ÷ Total Frequency. To find angle for a category: Multiply its frequency by degrees per item.



Degrees per 1 item:

360° ÷ Total Frequency

e.g. 360° ÷ 72 people = 5° each

⋆ **Golden Rule: Sum of all pie chart sector angles MUST equal exactly 360°!**

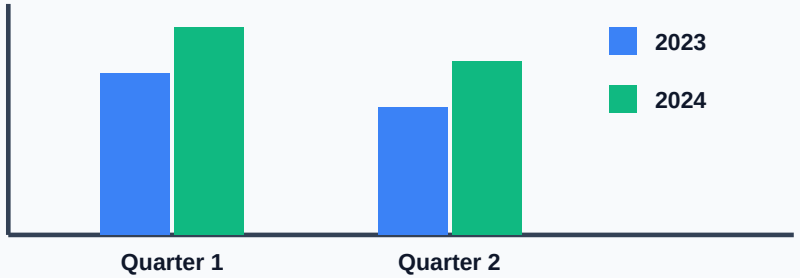
⚠️ **Watch Out: A pie chart only shows PROPORTIONS, not raw quantities, unless the total sample size is given!**

49. Dual & Grouped Bar Charts

HANDLING DATA

Side-by-Side Comparison with Key

Dual bar charts place two bars side-by-side for each category to compare groups (e.g. Male vs Female, 2023 vs 2024). Always check the legend key and vertical axis scale.



⋆ **Golden Rule: Read the key carefully to avoid mixing up years or cohorts!**

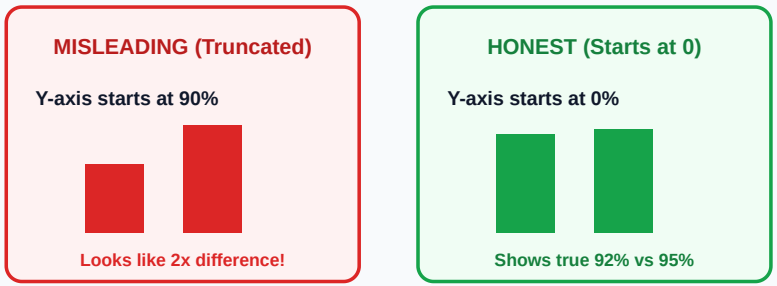
⚠️ **Watch Out: Check the vertical axis scale increments — each grid square may represent 2, 5, 10, or 20 units.**

50. Misleading Graphs & Traps

HANDLING DATA

Non-Zero Baselines & Uneven Scales

Common tricks used to deceive: 1. Broken vertical axis (not starting at 0) exaggerates tiny differences. 2. Unequal scale increments. 3. 3D distortion making front slices look bigger.



⋆ **Golden Rule: Always check if the vertical axis starts at ZERO (0)!**

⚠️ **Watch Out: A graph that looks dramatic often only represents a 1% or 2% actual difference.**

1. Place Value up to 1,000,000

NUMBER SYSTEM

Millions, Thousands & Ones

Numbers are grouped in 3-digit periods separated by commas or spaces: Millions (M), Hundred Thousands (HTh), Ten Thousands (TTh), Thousands (Th), Hundreds (H), Tens (T), Units (U).

482,519 (Four Hundred & Eighty-Two Thousand...)

4 HTh	8 TTh	2 Th	5 H	1 T	9 U

The digit '8' has a value of 80,000 (Eighty Thousand)

⋆ **Golden Rule: Read large numbers in groups of three digits from left to right!**

⚠️ **Watch Out: Don't confuse 'Eight Thousand' (8,000) with 'Eighty Thousand' (80,000).**

Challenge: Dual & Grouped Bar Charts

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L2.HD46

In Quarter 1: 2023 sales were £35k, 2024 sales were £45k. What was the percentage increase from 2023 to 2024?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Increase = £45k - £35k = £10k. % Increase = (10 ÷ 35) × 100 = 28.57% (or 28.6%).

Challenge: Pie Charts: Angles & Frequencies

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L2.HD45

In a survey of 120 hospital patients, 45 travelled by bus. What angle should represent bus travel on a pie chart?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Degrees per patient = 360° ÷ 120 = 3° per patient. Bus angle = 45 × 3° = 135°.

Challenge: Place Value up to 1,000,000

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L1.N1

What is the value of the digit 7 in the number 574,820?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

The digit 7 is in the ten thousands column, so its value is 70,000 (Seventy Thousand).

Challenge: Misleading Graphs & Traps

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L2.HD47

Explain one way a bar chart comparing company satisfaction could be misleading if one bar appears 4 times taller than another.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

The vertical axis does not start at zero (truncated/broken axis), which exaggerates the visual difference between the bars.

2. Four Operations with Whole Numbers

NUMBER SYSTEM

Column Methods & Checking

Addition & Subtraction: Line up unit digits strictly on the right. Carry over when sum ≥ 10; borrow from next column to the left for subtraction. Multiplication: Long column. Division: Short 'bus stop'.

SUBTRACTION

5 2 4
- 1 6 8
---
3 5 6

SHORT DIVISION

6) 8 5 ²4
---
1 4 2

★ **Golden Rule: Always check subtraction by adding your answer to the smaller number!**

⚠️ **Watch Out: Don't subtract smaller from larger in the same column! If top digit is smaller, borrow 1 from left.**

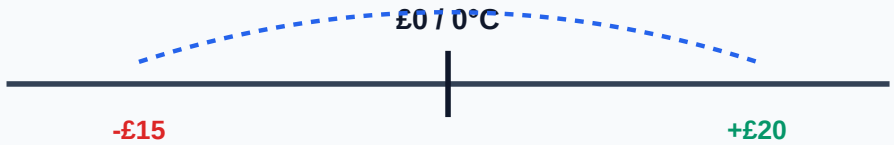
3. Negative Numbers (Temp & Debt)

NUMBER SYSTEM

Below Zero & Bank Overdrafts

Numbers below 0 have a minus sign (-). Moving RIGHT on the number line = getting warmer / paying in money (+). Moving LEFT = getting colder / spending money (-).

Difference = £35



-£15 is DEEPER in debt than -£5

★ **Golden Rule: -8°C is COLDER than -3°C!**

⚠️ **Watch Out: Overdraft balance: If balance is -£25 and you spend £10, new balance is -£35 (NOT -£15).**

4. Rounding to 10, 100, 1000

NUMBER SYSTEM

Look at the Neighbour on the Right

Find the rounding place: Look at the next digit to the right: If 5 or more (5, 6, 7, 8, 9), ROUND UP. If 4 or less (0, 1, 2, 3, 4), ROUND DOWN (stay same). Fill trailing places with 0.

Round 4,763 to the nearest 100:

Hundreds digit is 7. Neighbour is 6.

6 is ≥ 5 → Round up to 4,800

★ **Golden Rule: 5 to 9 climbs the vine; 0 to 4 stays on the floor!**

⚠️ **Watch Out: Don't delete digits! When rounding 3,428 to the nearest 10, it becomes 3,430, NOT 343.**

5. Decimals: Place Value & Ordering

NUMBER SYSTEM

Tenths, Hundredths & Thousandths

To compare decimals: Line up decimal points vertically and pad with trailing zeros so all numbers have the same number of digits after the dot (e.g. 0.4 becomes 0.400).

Which is bigger: 0.35 or 0.4?

Pad with a zero: 0.4 → 0.40

Compare 35 hundredths vs 40 hundredths

0.4 is BIGGER than 0.35!

★ **Golden Rule: Add trailing zeroes to level the playing field before comparing decimals!**

⚠️ **Watch Out: 0.35 has more digits than 0.4, but 0.4 is LARGER! (0.40 > 0.35).**

Challenge: Negative Numbers (Temp & Debt)

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L1.N3

A bank account is overdrawn by -£45. A deposit of £70 is paid in. What is the new bank balance?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

New balance = -45 + 70 = +£25.

Challenge: Four Operations with Whole Numbers

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L1.N2

Calculate: 735 - 289. Show your check using addition.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

735 - 289 = 446. Check: 446 + 289 = 735 ✓.

Challenge: Decimals: Place Value & Ordering

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L1.N5

Put in order from smallest to largest: 0.08, 0.8, 0.75, 0.079.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Pad to 3 decimal places: 0.080, 0.800, 0.750, 0.079. Order: 0.079, 0.08, 0.75, 0.8.

Challenge: Rounding to 10, 100, 1000

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L1.N4

Round 18,450: (a) To the nearest 1,000. (b) To the nearest 100.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

(a) To nearest 1,000: 4 is < 5 → 18,000. (b) To nearest 100: 5 is ≥ 5 → 18,500.

6. Operations with Decimals & Money

NUMBER SYSTEM

Line Up the Decimal Points

Always align the decimal points directly under each other when adding or subtracting money. Money must ALWAYS have TWO decimal places (e.g. £4.50, NEVER £4.5).

LINE UP THE DOTS

£14.50

+£3.85

£18.35

★ Golden Rule: Always write pounds with 2 decimal places: £6.20, NOT £6.2!

⚠ Watch Out: Don't mix pounds and pence without converting! £5 + 40p = £5.40, NOT £45.

7. FDP Common Benchmarks

NUMBER SYSTEM

Halves, Quarters & Tenths

1/2 = 0.5 = 50%. 1/4 = 0.25 = 25%. 3/4 = 0.75 = 75%. 1/10 = 0.1 = 10%. 1/5 = 0.2 = 20%. Memorising these visual equivalents saves time in exams.

Fraction	Decimal	Percentage
1/2	0.5	50%
1/4	0.25	25%
3/4	0.75	75%

★ Golden Rule: 1/5 is 20% (0.2), NOT 15%!

⚠ Watch Out: Don't confuse 0.1 (10%) with 0.01 (1%).

8. Simplifying Fractions

NUMBER SYSTEM

Divide Top and Bottom Equally

To simplify a fraction, divide both the numerator (top) and denominator (bottom) by their Highest Common Factor (HCF). Keep dividing until no common factors remain.

Simplify 12/18

Divide top and bottom by 6 (HCF):

12 ÷ 6 / 18 ÷ 6 = 2/3

★ Golden Rule: Whatever you do to the top, you MUST do to the bottom!

⚠ Watch Out: Never subtract numbers to simplify fractions — you can ONLY divide or multiply.

9. Fraction of an Amount (Basic)

NUMBER SYSTEM

Divide by Denominator, Times by Numerator

To find a unit fraction (1/n): Divide by n. To find a non-unit fraction (e.g. 3/4): 1. Divide by 4 to get 1 quarter. 2. Multiply result by 3 to get 3 quarters.

Find 3/4 of 48 kg

Step 1: 48 ÷ 4 = 12 kg (1/4)

Step 2: 12 × 3 = 36 kg

★ Golden Rule: Divide by bottom, multiply by top!

⚠ Watch Out: Don't stop after step 1! Dividing by 4 gives only ONE quarter (12 kg), you need THREE.

Challenge: FDP Common Benchmarks

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L1.N7

Which is greater: 0.3 or 1/4? Show your working.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

1/4 = 0.25. Since 0.30 > 0.25, 0.3 is greater.

Challenge: Operations with Decimals & Money

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L1.N6

You buy items costing £3.45, £8.20, and £0.95. You pay with a £20 note. How much change do you receive?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Total = £3.45 + £8.20 + £0.95 = £12.60. Change = £20.00 - £12.60 = £7.40.

Challenge: Fraction of an Amount (Basic)

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L1.N9

A community group has 60 members. 2/5 of them are volunteers. How many volunteers are there?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

60 ÷ 5 = 12 (1/5). 12 × 2 = 24 volunteers.

Challenge: Simplifying Fractions

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L1.N8

Write 24/36 in its simplest form.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Divide by 12: 24 ÷ 12 = 2. 36 ÷ 12 = 3. Simplest form = 2/3.

10. Percentages (Building Blocks)

NUMBER SYSTEM

10%, 5%, 1% Breakdown Method

Non-calculator percentage hacks: 10% = Divide by 10. 5% = Half of 10%. 1% = Divide by 100. 50% = Half (÷ 2). Combine blocks to find any percentage (e.g. 15% = 10% + 5%).

Find 15% of £80 without calculator

10% of £80 = £80 ÷ 10 = £8.00

5% of £80 = Half of £8 = £4.00

15% = £8.00 + £4.00 = £12.00

⋆ Golden Rule: Find 10% first — it unlocks 5%, 20%, 30%, and 15% instantly!

⚠️ Watch Out: Don't divide by 15 to find 15%! You find 10% and add 5%.

11. Simple Ratios (2 Parts)

NUMBER SYSTEM

Add Parts, Divide Total, Multiply

To share £60 in ratio 1 : 3: 1. Add parts: 1 + 3 = 4 parts. 2. Divide total: £60 ÷ 4 = £15 (value of 1 part). 3. Multiply: Share 1 = £15, Share 2 = 3 × £15 = £45.

Share £60 in ratio 1 : 3



Result = £15 and £45 (Total = £60 ✓)

⋆ Golden Rule: Ratios must always be simplified to lowest whole numbers!

⚠️ Watch Out: Keep the order of the ratio correct: Person A : Person B must match the order in the text.

12. Direct Proportion (Recipes)

NUMBER SYSTEM

Unitary Method: Find for 1 First

To scale a recipe: 1. Divide ingredients by the original number of servings to find the amount for 1 person. 2. Multiply by the new number of people needed.

Recipe for 4 people: 200g flour. Need for 6 people?

Step 1 (For 1 person): 200g ÷ 4 = 50g

Step 2 (For 6 people): 50g × 6 = 300g flour

⋆ Golden Rule: Find the quantity for ONE person first, then multiply!

⚠️ Watch Out: Don't just add 2 to the flour because 6 is 4 + 2 — you must scale proportionally!

13. Multiples, Factors & Primes

NUMBER SYSTEM

Definitions & Factor Pairs

Multiple = In the times table (Multiples of 5: 5, 10, 15...). Factor = Numbers that divide in exactly with no remainder. Prime = EXACTLY two factors: 1 and itself (2, 3, 5, 7, 11...).

FACTORS OF 12

1 × 12

2 × 6

3 × 4

PRIME NUMBERS < 20

2, 3, 5, 7, 11, 13, 17, 19

⚠️ 1 is NOT prime!

2 is the ONLY even prime

⋆ Golden Rule: 1 is NOT a prime number because it only has 1 factor!

⚠️ Watch Out: Factors are smaller (or equal); multiples are larger (or equal).

Challenge: Simple Ratios (2 Parts)

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L1.N11

Simplify the ratio 15 : 25 to its simplest form.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Divide both by 5: 15 ÷ 5 = 3, 25 ÷ 5 = 5. Simplified ratio = 3 : 5.

Challenge: Percentages (Building Blocks)

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L1.N10

Work out 35% of £240 without using a calculator.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

10% = £24. 30% = 24 × 3 = £72. 5% = half of £24 = £12. 35% = £72 + £12 = £84.

Challenge: Multiples, Factors & Primes

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L1.N13

List all the factors of 20.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Factor pairs: 1 × 20, 2 × 10, 4 × 5. Factors of 20: 1, 2, 4, 5, 10, 20.

Challenge: Direct Proportion (Recipes)

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L1.N12

A soup recipe for 8 people needs 600 ml of stock. How much stock is needed for 12 people?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

For 1 person: 600 ÷ 8 = 75 ml. For 12 people: 75 × 12 = 900 ml.

14. Inverse Operations for Checking

NUMBER SYSTEM

Reverse Your Calculation for 1 Mark

Pearson exams award a dedicated mark for 'Show a check of your answer': Addition check with SUBTRACTION. Subtraction check with ADDITION. Multiplication check with DIVISION. Division check with MULTIPLICATION.

PEARSON SECTION A CHECKING HACK

Main Calculation: 45 × 8 = 360

Written Check: 360 ÷ 8 = 45 ✓

★ **Golden Rule: Use reverse operations (add ↔ subtract, multiply ↔ divide) to check!**

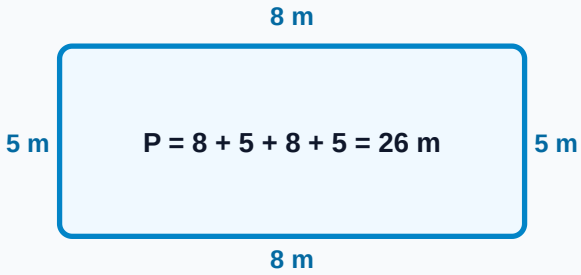
⚠️ **Watch Out: Don't repeat the same sum — examiners require the inverse operation!**

15. Perimeter of Rectangle & Square

MEASURES & SPACE

Add All Four Outside Edges

Perimeter is the total boundary distance around a 2D shape. Rectangle: Perimeter = Length + Width + Length + Width = 2l + 2w. Units are linear (cm, m).



★ **Golden Rule: Perimeter is the border fence around the shape!**

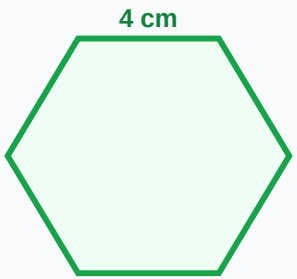
⚠️ **Watch Out: Don't just add 8 + 5 = 13! A rectangle has FOUR sides, so you must add all 4 sides.**

16. Perimeter of Regular Polygons

MEASURES & SPACE

Side Length × Number of Sides

A 'regular' polygon has sides that are ALL EQUAL in length: Equilateral Triangle = 3 × side. Pentagon = 5 × side. Hexagon = 6 × side. Octagon = 8 × side.



Regular Hexagon

6 equal sides of 4 cm

Perimeter = 6 × 4 = 24 cm

★ **Golden Rule: Regular means all sides and angles are identical!**

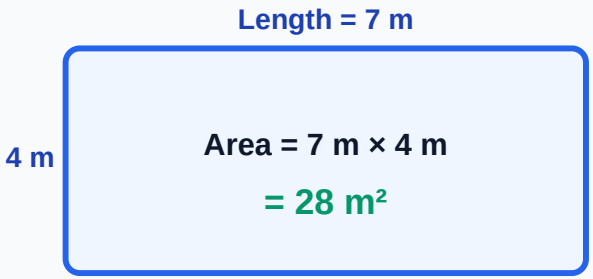
⚠️ **Watch Out: Don't count the sides incorrectly: Hexagon = 6, Octagon = 8, Pentagon = 5.**

17. Area of a Rectangle

MEASURES & SPACE

Area = Length × Width

Area measures the flat 2D space inside a shape. Multiply length by width. Answers are ALWAYS written in square units (cm², m², mm²).



★ **Golden Rule: Area is flat surface (carpet/grass), measured in square units (m²)!**

⚠️ **Watch Out: Don't confuse Area (L × W) with Perimeter (L + W + L + W).**

Challenge: Perimeter of Rectangle & Square

MEASURES & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L1.MSS15

A square garden has a side length of 7.5 metres. How many metres of fencing are needed to go all the way around?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

A square has 4 equal sides. Perimeter = 4 × 7.5 = 30 metres.

Challenge: Inverse Operations for Checking

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L1.N14

You calculated: 156 + 89 = 245. Write down a valid check of your working using inverse operations.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Check: 245 - 89 = 156 (or 245 - 156 = 89).

Challenge: Area of a Rectangle

MEASURES & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L1.MSS16

A living room floor is 6.5 m long and 4 m wide. How many square metres of carpet are needed to cover it?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Area = 6.5 × 4 = 26 m².

Challenge: Perimeter of Regular Polygons

MEASURES & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L1.MSS15

A stop sign is a regular octagon with each side measuring 25 cm. What is the total perimeter in metres?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

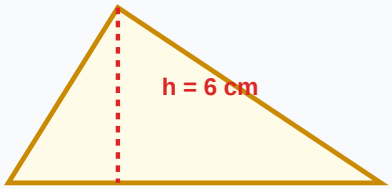
Octagon has 8 sides. Perimeter = 8 × 25 cm = 200 cm = 2.0 metres.

18. Area of a Triangle

MEASURES & SPACE

Area = (Base × Height) ÷ 2

1. Multiply base by perpendicular vertical height. 2. HALVE the answer (divide by 2) because a triangle is half of a rectangle.



Area = (10 × 6) ÷ 2 = 30 cm²

★ Golden Rule: Always remember to divide by 2 for a triangle!

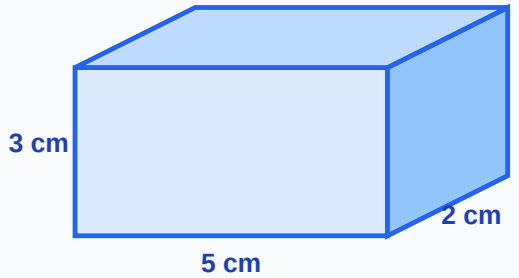
⚠️ Watch Out: Never use the slanted side length for area — only use vertical perpendicular height.

19. Volume of a Cuboid

MEASURES & SPACE

Volume = Length × Width × Height

Volume measures the 3D space inside a solid box. Multiply all three dimensions together: Length × Width × Height. Written in cubic units (cm³, m³).



V = 5 × 2 × 3

= 30 cm³

★ Golden Rule: Volume is 3D capacity, measured in cubic units (cm³ or m³)!

⚠️ Watch Out: Don't add the three dimensions — you must MULTIPLY them.

20. Metric Length Conversions

MEASURES & SPACE

mm • cm • m • km

1 cm = 10 mm. 1 m = 100 cm. 1 km = 1,000 m. Big to small unit = MULTIPLY (×). Small to big unit = DIVIDE (÷).

METRIC LENGTH STEPS

1 km = 1,000 m (e.g. 3.2 km = 3,200 m)

1 m = 100 cm (e.g. 4.5 m = 450 cm)

1 cm = 10 mm (e.g. 8 cm = 80 mm)

★ Golden Rule: Metres to cm: MULTIPLY by 100. cm to metres: DIVIDE by 100.

⚠️ Watch Out: Don't confuse mm and cm: 45 mm is 4.5 cm, NOT 450 cm.

21. Metric Weight Conversions

MEASURES & SPACE

1 kg = 1,000 g | 1 tonne = 1,000 kg

To convert kg to g: MULTIPLY by 1,000. To convert g to kg: DIVIDE by 1,000. For medicine: 1 g = 1,000 mg.

MASS: KILOGRAMS & GRAMS

kg → (× 1,000) → grams

grams → (÷ 1,000) → kg

Example: 2.5 kg = 2,500 g

★ Golden Rule: 'Kilo' always means 1,000!

⚠️ Watch Out: Half a kilogram is 500 grams, NOT 50 grams.

Challenge: Volume of a Cuboid

MEASURES & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L1.MSS18

A packing box is 40 cm long, 25 cm wide, and 20 cm high. What is its volume in cm³?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Volume = 40 × 25 × 20 = 1,000 × 20 = 20,000 cm³.

Challenge: Area of a Triangle

MEASURES & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L1.MSS17

A triangular banner has a base of 12 cm and a vertical height of 8 cm. What is its area?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Area = (12 × 8) ÷ 2 = 96 ÷ 2 = 48 cm².

Challenge: Metric Weight Conversions

MEASURES & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L1.MSS20

A postal parcel weighs 1,450 grams. What is its weight in kilograms?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

1,450 ÷ 1,000 = 1.45 kg.

Challenge: Metric Length Conversions

MEASURES & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L1.MSS19

Convert 275 cm into metres.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

275 ÷ 100 = 2.75 metres.

22. Metric Capacity Conversions

MEASURES & SPACE

1 Litre = 1,000 ml

Litres to millilitres: MULTIPLY by 1,000. Millilitres to litres: DIVIDE by 1,000. Common amounts: 1/2 Litre = 500 ml. 1/4 Litre = 250 ml. 3/4 Litre = 750 ml.

CAPACITY: LITRES & MILLILITRES

1 L = 1,000 ml

1/4 L = 250 ml • 1/2 L = 500 ml • 3/4 L = 750 ml

e.g. 1.8 Litres = 1,800 ml

⋆ Golden Rule: Milli means one-thousandth!

⚠ Watch Out: Don't divide by 100 — there are 1,000 millilitres in a litre.

23. Time: 12-Hour & 24-Hour Clock

MEASURES & SPACE

Converting & Timetables

To convert PM time to 24-hour: ADD 12 hours (e.g. 4:15 pm + 12 = 16:15). To convert 24-hour to PM: SUBTRACT 12 hours (e.g. 21:40 - 12 = 9:40 pm).

12-HOUR CLOCK

8:30 pm

Add 12 hours for PM

24-HOUR CLOCK

20:30

Always 4 digits, NO 'pm'

⋆ Golden Rule: Never write 'pm' after a 24-hour time (e.g. 20:30, NOT 20:30 pm)!

⚠ Watch Out: 12:30 pm is afternoon (lunchtime) = 12:30. 12:30 am is midnight = 00:30.

24. Money: Change & Best Buys

MEASURES & SPACE

Multi-Item Totals & Change

1. Multiply quantities by unit price. 2. Sum all items carefully. 3. Change = Amount paid - Total bill. Compare deals by finding price for the same amount.

Shopping Bill & Change

3 sandwiches @ £2.50 = £7.50

2 drinks @ £1.25 = £2.50 → Total = £10.00

Paid with £20 note → Change = £20 - £10 = £10.00

⋆ Golden Rule: Always write both decimal places for currency: £10.00, NOT £10!

⚠ Watch Out: Don't add items without multiplying by quantity first.

25. Properties of 2D Shapes

MEASURES & SPACE

Sides, Angles & Symmetry

Right Angle = 90°. Triangle angles sum to 180°. Quadrilateral angles sum to 360°. Parallel lines never meet. Perpendicular lines meet at 90°.

TRIANGLES (180°)

Equilateral: 3 equal sides

Isosceles: 2 equal sides

Scalene: 0 equal sides

QUADRILATERALS (360°)

Square: 4 equal sides, 90°

Rectangle: Opposite equal

Trapezium: 1 pair parallel

⋆ Golden Rule: All angles in ANY triangle add up to 180°!

⚠ Watch Out: A rhombus has 4 equal sides, but its angles are NOT 90° (a square has 90° angles).

Challenge: Time: 12-Hour & 24-Hour Clock

MEASURES & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L1.MSS22

A train departs at 14:45 and arrives at 16:10. How long was the journey in hours and minutes?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

From 14:45 to 15:00 is 15 mins. From 15:00 to 16:10 is 1 hr 10 mins. Total = 1 hour 25 minutes.

Challenge: Metric Capacity Conversions

MEASURES & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L1.MSS21

A bottle contains 2 Litres of water. 350 ml is poured out. How many ml remain in the bottle?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

2 Litres = 2,000 ml. Remaining = 2,000 - 350 = 1,650 ml (or 1.65 L).

Challenge: Properties of 2D Shapes

MEASURES & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L1.MSS24

Two angles in a triangle are 65° and 45°. What is the size of the third angle?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Sum of known angles = 65° + 45° = 110°. Third angle = 180° - 110° = 70°.

Challenge: Money: Change & Best Buys

MEASURES & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L1.MSS23

You buy 4 notebooks at £1.75 each and a pack of pens for £2.80. You pay with a £10 note. What change do you get?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Notebooks = 4 × £1.75 = £7.00. Total = £7.00 + £2.80 = £9.80. Change = £10.00 - £9.80 = £0.20 (20p).

26. Properties of 3D Solids

MEASURES & SPACE

Faces, Edges & Vertices

Face = Flat or curved surface. Edge = Line where two faces meet. Vertex (Vertices) = Corner point where edges meet. A cube/cuboid has 6 faces, 12 edges, 8 vertices.

CUBOID 3D PROPERTIES

Faces = 6 flat rectangular faces

Edges = 12 straight lines

Vertices = 8 corner points

★ Golden Rule: Vertices = Corners • Edges = Creases • Faces = Surfaces.

⚠️ Watch Out: A cylinder has 3 faces (2 flat circles, 1 curved), 2 edges, and 0 vertices!

27. Coordinates (x, y)

MEASURES & SPACE

Along the Corridor, Up the Stairs

Coordinates are written as (x, y): 1. First number (x) goes HORIZONTALLY along the bottom axis. 2. Second number (y) goes VERTICALLY up the side axis. Origin is (0, 0).



Point (4, 3):

4 along (x), 3 up (y)

★ Golden Rule: 'x' comes before 'y' in the alphabet, and in coordinates (x, y)!

⚠️ Watch Out: Don't mix up (2, 5) with (5, 2) — always read the bottom horizontal axis first.

28. Scale Drawings on Grid Paper

MEASURES & SPACE

Count Squares & Use the Key

Always check the scale key (e.g. 1 square = 1 metre, or 1 square = 2 metres). Real dimension = Number of squares counted × Scale key value.



Key: 1 sq = 2 metres

Room is 3 sq × 2 sq

Actual: 6 m by 4 m

★ Golden Rule: Always read the scale key: 1 square is rarely 1 cm in real life!

⚠️ Watch Out: If 1 square = 2 metres, 4 squares = 8 metres, NOT 4 metres.

29. Mean Average (Step-by-Step)

HANDLING DATA

Add All Up, Divide by Count

1. Add all numbers together to find total sum. 2. Count how many numbers there are. 3. Divide total sum by the count. Mean is the fair share.

Find Mean of: 6, 8, 12, 14

Step 1: Total = 6 + 8 + 12 + 14 = 40

Step 2: There are 4 numbers

Step 3: 40 ÷ 4 = 10

★ Golden Rule: Divide by how many items you added up!

⚠️ Watch Out: Don't forget to press equals (=) before dividing on your calculator, or use brackets.

Challenge: Coordinates (x, y)

MEASURES & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L1.MSS26

What are the coordinates of the origin where the two axes cross?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

The coordinates of the origin are (0, 0).

Challenge: Properties of 3D Solids

MEASURES & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L1.MSS25

How many vertices (corners) does a triangular prism have?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

A triangular prism has 6 vertices (3 at each triangular end).

Challenge: Mean Average (Step-by-Step)

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L1.HD28

Five friends score 14, 18, 12, 20, and 16 in a quiz. What is their mean score?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Total = 14 + 18 + 12 + 20 + 16 = 80. Count = 5. Mean = 80 ÷ 5 = 16.

Challenge: Scale Drawings on Grid Paper

MEASURES & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L1.MSS27

On a grid plan where 1 square = 0.5 metres, a desk is drawn 4 squares long. What is the real length of the desk?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Real length = 4 × 0.5 m = 2.0 metres.

30. Median (Middle Value)

HANDLING DATA

Line Up Low to High, Pick Middle

1. Put numbers in numerical order from smallest to largest. 2. Cross off one from each end until you reach the middle number. If two remain in middle, find their midpoint.

Raw Data: 9, 3, 7, 2, 8

1. Put in order: 2, 3, 7, 8, 9

Median = 7

⋆ Golden Rule: You MUST sort the numbers from smallest to largest first!

⚠ Watch Out: Never pick the middle number of the unsorted list (in 9, 3, 7, 2, 8, '7' happens to be middle, but never rely on luck!).

31. Mode (Most Frequent)

HANDLING DATA

The Most Common Item

Mode is the number or category that appears MOST OFTEN in the data. There can be one mode, more than one mode (bimodal), or no mode if all values appear once.

Shoe Sizes: 5, 6, 6, 7, 6, 8, 5

Size 5 appears 2 times • Size 6 appears 3 times

Mode = Size 6 ⋆

⋆ Golden Rule: 'MO'de = 'MO'st often!

⚠ Watch Out: The mode can be words (e.g. Blue, Coffee) — it is the only average that works for non-numbers.

32. Range (Spread of Data)

HANDLING DATA

Range = Highest - Lowest

Range is NOT an average — it measures how spread out the numbers are. Subtract the smallest number from the largest number. Smaller range = more consistent.

Temperatures: 12°C, 25°C, 18°C, 14°C

Highest = 25°C • Lowest = 12°C

Range = 25 - 12 = 13°C

⋆ Golden Rule: Range = Highest minus Lowest!

⚠ Watch Out: Don't write '12 to 25' — the range is a SINGLE number obtained by subtracting (13).

33. Tally Charts & Frequency

HANDLING DATA

Bundles of 5 (4 Vertical + 1 Diagonal Gate)

Tallies are grouped in bundles of five: four vertical marks (||||) with the fifth mark crossing diagonally as a gate across the bundle. Frequency is the total count written as a number.

TALLY SYSTEM (5-BAR GATE)

Item: Car  Freq: 8

Count tallies in 5s: 5 + 3 = 8

⋆ Golden Rule: The 5th tally stroke always gates diagonally across the bundle of 4!

⚠ Watch Out: Check your raw data twice by crossing off each item as you record its tally mark.

Challenge: Mode (Most Frequent)

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L1.HD30

What is the modal value of this dataset: 4, 7, 2, 7, 5, 9, 7, 2?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

7 appears three times (more than any other number). Mode = 7.

Challenge: Median (Middle Value)

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L1.HD29

Find the median of: 15, 6, 21, 9, 12.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Order: 6, 9, 12, 15, 21. Middle value is 12 (Median = 12).

Challenge: Tally Charts & Frequency

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L1.HD32

A tally chart shows three 5-bar tally gates and 2 single vertical strokes. What frequency number does this represent?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Three bundles of 5 plus 2: (3 × 5) + 2 = 15 + 2 = 17.

Challenge: Range (Spread of Data)

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L1.HD31

Weekly sales figures are: £180, £240, £150, £310, £220. Calculate the range.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

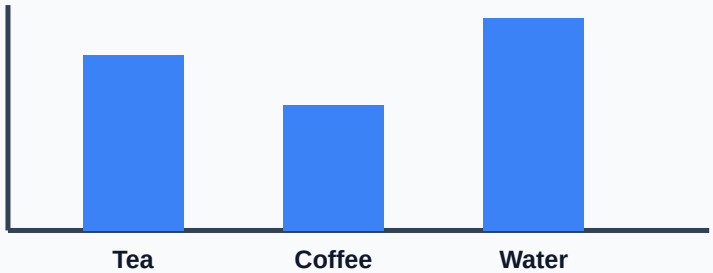
Highest = £310. Lowest = £150. Range = £310 - £150 = £160.

34. Bar Charts & Scales

HANDLING DATA

Equal Widths, Equal Gaps, Labelled Axes

To draw a bar chart for full marks: 1. Evenly spaced linear scale on vertical axis starting at 0. 2. Label both axes (e.g. 'Days' and 'Frequency'). 3. Equal width bars with equal gaps between them.



⋆ Golden Rule: Bars must NOT touch each other in a categorical bar chart!

⚠ Watch Out: Check what 1 grid square represents on the vertical axis (e.g. 1 square might be 2 or 5 units).

35. Pictograms (Using a Key)

HANDLING DATA

Always Check the Symbol Key

A pictogram uses pictures or symbols to represent quantities. Look at the key: 1 whole symbol = n items. A half symbol = n ÷ 2. A quarter symbol = n ÷ 4.

KEY: [★] = 4 BOOKS

Monday: [★] [★] = 2 × 4 = 8 books

Tuesday: [★] [★] [half] = 4 + 4 + 2 = 10 books

⋆ Golden Rule: Always check the KEY first — one picture is rarely just 1 item!

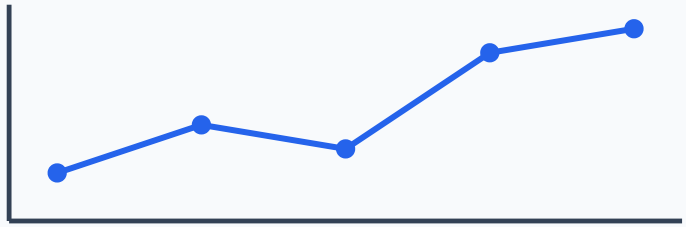
⚠ Watch Out: Don't count the number of symbols without multiplying by the key value.

36. Line Graphs (Trends over Time)

HANDLING DATA

Plot Points & Join with Straight Lines

Line graphs show how continuous data changes over time (temperature, sales, patient heart rate). Time is plotted along the horizontal axis. Read points by looking across to vertical axis.



Months (Jan to May) → Overall Upward Trend

⋆ Golden Rule: Time always goes along the bottom horizontal axis!

⚠ Watch Out: Join the plotted data points with neat straight ruler lines from point to point.

37. Probability Words & Scale

HANDLING DATA

Impossible to Certain

0 = Impossible (can never happen). 0.5 (1/2 or 50%) = Even Chance (e.g. flipping heads). 1 = Certain (guaranteed to happen). Unlikely is between 0 and 0.5; Likely is between 0.5 and 1.



⋆ Golden Rule: Probabilities on a number scale run strictly from 0 to 1!

⚠ Watch Out: Don't use words like 'Good chance' or 'Fifty-fifty' on an exam — use 'Likely' or 'Even chance'.

Challenge: Pictograms (Using a Key)

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L1.HD34

On a pictogram, a circle represents 8 cars. How many circles should you draw to represent 28 cars?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

28 ÷ 8 = 3.5. Draw 3 whole circles and 1 half circle.

Challenge: Bar Charts & Scales

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L1.HD33

State two things every bar chart must have to receive full marks in an exam.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

1. Equal bar widths with equal gaps. 2. Labelled axes with an evenly-spaced linear number scale starting at zero.

Challenge: Probability Words & Scale

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L1.HD36

Match the word to the event: (a) Tomorrow will be Wednesday if today is Tuesday. (b) Rolling an 8 on a standard 6-sided die.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

(a) Certain. (b) Impossible.

Challenge: Line Graphs (Trends over Time)

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L1.HD35

Between 9 am (14°C) and 1 pm (22°C), what was the rise in temperature?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Rise = 22°C - 14°C = 8°C.

38. Comparing Two Data Sets (L1)

HANDLING DATA

Compare One Average & One Range

To compare two groups for 2 marks: 1. State which group has a higher/lower mean or median. 2. State which group is more consistent by comparing their range (smaller range = more consistent).

LEVEL 1 EXAM COMPARISON TEMPLATE

1. "Team A scored higher on average (Mean 18 vs 14)."
2. "Team B was more consistent (smaller range 4 vs 10)."

⋆ Golden Rule: Always state the numbers AND the real-world context (sales, time)!

⚠ Watch Out: A larger range means LESS consistent results — not better performance!

1. Identifying Text Purpose

READING & ANALYSIS

P.A.F: Purpose • Audience • Form

Every text has a primary purpose: Persuade (sway opinions), Inform (factual details), Instruct (step-by-step directions), Explain (how/why something works), Advise (recommendations), or Entertain.

PRIMARY PURPOSE OF TEXT

PERSUADE | INFORM | INSTRUCT

Look at verbs, modal words, rhetorical questions & imperatives

⋆ Golden Rule: Pair purpose with evidence (e.g. 'persuades via rhetorical questions').

⚠ Watch Out: Identify the main purpose first — secondary intentions come second.

2. Audience & Tone Adaptation

READING & ANALYSIS

Who is the Reader?

Authors adapt their language, register, and layout for specific demographics: Professionals (formal jargon), Teenagers (casual, modern idioms), Parents, Patients, Customers.

FORMAL / PROFESSIONAL

- Passive voice
- Complex vocabulary
- No slang or contractions

INFORMAL / CASUAL

- Contractions (don't, it's)
- Direct address ('you')
- Conversational phrases

⋆ Golden Rule: Determine audience by looking at formality, vocabulary, and pronouns.

⚠ Watch Out: Avoid vague labels like 'general public' — name the exact audience.

3. Explicit vs Implicit (Inference)

READING & ANALYSIS

Fact Stated vs Reading Between Lines

Explicit = Directly and openly stated in the text word-for-word. Implicit = Hinted or suggested through clues, tone, and connotations without being said outright.

EXPLICIT (Stated)

"The door was locked with three heavy bolts."

Clear literal fact

IMPLICIT (Inferred)

"She checked over her shoulder before knocking."

Infers fear or secrecy

⋆ Golden Rule: Inference = Clues from text + Reader's logical deductions!

⚠ Watch Out: Don't guess wild theories — every inference MUST be grounded in textual clues.

Challenge: Identifying Text Purpose

READING & ANALYSIS

🎯 QUICK EXAM CHALLENGE — DfE L2.R1

An article uses phrases like 'You must act now before it is too late' and 'Statistics prove our service is superior'. What is its main purpose?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

The main purpose is to PERSUADE the reader to take urgent action using emotive language and direct address.

Challenge: Comparing Two Data Sets (L1)

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L1.HD37

Group 1: Mean = 24 kg, Range = 6 kg. Group 2: Mean = 20 kg, Range = 2 kg. Which group was more consistent in weight?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Group 2 was more consistent because its range was smaller (2 kg compared to 6 kg).

Challenge: Explicit vs Implicit (Inference)

READING & ANALYSIS

🎯 QUICK EXAM CHALLENGE — DfE L2.R3

Text: 'David stared at the empty petrol gauge and sighed heavily into the steering wheel on the quiet country lane.' What can you infer?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

We can infer that David has run out of petrol and is stranded/frustrated in a remote area.

Challenge: Audience & Tone Adaptation

READING & ANALYSIS

🎯 QUICK EXAM CHALLENGE — DfE L2.R2

A leaflet contains the sentence: 'All clinical practitioners must submit quarterly appraisal forms via the intranet.' Who is the intended audience?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

The intended audience is clinical medical staff / healthcare employees.

4. Fact vs Opinion vs Bias

READING & ANALYSIS

Verifiable Truth vs Subjective Belief

Fact: Can be proven true or false with objective evidence or data. Opinion: A personal view, feeling, or judgment. Bias: A one-sided presentation favouring one perspective.

FACT	OPINION	BIAS
Can be checked and verified with data/proof.	Belief or value (e.g. best, worst, greatest).	Unbalanced, one-sided view with vested interest.

⋆ Golden Rule: If it contains adjectives like 'finest', 'dreadful', or 'unacceptable', it is an OPINION!

⚠️ Watch Out: Watch out for opinions disguised as facts: 'Everyone knows London is the best city' is STILL an opinion.

5. Organisational Features & Effects

READING & ANALYSIS

Subheadings, Sidebars & Bullet Points

Always explain WHY an organisational feature is used: Subheadings (break text into chunks), Bullet points (isolate key items), Bold text (draws eye to essential warnings), Tables (organise data).

ORGANISATIONAL FEATURES & EFFECT

- Subheadings: Help reader navigate and find information quickly
- Bullet Points: Break complex steps into digestible chunks
- Pull-quotes / Sidebars: Emphasise critical statistics or takeaways

⋆ Golden Rule: Always explain HOW the feature guides the reader (not just its name)!

⚠️ Watch Out: Say 'draws reader's eye to safety steps', never just 'looks nice'.

6. Structural Signposting

READING & ANALYSIS

Chronological • Cause-Effect • Problem-Solution

Structure is how the writer organizes their points over time: Problem-Solution, Cause-and-Effect, Chronological narrative, or Thematic. Look for discourse markers (Firstly, Conversely, Therefore).

CORE TEXT STRUCTURES

1. Problem-Solution (Identifies crisis → gives remedy)
2. Cause & Effect (Action → consequence)
3. Comparative (Pros vs cons / Text A vs Text B)

⋆ Golden Rule: Track shifts in topic and tone from opening to conclusion!

⚠️ Watch Out: Structure means the ORDER and flow of ideas — not spelling or grammar.

7. Persuasive Devices (DAFOREST)

READING & ANALYSIS

The Master Persuasion Toolkit

D - Direct Address ('you'). A - Alliteration. F - Facts. O - Opinions. R - Rhetorical questions. E - Emotive language. S - Statistics (85%). T - Rule of Three (triples).

DAFOREST PERSUASION TOOLKIT

D - Direct address ('You') • A - Alliteration

F - Facts • O - Opinions • R - Rhetorical Questions

E - Emotive words • S - Statistics • T - Rule of 3

⋆ Golden Rule: Quote the exact phrase, name the DAFOREST device, and explain how it influences the reader!

⚠️ Watch Out: Don't just write 'it makes you want to read on' — explain that it creates guilt, urgency, or authority.

Challenge: Organisational Features & Effects

READING & ANALYSIS

🎯 QUICK EXAM CHALLENGE — DfE L2.R5

Explain how the use of bullet points helps the reader in a workplace safety notice.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Bullet points break the safety rules into clear, distinct steps so employees can absorb critical instructions quickly.

Challenge: Fact vs Opinion vs Bias

READING & ANALYSIS

🎯 QUICK EXAM CHALLENGE — DfE L2.R4

Identify whether this statement is fact or opinion: 'Over 65% of UK households own a pet, which proves pets are the greatest addition to family life.'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'Over 65% own a pet' is a FACT; 'proves pets are the greatest addition' is an OPINION.

Challenge: Persuasive Devices (DAFOREST)

READING & ANALYSIS

🎯 QUICK EXAM CHALLENGE — DfE L2.R7

Identify two DAFOREST devices in: 'Can you really ignore this heartbreaking crisis? Over 80% of families are struggling.'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

1. Rhetorical question ('Can you really ignore...?'). 2. Emotive language ('heartbreaking crisis') / Statistic ('Over 80%').

Challenge: Structural Signposting

READING & ANALYSIS

🎯 QUICK EXAM CHALLENGE — DfE L2.R6

What structural technique is used when an article starts with an alarming story and concludes by offering three practical solutions?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

A problem-and-solution structure (hooking with the crisis and resolving with actionable remedies).

8. Figurative Language Devices

READING & ANALYSIS

Metaphor • Simile • Hyperbole

Simile: Compares using 'like' or 'as' ('busy as a bee'). Metaphor: States one thing IS another ('the hospital was a warzone'). Hyperbole: Deliberate dramatic exaggeration.

FIGURATIVE LANGUAGE

- Simile: "Her smile was as bright as the sun" (uses like/as)
- Metaphor: "Time is a thief" (direct equivalence)
- Hyperbole: "I've waited a million years for this bus" (exaggeration)

★ Golden Rule: Metaphors create vivid mental images by transferring qualities from one concept to another!

⚠️ Watch Out: Don't confuse similes and metaphors: similes use 'like' or 'as'; metaphors do not.

9. Analysing Tone and Register

READING & ANALYSIS

The Writer's Attitude & Mood

Tone is the emotion or attitude behind the words: Sarcastic, Urgent, Critical, Humorous, Sympathetic, Objective, or Authoritative. Proved by diction (choice of words).

EXAM TONE VOCABULARY

- Critical / Indignant (Angry, disapproving)
- Pragmatic / Objective (Neutral, factual, balanced)
- Urgent / Passionate (Call to immediate action)

★ Golden Rule: Use precise tone adjectives instead of basic words like 'happy' or 'angry'!

⚠️ Watch Out: Tone can change within a text — look at the overall prevailing attitude.

10. Comparing Purpose & Perspective

READING & ANALYSIS

Pearson Section B Master Technique

To compare two texts for top marks: 1. Identify where both authors agree or share common ground. 2. Contrast how their viewpoints differ. 3. Use comparative connectives ('In contrast', 'Similarly').

TEXT A	TEXT B
Optimistic about AI	Cautionary / Skeptical
Quotes industry gains	Warns of job losses
Supports adoption	Urges regulation

★ Golden Rule: Always refer to BOTH Text A and Text B in every comparative paragraph!

⚠️ Watch Out: Writing two separate summaries without comparing them will cost you half the marks.

11. Comparing Layout Features

READING & ANALYSIS

Visual Presentation Differences

Contrast the presentation: Text A may use formal paragraph blocks and footnotes, whereas Text B uses subheadings, bullet points, callout boxes, and bold text.

Comparing Structural Formats

Text A: Dense, unbroken prose suited for academic reports.

Text B: Modern web layout with subheadings and numbered steps.

★ Golden Rule: Link the layout choice to how it helps the intended reader navigate the text!

⚠️ Watch Out: Don't just mention font colours unless it specifically relates to emphasis or branding.

Challenge: Analysing Tone and Register

READING & ANALYSIS

🎯 QUICK EXAM CHALLENGE — DfE L2.R9

Describe the tone of: 'Oh wonderful, another train cancellation. What exceptional service we pay thousands for!'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

The tone is sarcastic / ironic (using mock praise like 'wonderful' and 'exceptional' to express bitter dissatisfaction).

Challenge: Figurative Language Devices

READING & ANALYSIS

🎯 QUICK EXAM CHALLENGE — DfE L2.R8

What technique is used in: 'The wind screamed through the broken window'?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Personification (giving human actions like 'screamed' to non-human objects like the wind).

Challenge: Comparing Layout Features

READING & ANALYSIS

🎯 QUICK EXAM CHALLENGE — DfE L2.R11

Why might a public information website use bullet points instead of long continuous paragraphs?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

To make key information instantly scannable and easy to follow for busy mobile readers.

Challenge: Comparing Purpose & Perspective

READING & ANALYSIS

🎯 QUICK EXAM CHALLENGE — DfE L2.R10

Compare how Text A and Text B view remote working if Text A highlights productivity while Text B stresses employee loneliness.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

While both texts examine remote working, Text A has a positive perspective focusing on output, whereas Text B expresses concern over worker isolation.

12. Comparing Language Techniques

READING & ANALYSIS

Side-by-Side Technique Comparison

Examine how each author uses language to persuade: Author A relies on hard statistics and authoritative professional quotes. Author B relies on vivid metaphors and emotive storytelling.

COMPARATIVE LANGUAGE DRILL

- "Both authors use rhetorical questions to engage the reader..."
- "However, Text A employs objective statistics (92%)..."
- "...whereas Text B relies on emotive language ('disastrous')."

⋆ Golden Rule: Compare language side-by-side using 'similarly' or 'on the other hand'!

⚠️ Watch Out: Always provide exact quoted evidence from both texts.

13. Evaluating Source Credibility

READING & ANALYSIS

Authority • Date • Vested Interest

To evaluate reliability: 1. Author authority (doctor vs blogger). 2. Publication date (is it outdated?). 3. Commercial bias (is the writer selling a product?). 4. Corroboration (can claims be verified?).

SOURCE CREDIBILITY CHECKLIST

1. Who wrote it? (Expert credentials vs anonymous)
2. When was it published? (Recent vs obsolete)
3. Vested interest? (Sponsored ad vs independent study)

⋆ Golden Rule: Sources with commercial or personal interests are likely biased!

⚠️ Watch Out: Check author credentials — official-looking sites can still be biased.

14. Specialist Vocabulary & Jargon

READING & ANALYSIS

Deduce Meaning from Context

When encountering unfamiliar technical terms or jargon: 1. Read the surrounding sentences for clues. 2. Break down prefixes and root words. 3. Substitute a synonym to check if it makes sense.

Decoding Unfamiliar Jargon

"The patient's condition deteriorated rapidly..."

Clue: 'deteriorated' is paired with emergency actions.

Meaning: Became worse / declined in health.

⋆ Golden Rule: Use surrounding sentences to deduce the meaning of unfamiliar terms!

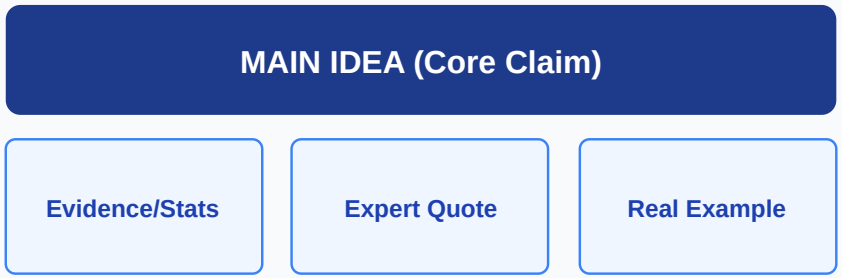
⚠️ Watch Out: Look for definitions nearby — don't skip technical terms.

15. Main Ideas vs Supporting Details

READING & ANALYSIS

Core Claim vs Examples

Main Idea: The central argument or topic sentence of a paragraph. Supporting Details: Statistics, quotes, illustrative anecdotes, and examples that prove the main claim.



⋆ Golden Rule: The topic sentence at the start of a paragraph usually states the main idea!

⚠️ Watch Out: Don't mistake a single supporting case study for the writer's overarching point.

Challenge: Evaluating Source Credibility

READING & ANALYSIS

🎯 QUICK EXAM CHALLENGE — DfE L2.R13

Why might a dental review written by the manufacturer of a new whitening toothpaste be considered biased?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

The manufacturer has a commercial vested interest in selling the product, making the claims biased rather than independent.

Challenge: Comparing Language Techniques

READING & ANALYSIS

🎯 QUICK EXAM CHALLENGE — DfE L2.R12

Text A says: 'Studies confirm 40% efficiency gains.' Text B says: 'It is a miraculous breath of fresh air.' Compare their language techniques.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Text A uses factual statistical evidence to present an objective argument, whereas Text B uses a poetic metaphor ('breath of fresh air') and hyperbole to create enthusiasm.

Challenge: Main Ideas vs Supporting Details

READING & ANALYSIS

🎯 QUICK EXAM CHALLENGE — DfE L2.R15

In a paragraph arguing that public transport reduces emissions, an author mentions that electric buses saved 200 tonnes of CO2 in Leeds. Which is the supporting detail?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

The 200 tonnes of CO2 saved in Leeds is a supporting detail (evidence) proving the main idea.

Challenge: Specialist Vocabulary & Jargon

READING & ANALYSIS

🎯 QUICK EXAM CHALLENGE — DfE L2.R14

What does the word 'mitigate' mean in: 'Measures were put in place to mitigate the risks of flooding'?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'Mitigate' means to reduce, lessen, or make less severe.

16. Summarising Multiple Paragraphs

READING & ANALYSIS

Synthesise in Your Own Words

To summarise: 1. Extract the key takeaways across paragraphs. 2. Condense into concise bullet points or sentences. 3. Paraphrase using YOUR OWN WORDS — never copy chunks verbatim.

SUMMARISING GOLDEN RULES

- ✓ Paraphrase in your own vocabulary
- ✓ Condense to essential core facts
- ✕ DO NOT copy word-for-word chunks!

⋆ Golden Rule: Exams penalise mindless copying — rewording proves you understood the text!

⚠️ Watch Out: Leave out minor descriptive details, anecdotes, and redundant adjectives.

17. Irony, Satire & Sarcasm

READING & ANALYSIS

Saying the Opposite of What is Meant

Verbal Irony: When an author says the opposite of their true meaning for humorous or critical effect. Understatement: Deliberately minimising seriousness ('it's just a scratch' on a wrecked car).

RECOGNISING VERBAL IRONY

Example: "Standing in the pouring rain with no coat is delightful."

The literal word is 'delightful', but the true meaning is miserable.

Examiners test whether you detect the sarcasm!

⋆ Golden Rule: Check if praise is sincere or sarcastic by looking at the broader context!

⚠️ Watch Out: Don't take ironic statements literally — that leads to picking the exact opposite answer.

18. Golden Reading Method: Q-E-L

READING & ANALYSIS

Quote • Explain • Link

To score full marks on 3-mark and 4-mark reading questions: 1. QUOTE exact textual phrase in quotation marks. 2. EXPLAIN the underlying meaning in your own words. 3. LINK directly to the author's purpose or question.

PEARSON 3-STEP FORMULA (Q - E - L)

1. QUOTE: "..." (Pick a precise word or short phrase)
2. EXPLAIN: Clarify what the quote shows or suggests
3. LINK: Connect directly back to the question's focus

⋆ Golden Rule: Keep quotes short (1 to 5 words) — never copy entire three-line sentences!

⚠️ Watch Out: A quote without explanation earns 0 or 1 mark; you must explain its impact.

19. Formal Letter Layout & Salutations

WRITING & FORMATS

Addresses, Date & Salutation Rules

Sender address top right. Recipient address top left below date. Rule: Dear Sir/Madam ends with 'Yours faithfully'. Named person (Dear Mr Smith) ends with 'Yours sincerely'.

Your Address (Top Right)
Date

Their Address (Top Left)

Dear Sir/Madam → Yours faithfully

Dear Mr Jones → Yours sincerely

⋆ Golden Rule: Named = Yours sincerely; Unnamed (Dear Sir/Madam) = Yours faithfully!

⚠️ Watch Out: Never mix them: Named = Sincerely; Unknown = Faithfully.

Challenge: Irony, Satire & Sarcasm

READING & ANALYSIS

🎯 QUICK EXAM CHALLENGE — DfE L2.R17

Explain the irony in: 'The fire station burned down while firefighters were attending a fire safety lecture.'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

It is situational irony because an institution dedicated to preventing fires was destroyed by one while learning about safety.

Challenge: Summarising Multiple Paragraphs

READING & ANALYSIS

🎯 QUICK EXAM CHALLENGE — DfE L2.R16

State two requirements of an effective Level 2 summary.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

1. It must cover the main points concisely. 2. It must be written in your own words without copying long quotes.

Challenge: Formal Letter Layout & Salutations

WRITING & FORMATS

🎯 QUICK EXAM CHALLENGE — DfE L2.W19

You are writing a formal complaint to customer services and do not know the recipient's name. How should you open and close the letter?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Open with: 'Dear Sir or Madam,' and close with: 'Yours faithfully,' followed by your full printed name.

Challenge: Golden Reading Method: Q-E-L

READING & ANALYSIS

🎯 QUICK EXAM CHALLENGE — DfE L2.R18

What are the three components of the Q-E-L reading response model?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

1. Quote (evidence). 2. Explain (analysis in your own words). 3. Link (tying back to the question/purpose).

20. Formal Report Structure

WRITING & FORMATS

Title • Background • Findings • Recommendations

A professional report requires 5 distinct sections: 1. Title. 2. Introduction / Purpose. 3. Current Findings / Key Issues (with subheadings). 4. Conclusion. 5. Actionable Recommendations.

REPORT FORMAT BLUEPRINT

1. Title: Report into Staff Parking Facilities
2. Introduction & Purpose
3. Key Findings (Subheading 1 & 2)
4. Conclusion & Actionable Recommendations

⋆ Golden Rule: Use four clear sections: Title, Intro, Findings, and Recommendations!

⚠️ Watch Out: Keep report tone strictly objective — state evidence, not emotions.

21. Article Writing (Newspaper/Magazine)

WRITING & FORMATS

Headline • Strapline • Byline • Hook

Articles must feature: 1. Punchy Headline (alliteration/pun). 2. Byline ('By [Name]'). 3. Engaging hook opening paragraph. 4. Balanced body exploring perspectives. 5. Concluding thought-provoking call to action.

HEADLINE: CATCHY & BOLD

Byline: By Alex Morgan, Health Correspondent

Hook opening: Introduce the debate with a question

Balanced body paragraphs exploring both viewpoints

Final call to action for the reader

⋆ Golden Rule: Every article must start with a captivating Headline and a Byline!

⚠️ Watch Out: Don't draw mock newspaper columns or pictures — just write the text with headers.

22. Speech & Presentation Writing

WRITING & FORMATS

Welcome • Passion • Rhetoric • Final Appeal

Speeches are written to be spoken aloud: 1. Greet audience ('Good morning fellow council members'). 2. State clear conviction. 3. Use rhetorical devices (rule of 3, pause cues). 4. Close with a memorable final rallying call.

SPEECH STRUCTURE

1. Direct Greeting: "Ladies and gentlemen..."
2. Rhetorical Triples: "We can, we must, we will."
3. Thank the audience: "Thank you for your time."

⋆ Golden Rule: Write in a spoken voice with direct address ('you', 'we', 'our community')!

⚠️ Watch Out: Remember to thank the audience at the very end of the speech.

23. Formal Workplace Email

WRITING & FORMATS

Subject Line • Professional Body • Sign-off

Email format: 1. Informative Subject line (concise, clear). 2. Professional salutation (Dear Mr Davis / Dear Team). 3. Purpose stated in opening sentence. 4. Clear bulleted action points. 5. Formal sign-off (Kind regards).

To: sarah.jenkins@healthtrust.nhs.uk

Subject: Urgent: Quarterly Clinical Audit Review - 14 Oct

Dear Ms Jenkins,

I am writing to confirm the schedule for Thursday's review...

Kind regards, / Best regards,

⋆ Golden Rule: Always include a clear Subject line and professional sign-off!

⚠️ Watch Out: Avoid casual greetings like 'Hey' or 'Cheers' in professional emails.

Challenge: Article Writing (Newspaper/Magazine)

WRITING & FORMATS

🎯 QUICK EXAM CHALLENGE — DfE L2.W21

Create an engaging alliterative headline for an article about the dangers of smartphone addiction.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Example: 'Screen Slavery: The Silent Struggle of Smartphone Addicts'.

Challenge: Formal Report Structure

WRITING & FORMATS

🎯 QUICK EXAM CHALLENGE — DfE L2.W20

What final section is mandatory at the end of every workplace formal report?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

The Recommendations section (proposing clear, actionable next steps based on the findings).

Challenge: Formal Workplace Email

WRITING & FORMATS

🎯 QUICK EXAM CHALLENGE — DfE L2.W23

Write an appropriate subject line for an email requesting two days of annual leave.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Subject: Annual Leave Request: 14th–15th November - [Your Name].

Challenge: Speech & Presentation Writing

WRITING & FORMATS

🎯 QUICK EXAM CHALLENGE — DfE L2.W22

Write the opening two sentences of a speech to high school students advocating for mental health days.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'Good morning students and staff. Have you ever felt so overwhelmed by exam pressure that you simply could not face another day?'

24. Writing an Online Review

WRITING & FORMATS

Balanced Evaluation & Verdict

Reviews must be balanced and credible: 1. Star rating or headline score. 2. What was experienced / purchased. 3. Positive features (pros). 4. Shortcomings or areas for improvement (cons). 5. Final verdict & recommendation.

★★★★★ (4/5 Stars) - Balanced Review

✓ Strengths: Excellent battery life & crisp screen

✕ Weaknesses: Charging cable feels flimsy

Final Verdict: Highly recommended for students.

⋆ Golden Rule: A top-level review must be balanced — mention at least one drawback even if you loved it!

⚠ Watch Out: Don't just rant emotionally; give specific evidence and examples.

25. Informative Leaflet / Guide

WRITING & FORMATS

Signposting • Chunks • Contact Box

Leaflet layout: 1. Engaging title with benefit for reader. 2. Brief introductory overview. 3. Section subheadings with bulleted practical tips. 4. Contact details / Where to find further help box.

LEAFLET FORMAT

Title: Healthy Eating on a Budget

Subheading: Top Tips for Smarter Shopping

• Plan meals weekly • Buy frozen produce

Contact: Visit www.nhs.uk/budget-meals

⋆ Golden Rule: Use subheadings, bullet points, and contact info for quick scanning!

⚠ Watch Out: Leaflets require broken-up sections, never solid walls of text.

26. Blog Post & Opinion Editorial

WRITING & FORMATS

Personal Voice & Reader Interaction

Blog features: 1. Catchy conversational title. 2. Date and author handle. 3. Strong first-person personal voice ('I couldn't believe it when...'). 4. Conversational tone with witty anecdotes. 5. Call for comments at the end.

BLOG POST FORMAT

Title: Confessions of a Commuter

Personal tone: "Like many of you, I dread the 7:15 train..."

Call to comments: "What do you think? Leave a reply!"

⋆ Golden Rule: Always invite reader comments or debate at the end of an online blog!

⚠ Watch Out: While conversational, your grammar and spelling must still be 100% accurate.

27. PEEL / TEEL Paragraphs

WRITING & FORMATS

Point • Evidence • Explanation • Link

Every body paragraph should follow PEEL: Point (topic sentence statement). Evidence (fact, statistic, or quote). Explanation (analyse the impact in detail). Link (tie back to overall argument/question).

PEEL PARAGRAPH TOWER

P - Point: State your main argument

E - Evidence: Support with data or examples

E - Explain: Deeply analyse the significance

L - Link: Connect directly back to the thesis

⋆ Golden Rule: Follow Point → Evidence → Explain → Link for every body paragraph!

⚠ Watch Out: Always explain your evidence — never leave a quote without analysis.

Challenge: Informative Leaflet / Guide

WRITING & FORMATS

🎯 QUICK EXAM CHALLENGE — DfE L2.W25

State two layout features that make a healthcare leaflet easy for patients to navigate.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

1. Informative subheadings dividing topics. 2. Bullet points highlighting key advice clearly.

Challenge: Writing an Online Review

WRITING & FORMATS

🎯 QUICK EXAM CHALLENGE — DfE L2.W24

Why is a review that includes both positive and negative aspects more persuasive than one with only praise?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

A balanced review appears objective, credible, and authentic, whereas uncritical praise looks like sponsored advertising.

Challenge: PEEL / TEEL Paragraphs

WRITING & FORMATS

🎯 QUICK EXAM CHALLENGE — DfE L2.W27

What does each letter in the PEEL acronym represent?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

P = Point, E = Evidence, E = Explanation, L = Link.

Challenge: Blog Post & Opinion Editorial

WRITING & FORMATS

🎯 QUICK EXAM CHALLENGE — DfE L2.W26

Write a closing sentence for a blog post discussing cycle lanes in cities.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'Do cycle lanes make your journey safer or cause traffic chaos? Share your thoughts in the comments below!'

28. Introduction: The 3-Sentence Hook

WRITING & FORMATS

Hook • Context • Thesis Statement

Craft a powerful opening in 3 sentences: 1. The Hook (surprising fact, rhetorical question, or provocative statement). 2. The Context (why this issue matters today). 3. The Thesis (your definitive stance/roadmap).

THE 3-SENTENCE INTRODUCTION

1. Hook: "Did you know that 8 million tonnes..."
2. Context: "Plastic pollution is now choking our oceans..."
3. Thesis: "This article will demonstrate why a ban is vital."

⋆ **Golden Rule: Hook the reader first, outline background, then state your purpose!**

⚠️ **Watch Out: Avoid weak openings like 'I am writing to...' on Level 2 tasks.**

29. Conclusion: Memorable Endings

WRITING & FORMATS

Re-state • Synthesise • Final Call

Never introduce brand new arguments in a conclusion! 1. Restate your main stance in fresh words. 2. Synthesise your strongest arguments. 3. End with a memorable final call to action or warning.

EFFECTIVE CONCLUSION BLUEPRINT

1. Re-affirm stance with authority
2. Summarise top two compelling reasons
3. Lasting final thought or call to action

⋆ **Golden Rule: Leave the examiner with a punchy, lingering final sentence!**

⚠️ **Watch Out: Never bring up brand-new evidence in the conclusion — it looks unplanned.**

30. Cohesive Devices & Connectives

WRITING & FORMATS

Linking Ideas Across Paragraphs

Use mature connective discourse markers to guide the reader: Adding (Furthermore, In addition). Contrasting (Conversely, Nevertheless, On the contrary). Cause/Result (Consequently, Therefore).

TRANSITION WORD BANK

Addition: Furthermore, Moreover, Additionally

Contrast: Conversely, However, Nevertheless

Result: Consequently, Therefore, As a result

⋆ **Golden Rule: Put a comma immediately after fronted connective words (e.g. 'Furthermore, ...')!**

⚠️ **Watch Out: Don't overuse 'Also', 'And', or 'But' to start every sentence.**

31. Varying Sentence Openers

WRITING & FORMATS

ISPACED Opener Variety

Avoid repeatedly starting sentences with 'The' or 'I': I - -Ing verb (Rushing to the ward...). S - Simile. P - Preposition (Under the table...). A - Adverb (Frantically,...). C - Conjunction (Although...). ED - Past participle (Exhausted,...).

VARY YOUR OPENERS (ISPACED)

- Adverb: 'Alarmingly, the waiting times doubled.'
- -Ing Word: 'Facing budget cuts, the clinic closed.'
- Conjunction: 'Although funding increased, staff left.'

⋆ **Golden Rule: Follow fronted openers with a comma to separate the clause!**

⚠️ **Watch Out: A piece where every sentence begins with 'The' or 'We' feels monotonous and loses grammar marks.**

Challenge: Conclusion: Memorable Endings

WRITING & FORMATS

🎯 QUICK EXAM CHALLENGE — DfE L2.W29

What is the primary mistake students make when writing conclusions?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Introducing new arguments or points that were never mentioned or explained in the main body.

Challenge: Introduction: The 3-Sentence Hook

WRITING & FORMATS

🎯 QUICK EXAM CHALLENGE — DfE L2.W28

What is the role of a thesis statement in an introduction?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

A thesis statement clearly declares the writer's central argument or stance on the topic.

Challenge: Varying Sentence Openers

WRITING & FORMATS

🎯 QUICK EXAM CHALLENGE — DfE L2.W31

Rewrite this sentence starting with an -ed participle: 'The nurses were exhausted by the night shift and took a break.'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'Exhausted by the night shift, the nurses took a break.'

Challenge: Cohesive Devices & Connectives

WRITING & FORMATS

🎯 QUICK EXAM CHALLENGE — DfE L2.W30

Replace 'also' with a higher-level transition word in: 'Also, local businesses will benefit.'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'Furthermore, local businesses will benefit' (or 'Moreover' / 'In addition').

32. Up-Levelling Vocabulary

WRITING & FORMATS

Swap Weak Words for Sophisticated Diction

Replace overused, simple words with mature academic language: 'Bad' → Detrimental / Inadequate. 'Good' → Beneficial / Exemplary. 'A lot of' → Substantial / Numerous. 'Think' → Contend / Assert.

UP-LEVELLING DICTION

Simple: "This is a very bad idea that will cause big problems."

Level 2: "This proposal is highly detrimental and will precipitate substantial operational challenges."

⋆ Golden Rule: Use sophisticated words accurately; don't force complex words if unsure of meaning!

⚠ Watch Out: Avoid colloquial slang: 'stuff', 'kids', 'gonna', 'loads of'.

33. Planning & Proofreading Strategy

WRITING & FORMATS

5-Minute Plan • Write • 5-Minute Sweep

Time allocation strategy: 5 mins: Spidergram plan (format, 4 key points, PAF). 35 mins: Sustained writing following PEEL. 5 mins: Proofreading sweep for capital letters, sentence boundaries, and homophones.

EXAM WRITING WORKFLOW (45 MINS)

1. Plan (5 mins): Check PAF, jot 4 paragraph bullets
2. Write (35 mins): Keep focus on paragraphing & tone
3. Check (5 mins): Sweep for capitals, full stops & commas

⋆ Golden Rule: Plan for 5 minutes and reserve 5 minutes at the end to proofread!

⚠ Watch Out: Always proofread — catching 3 SPaG errors can make or break a pass.

34. Sentence Boundaries & Splices

SPAG & GRAMMAR

Never Join Sentences with a Comma Alone

A comma splice occurs when two independent sentences are joined with only a comma. Fix it with: 1. A full stop. 2. A semicolon (;). 3. A coordinating conjunction (FANBOYS: for, and, nor, but, or, yet, so).

COMMA SPLICE ERROR

WRONG: "The weather was awful, we stayed indoors."

FIX 1: "The weather was awful. We stayed indoors."

FIX 2: "The weather was awful; we stayed indoors."

⋆ Golden Rule: A comma is a brief pause, NEVER a full stop!

⚠ Watch Out: Comma splices are the #1 penalized grammar error at Level 2.

35. The Comma Rulebook

SPAG & GRAMMAR

Lists • Adverbials • Parenthetical Clauses

Use commas for: 1. Separating items in a list. 2. After introductory adverbials ('Suddenly, ...'). 3. Around non-essential parenthetical information ('Dr Smith, who retired last year, attended').

3 ESSENTIAL COMMA RULES

1. List: 'Apples, oranges, and pears.'
2. Opener: 'Despite the delay, the flight landed safely.'
3. Parenthesis: 'Mr Patel, our head of finance, approved it.'

⋆ Golden Rule: Use commas after fronted adverbials and around non-essential clauses!

⚠ Watch Out: Never insert a comma between a subject and its main verb.

Challenge: Planning & Proofreading Strategy

WRITING & FORMATS

🎯 QUICK EXAM CHALLENGE — DfE L2.W33

What three things should you check for during your final 5-minute proofreading sweep?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

1. Capital letters (names, 'I', start of sentences). 2. Sentence boundary punctuation (avoiding comma splices). 3. Common homophones (there/their/they're).

Challenge: Up-Levelling Vocabulary

WRITING & FORMATS

🎯 QUICK EXAM CHALLENGE — DfE L2.W32

Up-level the word 'show' in: 'The figures show that attendance dropped.'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'The figures demonstrate / illustrate / indicate that attendance dropped.'

Challenge: The Comma Rulebook

SPAG & GRAMMAR

🎯 QUICK EXAM CHALLENGE — DfE L2.W35

Add commas to this sentence: 'Although it rained heavily the marathon runners who had trained for months completed the course.'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'Although it rained heavily, the marathon runners, who had trained for months, completed the course.'

Challenge: Sentence Boundaries & Splices

SPAG & GRAMMAR

🎯 QUICK EXAM CHALLENGE — DfE L2.W34

Correct the comma splice: 'The report is complete, it will be emailed tomorrow.'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'The report is complete; it will be emailed tomorrow.' (or with a full stop: 'The report is complete. It will be emailed tomorrow.')

36. Apostrophes: Omission vs Possession

SPAG & GRAMMAR

Missing Letters vs Ownership

Omission: Replaces missing letters in contractions (don't = do not, it's = it is). Singular possession: 's (the nurse's badge). Plural ending in s: s' (the nurses' station). Irregular plural: 's (children's ward).

IT'S vs ITS

It's = It is (Contraction)

"It's raining outside."

Its = Possessive pronoun

"The cat licked its paws."

PLURAL POSSESSION

One doctor: doctor's

"The doctor's car"

Many doctors: doctors'

"The doctors' room"

⋆ Golden Rule: 'Its' for possession NEVER has an apostrophe (like hers, his, ours)!

⚠ Watch Out: Never use an apostrophe to make a word plural! (e.g. 'Apple's for sale' is WRONG).

37. Colons & Semicolons

SPAG & GRAMMAR

Colon (Introduces) • Semicolon (Bridges)

Colon (:): Introduces a list, explanation, or direct quotation following a full independent clause. Semicolon (;): Connects two closely linked independent sentences without a conjunction.

COLON (:)

Introduces a list or quote:

"Please bring three items:
a pen, a ruler, and paper."

SEMICOLON (;)

Bridges two sentences:

"She loves surgery;
he prefers general practice."

⋆ Golden Rule: The clause before a colon MUST be a complete standalone sentence!

⚠ Watch Out: Don't put a colon after a verb like 'including' or 'are' (e.g. 'Items are: pen...' is INCORRECT).

38. Speech Marks & Quotations

SPAG & GRAMMAR

Punctuation Inside the Quotes

1. Put quotation marks around the spoken words only. 2. Punctuation (comma, full stop, exclamation mark) goes INSIDE the speech marks. 3. New speaker = start a new paragraph line.

DIRECT SPEECH PUNCTUATION

"We must act immediately," warned the director.

The nurse replied, "The ward is fully prepared."

⚠ Comma goes BEFORE the closing quote mark!

⋆ Golden Rule: Punctuation belongs INSIDE the speech marks!

⚠ Watch Out: Capitalise the first word inside the opening speech mark when starting dialogue.

39. Hyphens & Parenthetical Dashes

SPAG & GRAMMAR

Hyphen (Joins) vs Dash (Separates)

Hyphen (-): Short line joining words to form compound adjectives before nouns (well-known actor, part-time job). Dash (—): Longer pause used in pairs for dramatic emphasis or parenthetical side-thoughts.

HYPHEN (-)

Joins compound words:

- state-of-the-art facility
- thirty-five patients

DASH (—)

Dramatic pause/aside:

"The outcome — though unexpected — was positive."

⋆ Golden Rule: Hyphens glue words together; dashes push thoughts apart!

⚠ Watch Out: Notice the difference: 'twenty five-pound notes' (20 notes) vs 'twenty-five pound notes' (£25 notes).

Challenge: Colons & Semicolons

SPAG & GRAMMAR

🎯 QUICK EXAM CHALLENGE — DfE L2.W37

Insert a semicolon into this sentence: 'Call me tomorrow I will give you my final decision then.'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'Call me tomorrow; I will give you my final decision then.'

Challenge: Apostrophes: Omission vs Possession

SPAG & GRAMMAR

🎯 QUICK EXAM CHALLENGE — DfE L2.W36

Choose the correct word: 'The company reviewed (it's / its) annual policy.'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'its' (possessive pronoun showing ownership; 'it is annual policy' makes no sense).

Challenge: Hyphens & Parenthetical Dashes

SPAG & GRAMMAR

🎯 QUICK EXAM CHALLENGE — DfE L2.W39

Insert a hyphen where required: 'He submitted a well written application.'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'He submitted a well-written application.' ('well-written' is a compound adjective preceding 'application').

Challenge: Speech Marks & Quotations

SPAG & GRAMMAR

🎯 QUICK EXAM CHALLENGE — DfE L2.W38

Correct the punctuation: The manager said 'we will achieve our targets.'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

The manager said, "We will achieve our targets."

40. Subject-Verb Agreement

SPAG & GRAMMAR

Singular Subjects take Singular Verbs

Singular subject = singular verb (The student IS writing). Plural subject = plural verb (The students ARE writing). Trap: Ignore prepositional phrases (e.g. 'The box of chocolates IS on the desk').

PREPOSITIONAL PHRASE TRAP

TRAP: "The list of employees (is / are) on the desk."

The subject is 'The list' (singular), NOT 'employees'.

CORRECT: "The list of employees IS on the desk."

⋆ Golden Rule: Singular subjects take singular verbs (he runs; they run)!

⚠️ Watch Out: 'Everyone', 'each', and 'nobody' always take SINGULAR verbs.

41. Consistent Verb Tenses

SPAG & GRAMMAR

Maintain Tense Throughout

Do not drift accidentally between past and present tense within the same sentence or paragraph: Reports and recount narratives must stay strictly in the past tense.

TENSE DRIFT ERROR

WRONG: "She walked into the room and picks up the file."

CORRECT: "She walked into the room and picked up the file."

Keep both verbs in the simple past tense.

⋆ Golden Rule: Check all your verbs during proofreading to ensure they match in tense!

⚠️ Watch Out: Accidental tense drift into present tense is common when writing recount stories.

42. Active vs Passive Voice

SPAG & GRAMMAR

Active (Direct) vs Passive (Formal & Objective)

Active: Subject performs action ('The doctor administered the vaccine'). Direct and energetic. Passive: Action happens to subject ('The vaccine was administered'). Formal, impersonal, ideal for reports.

ACTIVE VOICE

"The committee approved the budget."

Subject does action

PASSIVE VOICE

"The budget was approved by the committee."

Formal & Objective

⋆ Golden Rule: Use active voice ('We improved...') for clear, direct writing!

⚠️ Watch Out: Overusing passive voice creates sluggish, bureaucratic sentences.

43. Complex & Compound Sentences

SPAG & GRAMMAR

Main Clauses & Subordinate Clauses

Compound: Joins two independent clauses with FANBOYS conjunction (and, but, so). Complex: Contains a main clause + subordinate clause joined with subordinating conjunction (although, because, while, since).

COMPLEX SENTENCE STRUCTURE

Subordinate Clause: 'Although funding was cut,' (cannot stand alone)

+ Main Clause: 'the project finished on time.' (stands alone)

Examiners look for a variety of simple, compound & complex!

⋆ Golden Rule: When a subordinate clause comes first, always put a comma after it!

⚠️ Watch Out: A subordinate clause alone is a fragment, NOT a complete sentence.

Challenge: Consistent Verb Tenses

SPAG & GRAMMAR

🎯 QUICK EXAM CHALLENGE — DfE L2.W41

Correct the tense in: 'The customer complained loudly and leaves the store.'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'The customer complained loudly and left the store.' (keeping both verbs in past tense).

Challenge: Subject-Verb Agreement

SPAG & GRAMMAR

🎯 QUICK EXAM CHALLENGE — DfE L2.W40

Choose the correct verb: 'Neither of the applicants (was / were) shortlisted.'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'was' ('Neither' is a singular indefinite pronoun taking a singular verb).

Challenge: Complex & Compound Sentences

SPAG & GRAMMAR

🎯 QUICK EXAM CHALLENGE — DfE L2.W43

Identify the subordinate clause in: 'Because traffic was congested, we arrived twenty minutes late.'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'Because traffic was congested' (it depends on the main clause to make sense).

Challenge: Active vs Passive Voice

SPAG & GRAMMAR

🎯 QUICK EXAM CHALLENGE — DfE L2.W42

Convert to passive voice: 'The inspector reviewed the fire doors.'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'The fire doors were reviewed by the inspector.'

44. Homophones: There / Their / They're

SPAG & GRAMMAR

Location • Ownership • Contraction

THERE: Place or fact ('Put it over there', 'There is a problem'). THEIR: Ownership / belonging ('Their coats'). THEY'RE: Short for 'they are' ('They're leaving now').

THERE

Place (contains 'here')

"Over there"

THEIR

Possession (their car)

"Their home"

THEY'RE

They are

(has apostrophe)

"They're late"

⋆ Golden Rule: If you can replace it with 'they are', use 'they're'!

⚠ Watch Out: Affect vs Effect: Affect is a Verb (Action), Effect is a Noun (End result).

45. Confusable Pairs: Practice vs Practise

SPAG & GRAMMAR

Noun (C) vs Verb (S)

In British English: Nouns end in 'ce' (Practice, Advice, Device, Licence). Verbs end in 'se' (Practise, Advise, Devise, License). Tip: Think of 'Ice is a noun' (-ice = noun).

NOUN (-CE)

- Practice (a doctor's surgery)
- Advice (useful advice)
- Licence (driving licence)

VERB (-SE)

- Practise (to practise guitar)
- Advise (I advise you...)
- License (licensed to sell)

⋆ Golden Rule: Test with 'Advice/Advise': You give advice (noun), you advise someone (verb)!

⚠ Watch Out: Loose vs Lose: 'Loose' rhymes with goose (untied shoelace). 'Lose' has one 'o' (to misplace something).

46. Suffix Rules (-able vs -ible)

SPAG & GRAMMAR

Root Word Test

-able: Used when the base word is a complete standalone word (depend → dependable, enjoy → enjoyable). -ible: Used when the base word is not a complete English word (horr → horrible, vis → visible).

-ABLE (Whole Word)

comfort + able = comfortable

rely + able = reliable

accept + able = acceptable

-IBLE (Incomplete Root)

poss + ible = possible

elig + ible = eligible

respons + ible = responsible

⋆ Golden Rule: If you take away the suffix and have a complete word left, use -able!

⚠ Watch Out: -tion vs -sion: Use -sion when base word ends in d/s (extend → extension; divide → division).

47. High-Frequency Exam Spellings

SPAG & GRAMMAR

The 10 Most Common Exam Misspellings

1. Accommodate (2 c's, 2 m's). 2. Definitely (de-finite-ly, NO 'a'). 3. Necessary (1 collar, 2 socks: 1 c, 2 s's). 4. Government (has 'n'). 5. Environment (has 'n'). 6. Colleague. 7. Separate.

SPELLING MNEMONICS

- Necessary: 1 Coffee (c) and 2 Sugars (ss)
- Definitely: Has the word 'FINITE' inside it (NO 'a')
- Separate: There is 'A RAT' in separate

⋆ Golden Rule: Break tricky words into syllables or mnemonics!

⚠ Watch Out: 'Definately' is always marked wrong — remember 'definitely'.

Challenge: Confusable Pairs: Practice vs Practise

SPAG & GRAMMAR

🎯 QUICK EXAM CHALLENGE — DfE L2.W45

Choose the correct spelling: 'Doctors must (practice / practise) medicine ethically.'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'practise' (it is the doing verb).

Challenge: Homophones: There / Their / They're

SPAG & GRAMMAR

🎯 QUICK EXAM CHALLENGE — DfE L2.W44

Fill the blanks: '(They're / Their) coats are hanging over (there / their).'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'Their coats are hanging over there.'

Challenge: High-Frequency Exam Spellings

SPAG & GRAMMAR

🎯 QUICK EXAM CHALLENGE — DfE L2.W47

Spell the word that means two people working together that begins with 'coll...'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'colleague'.

Challenge: Suffix Rules (-able vs -ible)

SPAG & GRAMMAR

🎯 QUICK EXAM CHALLENGE — DfE L2.W46

Add the correct suffix to form the adjective: 'aud___' (able to be heard).

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'audible' ('aud' is not a standalone English word, so it takes -ible).

48. Prefixes & Negative Antonyms

SPAG & GRAMMAR

Un-, In-, Im-, Dis-, Mis-, Ir-

Prefixes change word meaning without altering base word spelling: 'im-' before 'p/m' (patient → impatient). 'ir-' before 'r' (responsible → irresponsible). 'il-' before 'l' (legal → illegal). Root spelling is PRESERVED.

PREFIX PATTERNS

- **Words starting with P take IM-: im + perfect = imperfect**
- **Words starting with R take IR-: ir + regular = irregular**
- **Words starting with L take IL-: il + legible = illegible**

⋆ Golden Rule: Don't drop letters when adding prefixes: un + necessary = unnecessary (keep both n's)!

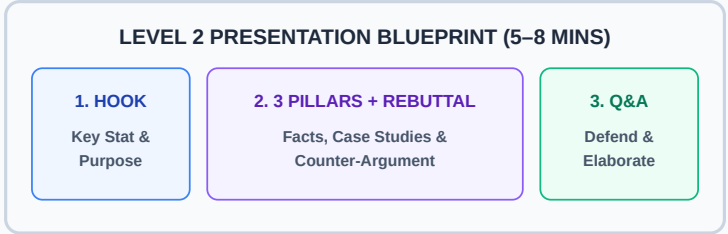
⚠ Watch Out: 'Disappointed' has ONE 's' and TWO 'p's (dis + appoint + ed).

49. SLC L2: Structuring a Formal Presentation

SPEAKING & LISTENING

Hook • 3 Evidence Pillars • Rebuttal • Q&A

A Level 2 presentation (5–8 mins + Q&A) requires: (1) Engaging Hook & Thesis, (2) 3 Structured Pillars supported by real statistics/examples, (3) Counter-Argument rebuttal, and (4) Call to Action.



⋆ Golden Rule: Rule of 3: Outline your points, present them, then summarise clearly!

⚠ Watch Out: Pause after key ideas — rushing through costs presentation marks.

50. SLC L2: Defending Points in Q&A

SPEAKING & LISTENING

Acknowledge • Evidence • Connect (A-E-C)

When challenged in Q&A, use A-E-C: (1) Acknowledge the question calmly, (2) Explain with specific evidence or a practical example, and (3) Conclude by linking back to your main proposal.

HANDLING CHALLENGING Q&A QUESTIONS

A — Acknowledge: "That is a valid concern regarding initial setup costs..."

E — Evidence: "...however, pilot schemes reduced waste by 30% in Year 1..."

C — Connect: "...meaning the project pays for itself within 14 months."

⋆ Golden Rule: Never give a one-sentence answer in Level 2 Q&A—always cite evidence!

⚠ Watch Out: Becoming defensive or saying "I don't know" without offering a reasoned response loses marks.

51. SLC L2: Adapting Spoken Register & Tone

SPEAKING & LISTENING

Formal Register vs Collaborative Discussion

Level 2 requires adapting spoken register to your audience and purpose: use formal discourse markers ("Consequently", "Furthermore") and specialist terminology in presentations, and collaborative warmth in peer discussions.



⋆ Golden Rule: Adapt vocabulary and formality to match professional discussions!

⚠ Watch Out: Avoid slang terms ('gonna', 'cheers') during formal assessments.

Challenge: SLC L2: Structuring a Formal Presentation

SPEAKING & LISTENING

🎯 QUICK EXAM CHALLENGE — DfE L2.SLC1

How should you structure a 5–8 minute Level 2 formal presentation to meet all Pearson Edexcel criteria?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

1. Hook & Objective (1 min). 2. Three Evidence-Backed Arguments + Counter-Argument (3–4 mins). 3. Summary Conclusion & Q&A Defence (2–3 mins).

Challenge: Prefixes & Negative Antonyms

SPAG & GRAMMAR

🎯 QUICK EXAM CHALLENGE — DfE L2.W48

Add the negative prefix to the word 'rational'.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'irrational' (prefix 'ir-' before 'r').

Challenge: SLC L2: Adapting Spoken Register & Tone

SPEAKING & LISTENING

🎯 QUICK EXAM CHALLENGE — DfE L2.SLC6

Adapt the statement "This new rota works way better" for a formal management presentation.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Formal boardroom pitch: "The pilot data demonstrates a 22% efficiency gain." Peer workshop: "Let's look at how this saves us time on shift."

Challenge: SLC L2: Defending Points in Q&A

SPEAKING & LISTENING

🎯 QUICK EXAM CHALLENGE — DfE L2.SLC2

During your Level 2 Q&A, an audience member challenges the cost of your proposal. How do you respond using the A-E-C method?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

1. Acknowledge the question. 2. Provide a specific reason/evidence. 3. Connect back to your core argument (e.g., "That is an important point; recent NHS data shows...").

52. SLC L2: Mediating & Reaching Consensus

SPEAKING & LISTENING

Synthesise • Mediate • Steer to Consensus

To excel in the Level 2 formal discussion (15 mins): synthesise different viewpoints, mediate disagreements constructively, challenge assumptions with evidence, and help steer the group to a shared conclusion.

STEERING A LEVEL 2 GROUP DISCUSSION

1. Synthesise: "So far, we've weighed up budget limits against staff safety..."
2. Probe Deeper: "What evidence do we have that Option B is faster?"
3. Consensus: "Can we agree to recommend a phased rollout starting in Q2?"

★ Golden Rule: Acknowledge opposing views fairly before proposing a compromise!

⚠️ Watch Out: Never dismiss arguments outright — validate before reframing.

1. Main Points and Ideas

READING SKILLS

What is the Big Picture?

The main point is the most important message the author wants you to understand. Read the title, opening sentence, and closing sentence to identify the central message.

FINDING THE MAIN POINT

1. Headline / Title: Gives the topic
2. First sentence: Introduces the central point
3. Ask yourself: "What is this text really telling me?"

★ Golden Rule: The main point is the overarching message, NOT a small illustrative detail!

⚠️ Watch Out: Don't confuse an example with the main point.

2. Text Types: Inform, Persuade, Instruct

READING SKILLS

Identify the Goal

Inform: Gives facts and data (news, timetables). Persuade: Wants you to buy or believe something (adverts, flyers). Instruct: Tells you how to do something step-by-step (recipes, user guides).

INFORM	PERSUADE	INSTRUCT
Facts, statistics, dates, reports, and notices	Adverts, reviews, speeches, and opinions	Numbered steps, recipes, bossy imperative verbs

★ Golden Rule: Look for imperative verbs (Chop, Turn, Press) = INSTRUCT!

⚠️ Watch Out: Adverts inform you about prices, but their PRIMARY goal is to PERSUADE you to buy.

3. Headings & Subheadings

READING SKILLS

Signposts for Quick Navigation

Main Heading: Tells you what the whole document is about. Subheadings: Break the text into topics so you can quickly scan and find specific details without reading every single word.

MAIN TITLE: COMMUNITY CENTRE

Subheading 1: Opening Hours

Monday to Friday 9am - 8pm

Subheading 2: Room Hire Fees

Main Hall: £25 per hour

★ Golden Rule: Read the subheadings first to find where your answer is hiding!

⚠️ Watch Out: Don't read the whole article line-by-line if a subheading tells you exactly where the topic is.

Challenge: Main Points and Ideas

READING SKILLS

🎯 QUICK EXAM CHALLENGE — DfE L1.R1

A text lists opening hours, emergency numbers, and ward rules for a hospital. What is the main point of the text?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

To inform visitors about essential hospital visiting rules and contact information.

Challenge: SLC L2: Mediating & Reaching Consensus

SPEAKING & LISTENING

🎯 QUICK EXAM CHALLENGE — DfE L2.SLC8

In a Level 2 group discussion, two peers are deadlocked on a decision. How can you move the group towards a consensus?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Summarise both sides fairly ("While Priya highlighted cost risks and Ben focused on speed..."), then propose a practical compromise ("...a 3-month trial balances both priorities").

Challenge: Headings & Subheadings

READING SKILLS

🎯 QUICK EXAM CHALLENGE — DfE L1.R3

Under which subheading would you look to find out how much a fitness class costs?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Under 'Prices', 'Fees', or 'Cost'.

Challenge: Text Types: Inform, Persuade, Instruct

READING SKILLS

🎯 QUICK EXAM CHALLENGE — DfE L1.R2

What type of text is a user manual for a washing machine?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

An instructional text (it gives step-by-step instructions on operating the appliance).

4. Bullet Points & Lists

READING SKILLS

Clear, Distinct Items

Bullet points: Separate key items in a list so they stand out clearly. Numbered lists: Indicate a strict order of steps that must be done in sequence (1st, 2nd, 3rd).

BULLETS vs NUMBERING

- **Bullets: Order does not matter (checklist, packing list)**

1. **Numbered: Sequence MATTERS (recipe, safety fire drill)**

Makes information fast and easy to digest

⋆ **Golden Rule: Use numbered lists when steps MUST be followed in exact sequence!**

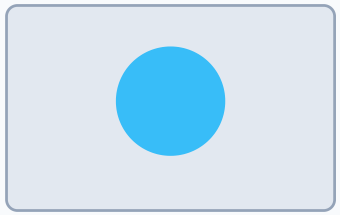
⚠️ **Watch Out: Keep each bullet point short and grammatically consistent.**

5. Images, Diagrams & Captions

READING SKILLS

Visual Evidence & Explanations

Images and diagrams reinforce the written text. Captions are the short descriptive sentences directly below an image explaining what it shows.



The Caption:

Explains the image and links to text

Figure 1: Solar panel installation

⋆ **Golden Rule: Always read the small print in the caption — it often contains exam answers!**

⚠️ **Watch Out: Don't ignore the pictures — exam questions often ask how the graphic supports the article.**

6. Fact vs Opinion (Core Test)

READING SKILLS

True / False vs Personal Belief

Fact: Can be proven true with dates, measurements, or official records ('The shop opens at 8 am'). Opinion: What someone thinks or feels ('The shop is wonderful').

FACT (Proven)

"The library has 12 computers."

Can be counted & verified

OPINION (Feeling)

"It is the best place to study."

Personal point of view

⋆ **Golden Rule: If it uses words like 'best', 'delicious', or 'terrible', it is an OPINION!**

⚠️ **Watch Out: Numbers and dates are almost always facts.**

7. Skimming for Overall Gist

READING SKILLS

Rapid 30-Second Glance

Skimming is fast reading to get the general idea: 1. Read the title and subheadings. 2. Glance at the first and last sentence of each paragraph. 3. Look at any pictures. Do NOT read every word.

SKIMMING TECHNIQUE

✓ **Read titles, subheadings & bold words**

✓ **Read the first sentence of each paragraph**

Result: Understand the general topic in 30 seconds!

⋆ **Golden Rule: Skimming gives you the big picture without getting bogged down!**

⚠️ **Watch Out: Don't read word-by-word when skimming — keep your eyes moving quickly down the page.**

Challenge: Images, Diagrams & Captions

READING SKILLS

🎯 QUICK EXAM CHALLENGE — DfE L1.R5

What is the purpose of a caption underneath a photograph in a news article?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

To describe what is happening in the photograph and provide context for the reader.

Challenge: Bullet Points & Lists

READING SKILLS

🎯 QUICK EXAM CHALLENGE — DfE L1.R4

Why are fire escape procedures presented as numbered lists rather than paragraphs?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

To ensure people follow the emergency steps in the correct chronological order without panic.

Challenge: Skimming for Overall Gist

READING SKILLS

🎯 QUICK EXAM CHALLENGE — DfE L1.R7

What is the difference between skimming and scanning?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Skimming is fast reading for the general overview; scanning is searching for a specific piece of information (like a number or name).

Challenge: Fact vs Opinion (Core Test)

READING SKILLS

🎯 QUICK EXAM CHALLENGE — DfE L1.R6

Is this a fact or an opinion: 'The train ticket costs £14.50'?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Fact (it is a verifiable price that can be proven true).

8. Scanning for Facts & Numbers

READING SKILLS

Search like a Metal Detector

Scanning is searching for specific targets: Look for capital letters (names, places), numbers (dates, times, prices), or symbols (£, %, @). Ignore everything else until you spot your target.

SCANNING TARGET CLUES

- **Looking for a price? Scan for: '£' and decimal points**
- **Looking for a person/place? Scan for: CAPITAL LETTERS**
- **Looking for a time? Scan for: 'am', 'pm', or colon (14:30)**

⋆ Golden Rule: Lock your target in your mind before your eyes touch the page!

⚠ Watch Out: Don't start reading the whole text from the beginning when answering 'What time...' questions.

9. Simple Inference (Reading Clues)

READING SKILLS

Detective Reading

Inference means working out meaning from hints in the text: If someone is wearing a heavy winter coat, gloves, and shivering, you infer that the weather is freezing.

INFERENCE DETECTIVE CLUES

Text: "She slammed her locker door and stomped away."

Inference: She is feeling very angry or upset.

⋆ Golden Rule: Inference is finding the hidden clue in the writer's description!

⚠ Watch Out: Make sure your answer is directly supported by words in the text.

10. Vocabulary: Using Context Clues

READING SKILLS

Guess the Word from the Sentence

If you don't know a word: 1. Read the words before and after it. 2. Look for examples or explanations in the sentence. 3. Try replacing it with a word you do know that makes sense.

CONTEXT CLUE STRATEGY

"The recipe requires several condiments, like ketchup and mustard."

Clue: Ketchup and mustard are sauces added to food.

Meaning: 'Condiments' means seasonings or sauces.

⋆ Golden Rule: Read the whole sentence before deciding an unknown word is impossible!

⚠ Watch Out: Don't panic when you see long words — look for root words you already recognise.

11. Alphabetical Order & Directories

READING SKILLS

Dictionaries & Index Pages

A to Z alphabetical order: Look at the 1st letter. If 1st letters are the same, look at the 2nd letter (Apple, Apricot). If 2nd letters match, look at the 3rd letter.

ALPHABETICAL ORDERING

Put in order: Brown, Baker, Bailey

1. All start with 'B'

2. Compare second letters: Ba... before Br...

3. Correct order: Bailey, Baker, Brown

⋆ Golden Rule: Look at the 3rd and 4th letter if the first two letters are identical!

⚠ Watch Out: Surnames in directories are listed Last Name first: 'Smith, John'.

Challenge: Simple Inference (Reading Clues)

READING SKILLS

🎯 QUICK EXAM CHALLENGE — DfE L1.R9

Text: 'Liam yawned and kept rubbing his eyes during the lecture.' What can you infer about Liam?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

You can infer that Liam is tired or bored.

Challenge: Scanning for Facts & Numbers

READING SKILLS

🎯 QUICK EXAM CHALLENGE — DfE L1.R8

If a question asks: 'Who is the event coordinator?', what visual clue will you scan for?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Scan for Capital Letters indicating a person's name or title (e.g. 'Coordinator' or 'Sarah Jenkins').

Challenge: Alphabetical Order & Directories

READING SKILLS

🎯 QUICK EXAM CHALLENGE — DfE L1.R11

Put in alphabetical order: Nurse, Needle, Next, Night.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Needle, Next, Night, Nurse.

Challenge: Vocabulary: Using Context Clues

READING SKILLS

🎯 QUICK EXAM CHALLENGE — DfE L1.R10

What does 'hazardous' mean in: 'Wear gloves when handling hazardous chemicals that burn the skin'?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'Hazardous' means dangerous or harmful.

12. Instructions & Step-by-Step

READING SKILLS

Chronological Sequence & Imperatives

Instructions must be followed in order. Look for sequence words: First, Next, Then, After that, Finally. Action verbs (bossy verbs) start each instruction ('Turn off power').

FOLLOWING INSTRUCTIONS

Step 1: First, wash hands thoroughly

Step 2: Next, open sterile packaging

Step 3: Finally, apply bandage securely

⋆ **Golden Rule: Instructions always use imperative 'bossy' verbs at the start of sentences!**

⚠️ **Watch Out: Never skip a step when answering questions about instructional texts.**

13. Formal vs Informal Language

READING SKILLS

Professional vs Casual Speech

Formal: Used with managers, council, doctors ('I am writing to enquire', no slang, full words). Informal: Used with friends, family ('Hi mate', contractions, emojis, casual phrases).

FORMAL

- **"Dear Sir or Madam"**
- **"I am writing to..."**
- **"Yours sincerely"**

INFORMAL

- **"Hi Alex"**
- **"Can't wait to see you!"**
- **"Cheers / Bye"**

⋆ **Golden Rule: Avoid contractions (don't, can't) when writing formal letters or reports!**

⚠️ **Watch Out: Don't use informal slang like 'gonna' or 'cool' in workplace emails.**

14. Comparing Two Simple Texts

READING SKILLS

What is the Same? What is Different?

To compare two texts at Level 1: 1. State one piece of information that BOTH texts agree on. 2. State one difference between the two texts (e.g. price, opening time, or opinion).

LEVEL 1 TEXT COMPARISON

Agreement: "Both texts state the gym has a swimming pool."

Difference: "Text A costs £30, while Text B costs £35."

⋆ **Golden Rule: Always answer both parts: one similarity AND one difference!**

⚠️ **Watch Out: Make sure you name both Text A and Text B in your answer.**

15. Target Audience (Who is it for?)

READING SKILLS

Spotting Who the Text is Written For

To find the audience, look at who benefits from the information: Parents (school notices), Motorists (parking leaflet), Dog owners, Job seekers, Patients.

IDENTIFYING TARGET AUDIENCE

- **Look at vocabulary and images**
- **Notice who is directly addressed ('Dear Residents')**
- **Be specific: 'New parents', not just 'people'**

⋆ **Golden Rule: Be specific when naming the audience (e.g. 'Gym members', NOT just 'adults')!**

⚠️ **Watch Out: Don't just say 'everyone' — almost every text has a specific intended reader.**

Challenge: Formal vs Informal Language

READING SKILLS

🎯 QUICK EXAM CHALLENGE — DfE L1.R13

Rewrite this informally for a friend: 'I should be delighted to accompany you to the cinema.'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'I'd love to come to the cinema with you!' (or 'I'd love to go to the cinema!').

Challenge: Instructions & Step-by-Step

READING SKILLS

🎯 QUICK EXAM CHALLENGE — DfE L1.R12

Identify the imperative verb in: 'Carefully insert the key and turn clockwise.'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'insert' and 'turn'.

Challenge: Target Audience (Who is it for?)

READING SKILLS

🎯 QUICK EXAM CHALLENGE — DfE L1.R15

A leaflet gives advice on preparing babies for weaning onto solid food. Who is the target audience?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Parents and caregivers of young babies.

Challenge: Comparing Two Simple Texts

READING SKILLS

🎯 QUICK EXAM CHALLENGE — DfE L1.R14

Text 1 says a cafe is open 7 days a week. Text 2 says it is open Monday to Saturday only. What is the difference?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Text 1 says the cafe opens on Sundays (7 days), while Text 2 says it is closed on Sundays (6 days).

16. Informal Email to a Friend

WRITING SKILLS

Friendly Greeting & Clear Points

Format: 1. Subject line. 2. Friendly greeting (Hi Sam / Dear Chloe). 3. Polite friendly opener ('Hope you are well'). 4. Clear details in 2 paragraphs. 5. Warm sign-off (Best wishes / See you soon).

Subject: Catching up / Weekend plans

Hi Marcus,

Hope you are having a good week. Would you like to meet...

See you soon, / Best wishes,

⋆ Golden Rule: Always include a relevant Subject Line, even for friendly emails!

⚠ Watch Out: Even informal emails must have capital letters and full stops in exams.

17. Formal Email (Enquiry/Absence)

WRITING SKILLS

Polite Workplace Message

Format: 1. Clear subject line ('Absence Notification - 12 Oct'). 2. Polite greeting (Dear Mr Davies). 3. Factual reason for email. 4. Request for confirmation or action. 5. Formal sign-off (Kind regards).

Subject: Absence Notification: [Your Name]

Dear Ms Taylor,

I am writing to inform you that I am unwell today and will be unable to attend work...

Kind regards,

⋆ Golden Rule: Keep workplace emails concise, polite, and professional!

⚠ Watch Out: Don't write text-speak (e.g. 'u', 'thx', 'c u') in any exam email.

18. Letter of Complaint (L1)

WRITING SKILLS

Addresses • Clear Issue • Expected Remedy

Layout: Addresses and date at top. Opening: 'I am writing to complain about...'. Body: Factual details (date, receipt number, defect). Closing: What you want done ('I expect a full refund or replacement').

Dear Customer Services,

1. Reason: I am writing to complain about my recent order...

2. Problem: The item arrived damaged on 14th May...

3. Solution: I request a replacement or full refund...

Yours faithfully,

⋆ Golden Rule: Be polite and factual — never use rude or abusive language when complaining!

⚠ Watch Out: Always state what resolution you expect (refund, apology, repair).

19. Community Newsletter Article

WRITING SKILLS

Catchy Title • Positive News • Details

Features: 1. Interesting Title. 2. Introduction: What, where, and when. 3. Body: Highlights, quotes, and benefits for residents. 4. Call to join or participate.

NEWSLETTER ARTICLE

Title: New Community Garden Opens!

On Saturday, local volunteers transformed the park...

Come along this Saturday at 10 am to help plant seeds!

⋆ Golden Rule: Articles must have a headline at the top!

⚠ Watch Out: Organise into at least two or three paragraphs rather than one solid block of text.

Challenge: Formal Email (Enquiry/Absence)

WRITING SKILLS

🎯 QUICK EXAM CHALLENGE — DfE L1.W17

Write a professional closing sign-off for a formal email to your tutor.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'Kind regards,' or 'Best regards,' followed by your full name.

Challenge: Informal Email to a Friend

WRITING SKILLS

🎯 QUICK EXAM CHALLENGE — DfE L1.W16

What are two appropriate sign-offs for an informal email to a friend?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'Best wishes,' or 'See you soon,'.

Challenge: Community Newsletter Article

WRITING SKILLS

🎯 QUICK EXAM CHALLENGE — DfE L1.W19

Write an engaging title for an article about a local fun run.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Example: 'Lace Up Your Trainers for the Annual Riverside Fun Run!'

Challenge: Letter of Complaint (L1)

WRITING SKILLS

🎯 QUICK EXAM CHALLENGE — DfE L1.W18

What should always be stated in the final paragraph of a complaint letter?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

The action or remedy you expect the company to take (e.g. a refund or replacement).

20. Writing an Event Flyer / Leaflet

WRITING SKILLS

Who • What • Where • When • Cost

A flyer must include the 5 W's: What the event is. When (date & time). Where (venue). Cost (ticket price / free). Who to contact (email / telephone / website).

COMMUNITY BAKE SALE

📅 Date: Saturday 18th November | ⏰ Time: 10am - 2pm

📍 Venue: St Mark's Community Hall

🎟️ Entry: FREE (All proceeds to charity)

📞 Contact: bakesale@email.com / 01865 123456

⋆ Golden Rule: Make dates, times, and venue stand out with bullet points!

⚠️ Watch Out: Don't forget contact details — visitors need to know how to book or ask questions.

21. Writing a Review (L1)

WRITING SKILLS

What You Liked & What Could Improve

Review structure: 1. Star rating (e.g. 4/5). 2. What you bought or experienced. 3. Positive features. 4. One thing that could be improved. 5. Overall recommendation.

CAFE REVIEW ⋆⋆⋆⋆⋆

✓ Positives: Delicious food and friendly staff.

✕ Drawback: The service was slightly slow at lunch.

Recommendation: I would definitely return!

⋆ Golden Rule: Give specific reasons for your opinion rather than just saying 'it was good'!

⚠️ Watch Out: Mention both pros and at least one small con to show balanced judgment.

22. Incident & Accident Report

WRITING SKILLS

Strictly Factual & Clear

Workplace report format: 1. Date, time, and location of incident. 2. Persons involved and witnesses. 3. Factual description of what occurred (no guessing). 4. First aid given or action taken.

ACCIDENT REPORT FORM

Date: 12 October | Time: 14:15 | Location: Warehouse Bay 3

Details: Employee slipped on spilled water near the sink.

Action Taken: First aid ice pack applied; area cordoned off.

Reported by: [Staff Name]

⋆ Golden Rule: Incident reports must be 100% factual — state only what happened!

⚠️ Watch Out: Don't assign blame or write emotional rants in an incident report.

23. Basic Paragraphing (TiP ToP)

WRITING SKILLS

New Idea = New Paragraph

Start a new paragraph whenever there is a change in: Time (later that day), Place (back at the office), Topic (turning to the cost), or Person (a new speaker). Leave a blank line between paragraphs.

WHEN TO START A NEW PARAGRAPH

Ti - Time (Next morning, Later)

P - Place (In the hospital, At home)

To - Topic (Discussing the cost, Looking at safety)

P - Person (A different speaker or character)

⋆ Golden Rule: Leave a clear blank line between paragraphs in your exam!

⚠️ Watch Out: One giant unspaced block of writing loses marks for organisation.

Challenge: Writing a Review (L1)

WRITING SKILLS

🎯 QUICK EXAM CHALLENGE — DfE L1.W21

Write a sentence recommending a book or film you enjoyed.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'I would highly recommend this film because the storyline was exciting and kept me guessing until the very end.'

Challenge: Writing an Event Flyer / Leaflet

WRITING SKILLS

🎯 QUICK EXAM CHALLENGE — DfE L1.W20

What five essential details must every event flyer include?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

1. Event name (what). 2. Date and time (when). 3. Location (where). 4. Price/ticket cost. 5. Contact information.

Challenge: Basic Paragraphing (TiP ToP)

WRITING SKILLS

🎯 QUICK EXAM CHALLENGE — DfE L1.W23

What does the 'TiP ToP' paragraph rule stand for?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Ti = Time, P = Place, To = Topic, P = Person (start a new paragraph when any of these change).

Challenge: Incident & Accident Report

WRITING SKILLS

🎯 QUICK EXAM CHALLENGE — DfE L1.W22

Why should personal opinions be excluded from an workplace accident report?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Accident reports must be objective legal records based entirely on verified facts.

24. Using Linking Connectives

WRITING SKILLS

Joining Sentences Smoothly

Use connectives to connect ideas: Adding (and, also, as well as). Contrasting (but, however, although). Explaining cause (because, so, therefore). Sequencing (then, after that, finally).

ESSENTIAL CONNECTIVES

Cause: "I was late BECAUSE the bus broke down."

Contrast: "The car was old, HOWEVER it ran well."

Addition: "We need bread AND we also need milk."

⋆ **Golden Rule: Linking words turn short choppy sentences into flowing prose!**

⚠️ **Watch Out: Don't start every sentence with 'And' or 'Because'.**

25. Writing Instructions (Bossy Verbs)

WRITING SKILLS

Start with Imperative Verbs

When writing instructions, start each point with a clear command verb (imperative): 'Chop', 'Wash', 'Turn', 'Press', 'Check'. Keep sentences direct, short, and numbered.

IMPERATIVE COMMAND VERBS

1. **Wash hands with warm soap and water.**
2. **Remove the plastic safety cap.**
3. **Press the red button to power on.**

⋆ **Golden Rule: Put the action verb first: 'Press button', not 'You should press button'!**

⚠️ **Watch Out: Avoid polite padding like 'If you wouldn't mind pressing...' in user instructions.**

26. Varying Sentence Lengths

WRITING SKILLS

Mix Short and Long Sentences

Use a mix: Short sentences create impact, suspense, or clarity ('Stop immediately.'). Longer compound sentences explain reasons and add detail smoothly.

SENTENCE VARIETY

Short: "Help arrived just in time." (Impact)

Long: "The team worked together through the night, and they successfully restored full electrical power."

⋆ **Golden Rule: A short sentence at the end of a paragraph creates drama and punch!**

⚠️ **Watch Out: Don't write endless rambling sentences joined by 'and... and... and...'.**

27. Tone: Polite and Constructive

WRITING SKILLS

Professional Courtesy

Always maintain a polite, respectful tone even when making a complaint: Use 'Please could you look into this matter' rather than 'You need to sort this out now'.

CONSTRUCTIVE POLITE TONE

Rude: "Your staff are useless and I want my money back."

Polite: "I was disappointed with the service received, and would appreciate a full refund at your convenience."

⋆ **Golden Rule: Polite firmness achieves faster results than angry hostility!**

⚠️ **Watch Out: Never write in ALL CAPITALS in an email — it looks like shouting.**

Challenge: Writing Instructions (Bossy Verbs)

WRITING SKILLS

🎯 QUICK EXAM CHALLENGE — DfE L1.W25

Rewrite as a clear command instruction: 'You need to make sure the door is locked before leaving.'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'Lock the door securely before leaving.'

Challenge: Using Linking Connectives

WRITING SKILLS

🎯 QUICK EXAM CHALLENGE — DfE L1.W24

Join these two sentences with a suitable connective: 'It was raining heavily. We decided to walk to work.'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'Although it was raining heavily, we decided to walk to work' (or 'It was raining heavily, but we decided to walk to work').

Challenge: Tone: Polite and Constructive

WRITING SKILLS

🎯 QUICK EXAM CHALLENGE — DfE L1.W27

Make this sentence more polite: 'Give me the report by 3 pm.'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'Please could you provide the report by 3 pm?'

Challenge: Varying Sentence Lengths

WRITING SKILLS

🎯 QUICK EXAM CHALLENGE — DfE L1.W26

Why should you avoid having all sentences the same length?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Writing with varied sentence lengths sounds natural, engaging, and prevents the text from becoming repetitive.

28. Proofreading Checklist

WRITING SKILLS

Final 3-Minute Sweep

Before submitting: 1. Capitals on names and 'I'. 2. Full stop at every sentence end. 3. Correct homophones (there/their/they're). 4. Clear paragraphs. 5. Does every sentence make sense?

5-POINT PROOFREADING SWEEP

- [✓] 1. Every sentence starts with a CAPITAL letter
- [✓] 2. Every sentence ends with . or ?
- [✓] 3. The pronoun 'I' is always capitalized
- [✓] 4. Clear paragraph breaks

★ Golden Rule: Read your work slowly in your head to check for missing words!

⚠️ Watch Out: Don't rush to finish early — use the last minutes to collect easy SPaG marks.

29. Capital Letters: 5 Golden Rules

SPAG ESSENTIALS

M.I.N.T.S Rules

Always use a capital letter for: M - Months & Days (Monday, July). I - The word 'I'. N - Names of people & places (Sarah, London). T - Titles of books/films. S - Start of every sentence.

M.I.N.T.S FOR CAPITAL LETTERS

- M - Months & Days: Friday, October
- I - The pronoun: 'I' (NEVER lower case 'i')
- N - Names: David, Manchester, NHS
- S - Start of every new sentence

★ Golden Rule: The word 'I' is ALWAYS a capital letter, no matter where it is in a sentence!

⚠️ Watch Out: Days and months must have capitals: 'wednesday' and 'august' are WRONG.

30. Full Stops & Question Marks

SPAG ESSENTIALS

Sentence Endings

Every complete sentence must end with an end mark: Full stop (.) for statements. Question mark (?) for direct questions starting with Who, What, Where, When, Why, How, or Can.

SENTENCE ENDINGS

Statement (.): "The meeting begins at 10 am."

Question (?): "What time does the meeting start?"

★ Golden Rule: Direct questions always demand a question mark (?)!

⚠️ Watch Out: Never use multiple question marks (???) or full stops (...!) in exam writing.

31. Exclamation Marks (!)

SPAG ESSENTIALS

Urgency, Surprise & Warning

Use exclamation marks (!) sparingly to show strong emotion, surprise, danger, or urgency ('Watch out!', 'What wonderful news!'). Avoid in formal letters and reports.

EXCLAMATION MARK RULES

- ✓ Use for emergency warnings: 'Fire! Evacuate now!'
- ✓ Use in friendly flyers: 'Join our summer gala!'
- ✗ DO NOT use in formal reports or complaint letters!

★ Golden Rule: One exclamation mark is enough — never write '!!' or '!!!'!

⚠️ Watch Out: Overusing exclamation marks makes writing look childish.

Challenge: Capital Letters: 5 Golden Rules

SPAG ESSENTIALS

🎯 QUICK EXAM CHALLENGE — DfE L1.W29

Correct the capital letters: 'last monday, i went to birmingham with my sister anna.'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'Last Monday, I went to Birmingham with my sister Anna.'

Challenge: Proofreading Checklist

WRITING SKILLS

🎯 QUICK EXAM CHALLENGE — DfE L1.W28

What punctuation mark is missing from: 'How much does a weekly train ticket cost'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

A question mark (? at the end of the sentence).

Challenge: Exclamation Marks (!)

SPAG ESSENTIALS

🎯 QUICK EXAM CHALLENGE — DfE L1.W31

Is an exclamation mark appropriate in a formal letter to your local council about council tax?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

No, formal letters should maintain a calm, professional tone using full stops.

Challenge: Full Stops & Question Marks

SPAG ESSENTIALS

🎯 QUICK EXAM CHALLENGE — DfE L1.W30

Add the correct punctuation mark: 'Could you please confirm your email address'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'Could you please confirm your email address?'

32. Commas in Lists

SPAG ESSENTIALS

Separating 3 or More Items

Use commas to separate items in a list of three or more things. Do NOT put a comma before the word 'and' at the end of a simple list.

COMMA LIST RULE

"We bought milk, eggs, bread and butter."

Commas between first items; 'and' before the last item.

⋆ Golden Rule: Comma between items, 'and' before the final item!

⚠ Watch Out: Don't put a comma after the final item in the list.

33. Apostrophes for Contractions

SPAG ESSENTIALS

Replacing Missing Letters

An apostrophe replaces missing letters when two words are joined into one: do not → don't (o is missing). cannot → can't (no is missing). it is → it's (i is missing).

CONTRACTION APOSTROPHES

do not → don't (replaces 'o')

could not → couldn't (replaces 'o')

they have → they've (replaces 'ha')

⋆ Golden Rule: The apostrophe sits exactly where the missing letter used to be!

⚠ Watch Out: Don't write 'dont' or 'cant' without an apostrophe — examiners mark these as spelling errors.

34. Apostrophes for Possession

SPAG ESSENTIALS

Showing Who Owns What

Singular owner: Add 's after the owner's name (the boy's bike = bike belongs to one boy). Plural owners: If word already ends in s, just add apostrophe after the s (the boys' bikes = bikes belong to many boys).

POSSESSIVE APOSTROPHE

Singular: John's bag (1 person owns it)

Plural: The teachers' lounge (many teachers)

⚠ Never use an apostrophe for simple plurals ('two cats')!

⋆ Golden Rule: Ask: 'Who does it belong to?' Then put the apostrophe after the owner!

⚠ Watch Out: 'CDs' and '1990s' do NOT have apostrophes (they are simple plurals).

35. Plural Spelling Rules

SPAG ESSENTIALS

-s, -es, -ies, -ves

Standard: Add -s (book → books). Ending in ch, sh, s, x: Add -es (box → boxes, watch → watches). Ending in consonant + y: Drop y, add -ies (baby → babies). Ending in f/fe: Change to -ves (leaf → leaves).

PLURAL SPELLING RULES

box → boxes (ends in x, s, ch, sh → add -es)

city → cities (consonant + y → change to -ies)

knife → knives (ends in f/fe → change to -ves)

⋆ Golden Rule: If there's a vowel before 'y' (boy, day), just add 's' (boys, days)!

⚠ Watch Out: Don't write 'babys' or 'boxs' — apply the rules.

Challenge: Apostrophes for Contractions

SPAG ESSENTIALS

🎯 QUICK EXAM CHALLENGE — DfE L1.W33

Write the contracted form with an apostrophe for: 'have not'.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'haven't'.

Challenge: Commas in Lists

SPAG ESSENTIALS

🎯 QUICK EXAM CHALLENGE — DfE L1.W32

Punctuate with commas: 'The stall sold cakes pies pastries and fresh bread.'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'The stall sold cakes, pies, pastries and fresh bread.'

Challenge: Plural Spelling Rules

SPAG ESSENTIALS

🎯 QUICK EXAM CHALLENGE — DfE L1.W35

What is the correct plural form of 'match'?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'matches' (ends in 'ch', so add '-es').

Challenge: Apostrophes for Possession

SPAG ESSENTIALS

🎯 QUICK EXAM CHALLENGE — DfE L1.W34

Add the possessive apostrophe: 'The nurse coat was hanging on the door.'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'The nurse's coat was hanging on the door.'

36. Irregular Plurals

SPAG ESSENTIALS

Words That Change Internally

Some words change their internal vowels instead of adding 's': man → men, woman → women, child → children, person → people, tooth → teeth, foot → feet, mouse → mice. Sheep and fish stay the same.

IRREGULAR PLURALS

child → children (NOT childs)

person → people (NOT persons)

tooth → teeth • foot → feet

⋆ Golden Rule: Never add an 's' to irregular plurals: 'childrens' is WRONG!

⚠ Watch Out: 'Women' is pronounced 'wim-in' (plural); 'woman' is singular.

37. Subject-Verb Agreement (L1)

SPAG ESSENTIALS

Singular vs Plural Matching

I go / He goes. We are / She is. They were / He was. You do / She does. The verb must agree with who is doing the action.

AGREEMENT MATCHING

He WAS (singular) | They WERE (plural)

She DOES (singular) | They DO (plural)

✕ "We was going..." → ✓ "We were going..."

⋆ Golden Rule: 'We was' is always marked wrong — it must be 'We were'!

⚠ Watch Out: Watch out for slang habits in speaking that creep into written work.

38. Past Tense (-ed Rules)

SPAG ESSENTIALS

Regular Past Tense Suffixes

Most regular verbs add -ed (walk → walked). If verb ends in 'e', just add 'd' (like → liked). If short vowel + consonant, double consonant (stop → stopped, plan → planned).

REGULAR PAST TENSE (-ED)

walk + ed = walked

hope + d = hoped (drop redundant e)

stop → stopped (double the 'p')

⋆ Golden Rule: Double the consonant if the vowel before it is short (hop → hopped)!

⚠ Watch Out: 'Hoped' (from hope) has one 'p'; 'hopped' (from hop) has two 'p's.

39. Irregular Past Tense Verbs

SPAG ESSENTIALS

Saw • Went • Bought • Came

Irregular verbs do NOT add -ed in the past tense: See → saw (NOT seened or sawed). Go → went (NOT goed). Buy → bought (NOT buyed). Come → came (NOT comed). Write → wrote.

IRREGULAR VERBS IN PAST TENSE

see → SAW | go → WENT

buy → BOUGHT | catch → CAUGHT

⚠ "I seen it" is WRONG → "I SAW it"

⋆ Golden Rule: 'I saw it' is correct; 'I seen it' is incorrect without 'have'!

⚠ Watch Out: Don't add -ed to words that already changed: 'went', not 'wented'.

Challenge: Subject-Verb Agreement (L1)

SPAG ESSENTIALS

🎯 QUICK EXAM CHALLENGE — DfE L1.W37

Choose the correct verb: 'The nurses (was / were) very supportive.'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'were' ('nurses' is plural, so it takes 'were').

Challenge: Irregular Plurals

SPAG ESSENTIALS

🎯 QUICK EXAM CHALLENGE — DfE L1.W36

Correct the error: 'Several childs were playing in the park.'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'Several children were playing in the park.'

Challenge: Irregular Past Tense Verbs

SPAG ESSENTIALS

🎯 QUICK EXAM CHALLENGE — DfE L1.W39

Correct the past tense verb: 'Yesterday, she goed to the supermarket and buyed milk.'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'Yesterday, she went to the supermarket and bought milk.'

Challenge: Past Tense (-ed Rules)

SPAG ESSENTIALS

🎯 QUICK EXAM CHALLENGE — DfE L1.W38

Write the past tense of the verb 'drop'.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'dropped' (double the final consonant).

40. Confusables: To / Too / Two

SPAG ESSENTIALS

Direction • Also/Excessive • Number

TO: Direction or verb ('Going to school', 'to sleep'). TOO: Means 'also' or 'too much' ('Can I come too?', 'too hot'). TWO: The number 2 ('two tickets').

TO

Direction / Action

"Go to work"

TOO

Also / Excessive

"Too expensive"

TWO

Number 2

"Two cups"

★ Golden Rule: 'Too' has an extra 'o' because it has too much of something!

⚠ Watch Out: 'I want to go to' is WRONG — it must be 'I want to go too'.

41. There / Their / They're (L1)

SPAG ESSENTIALS

Location • Belonging • They Are

THERE has 'here' in it = place ('over there'). THEIR has 'heir' in it = ownership ('their coats'). THEY'RE has an apostrophe = 'they are' ('they're coming').

HOMOPHONE MEMORY TRICK

tHERE = Has the word 'HERE' (place)

tHEIR = Belongs to them

THEY'RE = Short for 'THEY ARE'

★ Golden Rule: Test with 'they are' — if it fits, use 'they're'!

⚠ Watch Out: This is tested in almost every Functional Skills English exam.

42. Essential Level 1 Spellings

SPAG ESSENTIALS

The 10 Golden Words

1. Because (Big Elephants Can Always Upset Small Elephants). 2. Beautiful. 3. Believe (never be-LIE-ve a lie). 4. Friend (ends in end). 5. Tomorrow (1 'm', 2 'r's). 6. Wednesday. 7. People.

ESSENTIAL SPELLING MNEMONICS

- FriEND: A true friend stays to the END
- Wed-nes-day: Say it as 'Wed-Nes-Day'
- Tomorrow: 1 Morning, 2 Roses (1 m, 2 r's)

★ Golden Rule: Spell 'friend' with 'i' before 'e', ending in 'end'!

⚠ Watch Out: 'Becos' and 'tommorow' are immediate red pen errors.

43. SLC Overview: The Two Assessments

SPEAKING & LISTENING

Task 1: Short Talk (4-5m) | Task 2: Group Discussion

Level 1 SLC consists of two parts: (1) A 4–5 minute formal talk or explanation on a prepared topic followed by questions, and (2) A 10–15 minute group discussion on an everyday topic.

TASK 1: FORMAL TALK

- 4–5 mins prepared talk
- Clear Intro, 3 Points, End
- Answer audience Q&A

TASK 2: DISCUSSION

- 10–15 mins (3–5 peers)
- Share views & listen
- Take turns & invite others

★ Golden Rule: Use bullet-point cue cards — never read a full script word-for-word!

⚠ Watch Out: Reading aloud head-down from an A4 essay script prevents eye contact and caps your grade.

Challenge: There / Their / They're (L1)

SPAG ESSENTIALS

🎯 QUICK EXAM CHALLENGE — DfE L1.W41

Which word completes the sentence: 'The team forgot to submit _____ timesheets.'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'their' (possession/belonging to the team).

Challenge: Confusables: To / Too / Two

SPAG ESSENTIALS

🎯 QUICK EXAM CHALLENGE — DfE L1.W40

Fill the blanks: 'We bought (to / too / two) tickets (to / too / two) London.'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'We bought two tickets to London.'

Challenge: SLC Overview: The Two Assessments

SPEAKING & LISTENING

🎯 QUICK EXAM CHALLENGE — DfE L1.SLC1

What are the two mandatory tasks in the Pearson Edexcel Level 1 Speaking, Listening & Communicating assessment?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Task 1: Formal 4–5 minute Talk/Presentation + Q&A. Task 2: Informal 10–15 minute Group Discussion (3–5 learners).

Challenge: Essential Level 1 Spellings

SPAG ESSENTIALS

🎯 QUICK EXAM CHALLENGE — DfE L1.W42

Spell the day of the week that comes after Tuesday.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'Wednesday'.

44. Asking Open & Follow-Up Questions

SPEAKING & LISTENING

Open Questions (How/Why) vs Closed (Yes/No)

Closed questions (Do you...? Is it...?) get one-word Yes/No answers that stall discussions. Open questions (How would...? Why do you think...?) invite detailed opinions and examples.

× CLOSED (STALLS TALK)

"Do you agree with this?"

Reply: "Yes." (Silence)

✓ OPEN (SPARKS DEBATE)

"How would this change impact local commuters?"

Invites full explanation!

⋆ **Golden Rule: Use HOW, WHY, or WHAT to ask open questions and invite full answers!**

⚠ **Watch Out: Ask at least two relevant questions to meet the passing criteria.**

45. Turn-Taking & Collaborative Talk

SPEAKING & LISTENING

Acknowledge • Build • Invite (A-B-I)

Assessors grade how well you take turns: (1) Acknowledge ("Building on Ana's point..."), (2) Add your evidence, and (3) Invite quieter members ("What is your view on this, David?").

TURN-TAKING STEMS FOR FULL MARKS

1. Agree & Build: "Following on from what Sarah said..."
2. Polite Challenge: "I see your point, but another angle is..."
3. Invite Others: "We haven't heard from Leo yet—what do you think?"

⋆ **Golden Rule: Acknowledge the last speaker before introducing your own point!**

⚠ **Watch Out: Never interrupt or dominate — invite quieter group members to speak.**

46. Active Listening & Body Language

SPEAKING & LISTENING

Eye Contact • Posture • Measured Pace

Non-verbal communication carries half your impact: maintain steady eye contact across the room, nod to show active listening, speak at a measured pace, and avoid slang in formal talks.

NON-VERBAL SIGNALS

- ✓ Eye contact with all
- ✓ Nodding when listening
- ✓ Open hand gestures

VOCAL DELIVERY

- ✓ Clear, steady pace
- ✓ Varied pitch & emphasis
- ✓ Professional register

⋆ **Golden Rule: Match your tone and body language to the formality of the setting!**

⚠ **Watch Out: Crossing your arms, looking at your phone, or staring only at the assessor loses engagement marks.**

1. OS Settings & Accessibility

DEVICES & HANDLING

Tailoring Your Device

Accessibility settings assist users with physical or cognitive impairments: High Contrast Mode (visual clarity), Text Scaling / Magnifier (enlarging text), Screen Reader (Narrator), Sticky Keys (motor support).

ACCESSIBILITY SETTINGS

- • High Contrast & Dark Mode (Low vision)
- • Screen Magnifier / Zoom (Large font scaling)
- • Screen Reader / Text-to-Speech (Dyslexia & blind)

⋆ **Golden Rule: Accessibility features ensure digital inclusion for all users!**

⚠ **Watch Out: Don't confuse display brightness with high-contrast accessibility themes.**

Challenge: Turn-Taking & Collaborative Talk

SPEAKING & LISTENING

🎯 QUICK EXAM CHALLENGE — DfE L1.SLC5

Give a polite, collaborative phrase to disagree with a colleague named Sam about budget costs.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

"I understand your point about the cost, Sam; however, have we considered how much energy the new lighting will save over five years?"

Challenge: Asking Open & Follow-Up Questions

SPEAKING & LISTENING

🎯 QUICK EXAM CHALLENGE — DfE L1.SLC2

Turn this closed question into an open question: "Do you think flexible working hours are a good idea?"

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Open question: "How do you think flexible working hours would affect staff morale?" (requires a detailed explanation, not just yes/no).

Challenge: OS Settings & Accessibility

DEVICES & HANDLING

🎯 QUICK EXAM CHALLENGE — DFS 1.1

Which accessibility feature allows a user with visual impairment to hear text read aloud from the screen?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

A screen reader (or text-to-speech / Narrator).

Challenge: Active Listening & Body Language

SPEAKING & LISTENING

🎯 QUICK EXAM CHALLENGE — DfE L1.SLC7

State two non-verbal body language signals that show assessors you are actively listening.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

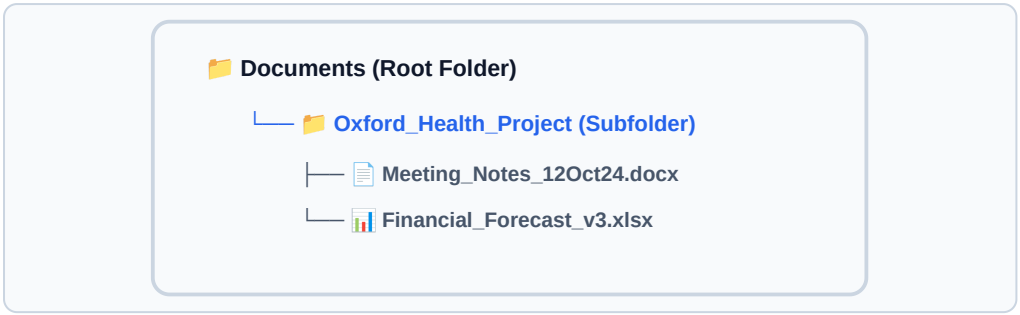
Maintain regular eye contact with the group, nod to show active listening, and keep an open, upright posture.

2. File Management & Naming

DEVICES & HANDLING

Hierarchical Folders & Version Control

Organise files into logical tree hierarchies: Root Folder → Subfolder → Specific file. Meaningful naming convention: Topic_Project_Date_Version (e.g. Budget_Q3_2024_v2.xlsx). Avoid 'Doc1.docx'.



⋆ Golden Rule: Include the date (YYYY-MM-DD) and version number (v1, v2) in filename!

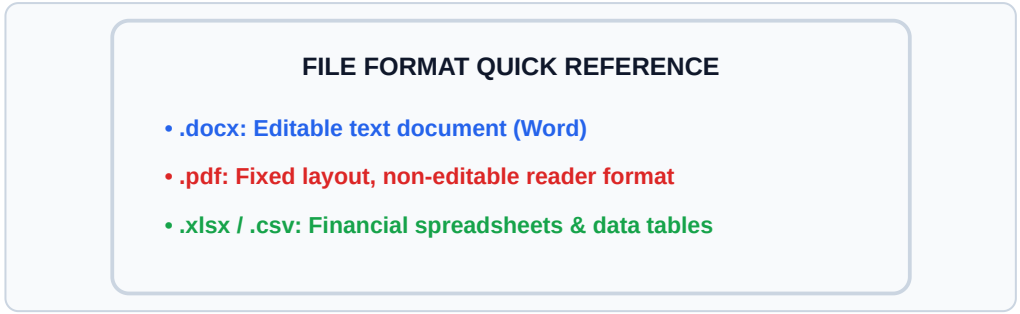
⚠ Watch Out: Avoid saving everything to your Desktop — it causes system clutter and data loss risk.

3. File Extensions & Formats

DEVICES & HANDLING

Recognising Common File Types

.docx = Word processing. .pdf = Portable read-only document (universal). .xlsx / .csv = Spreadsheets / raw comma-separated data. .pptx = Slideshow. .png / .jpg = Images. .mp4 = Video. .zip = Compressed folder.



⋆ Golden Rule: Export to PDF before sharing to keep layout and formatting intact!

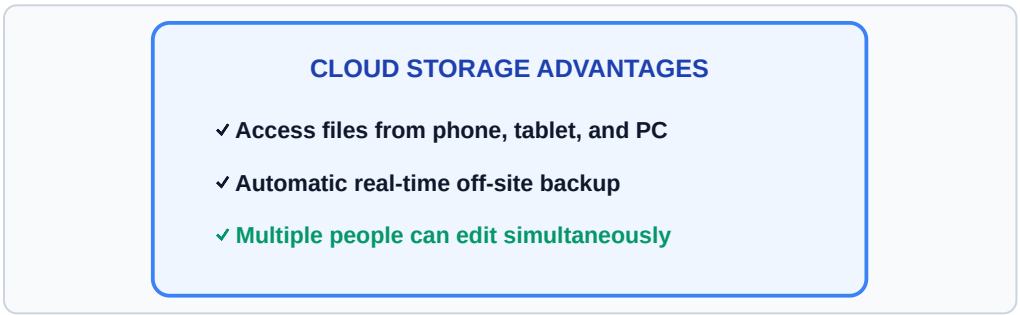
⚠ Watch Out: A .csv file saves raw data only — all formulas and styling are lost.

4. Cloud Storage & Syncing

DEVICES & HANDLING

Remote Servers & Multi-Device Access

Cloud storage (OneDrive, Google Drive, iCloud) stores files on secure remote servers over the Internet. Benefits: Automatic backup, access from any device, easy real-time collaboration. Requires internet connectivity.



⋆ Golden Rule: Cloud files protect you against hardware crashes, fires, or stolen laptops!

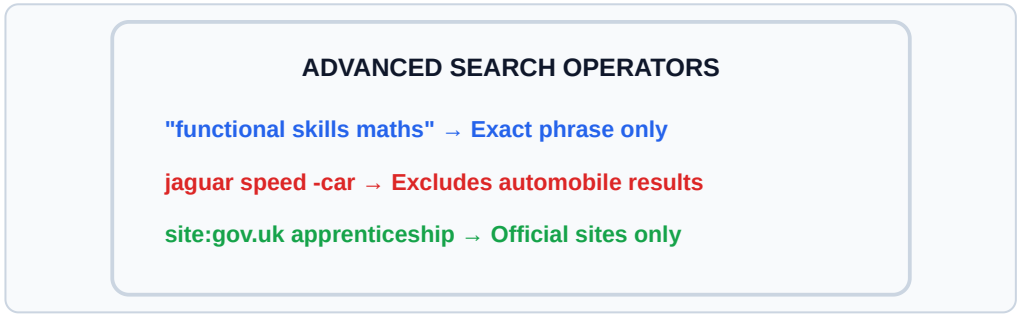
⚠ Watch Out: If you don't enable 'Keep on this device' (offline mode), you cannot open cloud files without Wi-Fi.

5. Search Engine Operators

DEVICES & HANDLING

Boolean & Precision Search Tricks

Quotation Marks ("..."): Searches for the EXACT phrase in that precise order. Minus Sign (-word): Excludes unwanted results. site:gov.uk: Restricts search to official government websites.



⋆ Golden Rule: Use quotation marks to search for exact multi-word terms!

⚠ Watch Out: Don't put a space after the minus sign (type '-car', NOT '- car').

Challenge: File Extensions & Formats

DEVICES & HANDLING

🎯 QUICK EXAM CHALLENGE — DFS 1.3

Which file format should you choose to ensure a formal report cannot be accidentally altered by the recipient?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

PDF (.pdf - Portable Document Format).

Challenge: File Management & Naming

DEVICES & HANDLING

🎯 QUICK EXAM CHALLENGE — DFS 1.2

Suggest a professional filename for the second draft of a staff training plan written in October 2024.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Staff_Training_Plan_Oct2024_v2.docx.

Challenge: Search Engine Operators

DEVICES & HANDLING

🎯 QUICK EXAM CHALLENGE — DFS 1.5

What search query would you type to find exact information on NHS dental fees while excluding private dentists?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

"NHS dental fees" -private (or site:nhs.uk "dental fees").

Challenge: Cloud Storage & Syncing

DEVICES & HANDLING

🎯 QUICK EXAM CHALLENGE — DFS 1.4

State two benefits of saving business documents to cloud storage rather than a local USB flash drive.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

1. Files are backed up remotely if the physical device is lost/damaged. 2. Documents can be accessed from any location or device with internet.

6. Evaluating Online Information

DEVICES & HANDLING

Domains & Sponsored Ads

Top-level domains: .gov.uk = Official UK government. .ac.uk = Universities. .org = Non-profit organisations. .com / .co.uk = Commercial businesses. Look out for 'Sponsored / Ad' tags at top of search results.

CHECKING WEBSITE RELIABILITY

.gov.uk & .nhs.uk: Highly authoritative

.ac.uk: Academic university research

Sponsored / Ad tags: Commercial promotion

★ Golden Rule: Cross-check web information across at least two trusted sources!

⚠️ Watch Out: Watch for lookalike URLs and typosquatting domains (e.g. paypa1.com).

7. Device Storage vs RAM

DEVICES & HANDLING

Storage (SSD/Hard Drive) vs RAM

Storage (GB/TB): Permanent memory holding files and apps when PC is turned off. RAM (Random Access Memory): Temporary working memory used while apps are open. Free up storage by deleting temp files and emptying Recycle Bin.

STORAGE (SSD/HDD)

- Permanent file storage
- 256 GB / 1 TB
- Survives restarts

MEMORY (RAM)

- Active app workspace
- 8 GB / 16 GB
- Clears on restart

★ Golden Rule: Empty your Recycle Bin / Trash to permanently reclaim disk storage!

⚠️ Watch Out: Running out of RAM makes your computer freeze; running out of Storage prevents saving new files.

8. Compressing & Extracting (.zip)

DEVICES & HANDLING

Reducing File Size for Email

Compress (Zip): Shrinks file size and bundles multiple documents into one single .zip folder for emailing. Extract: Unpacks the zipped files into normal editable documents.

ZIPPING & EXTRACTING FILES

Compress: Right-click → 'Send to compressed folder'

Combines 10 files into 1 small .zip attachment

Extract: Right-click .zip → 'Extract All...' to view

★ Golden Rule: You must EXTRACT files before you can properly edit and save them!

⚠️ Watch Out: If you edit files directly inside an unextracted zip, changes may be lost.

9. Basic Troubleshooting

DEVICES & HANDLING

First Steps When Tech Fails

App frozen: Open Task Manager (Ctrl + Shift + Esc) and select 'End Task'. Wi-Fi dropped: Check flight mode, restart router, toggle Wi-Fi off and on. Total system freeze: Hold power button for 10 seconds to reboot.

FIRST RESPONSE TROUBLESHOOTING

1. App Frozen? Press Ctrl+Shift+Esc → 'End Task'
2. No Internet? Check Airplane mode & restart Wi-Fi router
3. The Golden Rule: Restart the computer before calling IT!

★ Golden Rule: Restarting clears memory caches — try it first before calling support!

⚠️ Watch Out: Never unplug a PC while it is writing data or installing updates.

Challenge: Device Storage vs RAM

DEVICES & HANDLING

🎯 QUICK EXAM CHALLENGE — DFS 1.7

State two actions you can take if your smartphone displays a 'Storage Full' warning.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

1. Delete unused apps or clear app caches. 2. Back up photos/videos to cloud storage or delete large video files.

Challenge: Evaluating Online Information

DEVICES & HANDLING

🎯 QUICK EXAM CHALLENGE — DFS 1.6

Why is information found on a .gov.uk website generally more reliable than a .com website for renewing a passport?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

.gov.uk is the official UK government portal, whereas .com sites are commercial agencies that often charge extra fees.

Challenge: Basic Troubleshooting

DEVICES & HANDLING

🎯 QUICK EXAM CHALLENGE — DFS 1.9

What keyboard shortcut opens Windows Task Manager directly to close a frozen program?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Ctrl + Shift + Esc (or Ctrl + Alt + Delete).

Challenge: Compressing & Extracting (.zip)

DEVICES & HANDLING

🎯 QUICK EXAM CHALLENGE — DFS 1.8

Why is it beneficial to zip five large image files before sending them by email?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Zipping reduces the overall file size to fit within email attachment limits and bundles all images into one file.

10. Data Protection & UK GDPR

DEVICES & HANDLING

Handling Personal Data Lawfully

Personal data relates to an identifiable living person (names, NHS numbers, addresses). Under UK GDPR: 1. Process lawfully, fairly, and transparently. 2. Keep accurate and secure. 3. Delete when no longer required.

UK GDPR KEY PRINCIPLES

- Collect only the data that is strictly necessary
- Keep data secure with encryption & passwords
- Never share patient/client records without consent

★ Golden Rule: Always lock your computer screen (Win + L) when leaving your desk!

⚠ Watch Out: Leaving customer spreadsheets visible on an unattended screen is a serious GDPR data breach.

11. Word Processing: Headings & Styles

CREATING CONTENT

H1, H2 & Consistent Hierarchy

Use built-in Heading Styles (Heading 1 for main titles, Heading 2 for subheadings) instead of manually bolding and enlarging text. This creates an automatic table of contents and screen-reader structure.

BUILT-IN HEADING STYLES

Heading 1: Major Document Title

Heading 2: Section Subheading

Normal: Body text with 1.15 line spacing

★ Golden Rule: Use Heading styles to automatically create clickable Contents pages!

⚠ Watch Out: Use Ctrl + Enter for new pages — never press Enter multiple times.

12. Word Processing: Tables & Lists

CREATING CONTENT

Organising Information into Columns

Tables: Insert rows and columns to align comparative data neatly. Merge cells to create spanning banner headers. Set header rows to repeat across page breaks.

Merged Header Cell	
Column A	Column B

★ Golden Rule: Use tables to align numbers and facts rather than pressing the Spacebar or Tab key!

⚠ Watch Out: Never align text by tapping the spacebar repeatedly — it falls out of alignment on other screens.

13. Images & Text Wrapping

CREATING CONTENT

Square, Tight & Aspect Ratio

Always resize images using CORNER handles to maintain original aspect ratio (prevent squishing). Text Wrapping: 'Square' or 'Tight' wraps text smoothly around the picture. Always add Alt Text for accessibility.



★ Golden Rule: Drag corner handles only so images keep their proper proportions!

⚠ Watch Out: Always add Alt Text to images for accessibility and screen readers.

Challenge: Word Processing: Headings & Styles

CREATING CONTENT

🎯 QUICK EXAM CHALLENGE — DFS 2.1

What keyboard shortcut inserts a new page break instantly in Microsoft Word or Google Docs?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Ctrl + Enter.

Challenge: Data Protection & UK GDPR

DEVICES & HANDLING

🎯 QUICK EXAM CHALLENGE — DFS 1.10

State two ways you can protect sensitive customer data on a workplace laptop.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

1. Use strong password protection / encryption. 2. Lock your screen whenever stepping away from the device.

Challenge: Images & Text Wrapping

CREATING CONTENT

🎯 QUICK EXAM CHALLENGE — DFS 2.3

Why should you provide Alt Text (alternative text) when inserting an image into a document or website?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Alt Text allows screen readers to describe the image to visually impaired users.

Challenge: Word Processing: Tables & Lists

CREATING CONTENT

🎯 QUICK EXAM CHALLENGE — DFS 2.2

What table feature combines two or more adjacent cells into a single large cell?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

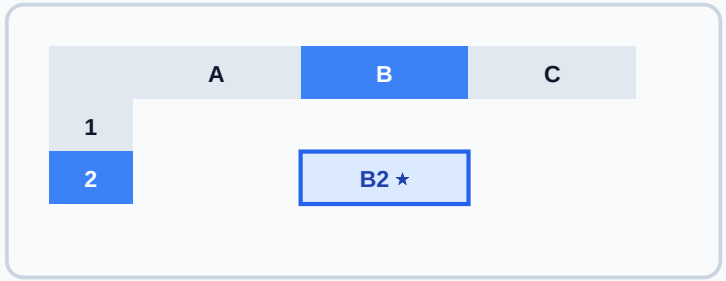
Merge Cells.

14. Spreadsheets: Anatomy of a Sheet

CREATING CONTENT

Columns, Rows & Cell References

Columns: Lettered vertically (A, B, C...). Rows: Numbered horizontally (1, 2, 3...). Cell Reference: The intersection letter + number (e.g. B4 = column B, row 4). Format cells as Currency (£), Percentage (%), or Date.



★ Golden Rule: Column Letter ALWAYS comes before the Row Number (e.g. C5, NEVER 5C)!

⚠ Watch Out: Format raw numbers as Currency (£) so the pound sign appears automatically.

15. Spreadsheets: Basic Arithmetic

CREATING CONTENT

Always Start with an Equals Sign (=)

Every formula in Excel / Google Sheets MUST begin with '=': Add: =A1+B1. Subtract: =A1-B1. Multiply: =A1*B1 (asterisk). Divide: =A1/B1 (forward slash). Never type hardcoded numbers into formulas.

SPREADSHEET OPERATORS

Addition: =B2 + C2

Subtraction: =B2 - C2

Multiplication: =B2 * C2 (Asterisk *)

Division: =B2 / C2 (Slash /)

★ Golden Rule: Always begin spreadsheet formulas with '=' (e.g. =A1+B1)!

⚠ Watch Out: Use '*' to multiply and '/' to divide — never type the letter 'x'.

16. Spreadsheets: SUM & AVERAGE

CREATING CONTENT

Ranges with Colons (e.g. B2:B10)

=SUM(B2:B10): Adds all numbers in range. =AVERAGE(B2:B10): Calculates mean. =MAX(B2:B10): Finds highest value. =MIN(B2:B10): Finds lowest value. Colon (:) means 'through to'.

CORE EXCEL FUNCTIONS

- =SUM(C2:C20) → Total of all 19 cells
- =AVERAGE(C2:C20) → Mean average score
- =MAX(C2:C20) • =MIN(C2:C20)

★ Golden Rule: The colon (:) indicates a contiguous cell range from start to finish!

⚠ Watch Out: Don't write '=SUM(A1+A2)' — either write '=SUM(A1:A2)' or '=A1+A2'.

17. Spreadsheets: COUNT vs COUNTA

CREATING CONTENT

Count Numbers vs Count Non-Empty Cells

=COUNT(range): Counts ONLY cells containing NUMBERS. =COUNTA(range): Counts ALL non-empty cells (numbers, text, symbols). Think 'Count All' (COUNTA).

=COUNT(A1:A10)

Counts NUMBERS only

Ignores names and text

Numerical tallies

=COUNTA(A1:A10)

Counts ALL non-empty

Counts names, codes & text

Attendee rosters

★ Golden Rule: Use COUNTA to count attendee names or text entries!

⚠ Watch Out: If you use COUNT on a column of patient names, the answer will be 0!

Challenge: Spreadsheets: Basic Arithmetic

CREATING CONTENT

🎯 QUICK EXAM CHALLENGE — DFS 2.5

Write the formula to multiply the quantity in cell B4 by the unit price in cell C4.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

=B4*C4.

Challenge: Spreadsheets: Anatomy of a Sheet

CREATING CONTENT

🎯 QUICK EXAM CHALLENGE — DFS 2.4

What cell reference is located at the intersection of column D and row 14?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

D14.

Challenge: Spreadsheets: COUNT vs COUNTA

CREATING CONTENT

🎯 QUICK EXAM CHALLENGE — DFS 2.7

Which function should you use to count how many employees signed in on a text attendance sheet in cells B2 to B30?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

=COUNTA(B2:B30).

Challenge: Spreadsheets: SUM & AVERAGE

CREATING CONTENT

🎯 QUICK EXAM CHALLENGE — DFS 2.6

Write the function to calculate the average of sales in cells D3 to D15.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

=AVERAGE(D3:D15).

18. Spreadsheets: Simple IF Statements

CREATING CONTENT

=IF(condition, value_if_true, value_if_false)

Syntax: 1. Test condition (e.g. B2>=50). 2. Value if TRUE in quotes ("Pass"). 3. Value if FALSE in quotes ("Fail"). Example: =IF(B2>=50, "Pass", "Fail").

IF STATEMENT LOGIC

`=IF(C2 >= 100, "Bonus", "No")`

If C2 is 100+, outputs 'Bonus'; otherwise outputs 'No'

★ Golden Rule: Text outputs in IF statements MUST be enclosed in quotation marks ("...")!

⚠️ Watch Out: Don't forget the commas separating the three arguments.

19. Spreadsheets: Sorting & Filtering

CREATING CONTENT

Order vs Isolate

Sort: Reorders rows (A–Z alphabetical, or Largest to Smallest numerical). Filter: Temporarily hides rows that do not match criteria (e.g. show ONLY customers in 'Oxford').

SORT (Reorder)

- A to Z (Alphabetical)
- High to Low (Sales)
- Keeps all records visible

FILTER (Isolate)

- Funnel icon on headers
- Hides non-matching rows
- e.g. Show 'London' only

★ Golden Rule: Filtering hides rows; sorting changes their sequence!

⚠️ Watch Out: Make sure all data columns are selected before sorting, or columns will become mismatched!

20. Spreadsheets: Creating Charts

CREATING CONTENT

Titles, Axes & Legends

To score full marks on spreadsheet charts: 1. Select category labels AND numerical data together. 2. Choose appropriate chart type (Bar for categories, Line for trends). 3. Add Chart Title, Axis Titles, and Legend.

CHART LABELLING CHECKLIST

[✓] Meaningful Chart Title at top

[✓] Horizontal Axis Title (e.g. Month / Department)

[✓] Vertical Axis Title (e.g. Revenue (£) / Patient Count)

★ Golden Rule: Every chart must have a clear descriptive Title and labelled X and Y axes!

⚠️ Watch Out: Don't include the 'Total' row when highlighting data for a pie chart — it distorts the slices.

21. Presentations: Slide Design

CREATING CONTENT

The 6 × 6 Rule & Slide Masters

Best practice: 6 × 6 Rule: Maximum 6 bullet points per slide, with 6 words per bullet. Use consistent Slide Master layouts, high-contrast fonts (dark on light), and avoid distracting animations.

SLIDE PRESENTATION RULES

- 6 × 6 Rule: Avoid dense walls of text
- High-contrast colours for legibility
- Consistent corporate theme across all slides

★ Golden Rule: Slides are visual prompts for the speaker, not essays to read aloud!

⚠️ Watch Out: Don't use low-contrast text (e.g. yellow font on white background).

Challenge: Spreadsheets: Sorting & Filtering

CREATING CONTENT

🎯 QUICK EXAM CHALLENGE — DFS 2.9

What spreadsheet tool would you use to view only employees who work in the 'Finance' department?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

The Filter tool (filtering the Department column by 'Finance').

Challenge: Spreadsheets: Simple IF Statements

CREATING CONTENT

🎯 QUICK EXAM CHALLENGE — DFS 2.8

Write an IF formula for cell D2 that displays 'Yes' if cell B2 is greater than 1000, and 'No' otherwise.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

=IF(B2>1000, "Yes", "No").

Challenge: Presentations: Slide Design

CREATING CONTENT

🎯 QUICK EXAM CHALLENGE — DFS 2.11

State two design principles that make a presentation slide easy for an audience to read.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

1. Using bullet points instead of dense paragraphs. 2. Maintaining high contrast between text and background.

Challenge: Spreadsheets: Creating Charts

CREATING CONTENT

🎯 QUICK EXAM CHALLENGE — DFS 2.10

Which chart type is best for showing sales figures over 12 consecutive months?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

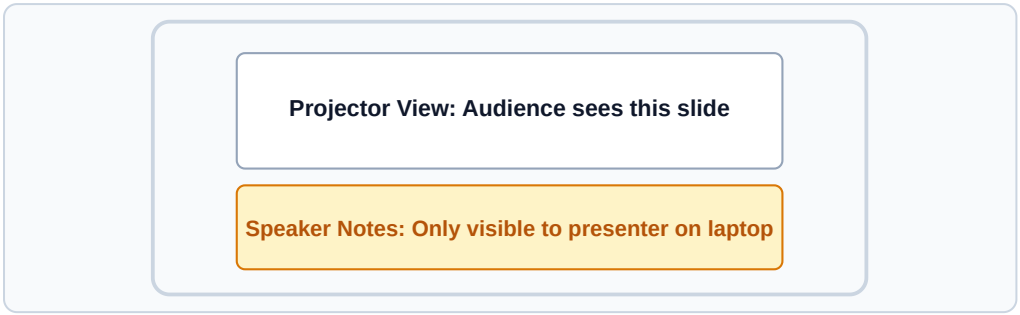
A line chart (ideal for showing continuous trends over time).

22. Presentations: Speaker Notes

CREATING CONTENT

Visible to Speaker Only

Speaker Notes Pane: Area underneath each slide where the presenter writes their talking script. In 'Presenter View', notes are visible on the speaker's laptop screen while the audience only sees the slides.



⋆ Golden Rule: Put detailed speech text in the Speaker Notes, NOT on the slide itself!

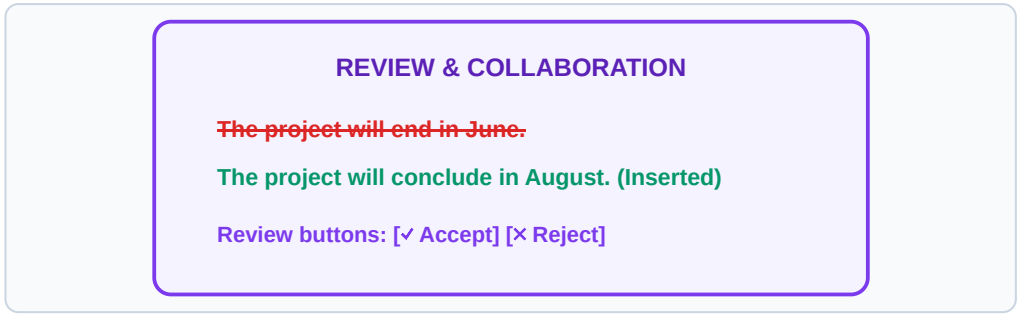
⚠️ Watch Out: Make sure your display is set to 'Extend' rather than 'Duplicate' to use Presenter View.

23. Collaboration: Track Changes

CREATING CONTENT

Comments & Suggesting Edits

Track Changes records every edit (strikethroughs, insertions in coloured ink). Other reviewers can: 1. Add Comments to discuss points. 2. 'Accept' or 'Reject' individual changes before finalizing.



⋆ Golden Rule: Always accept or reject all tracked changes before sending documents to clients!

⚠️ Watch Out: Leaving tracked changes active can leak internal confidential comments.

24. Exporting & Converting to PDF

CREATING CONTENT

File → Export → Create PDF

Why export to PDF: 1. Universal compatibility (opens on any phone/PC without needing Word). 2. Preserves exact layout, fonts, and images. 3. Prevents accidental editing. 4. Smaller file size.



⋆ Golden Rule: Save as PDF to lock your formatting before printing or emailing!

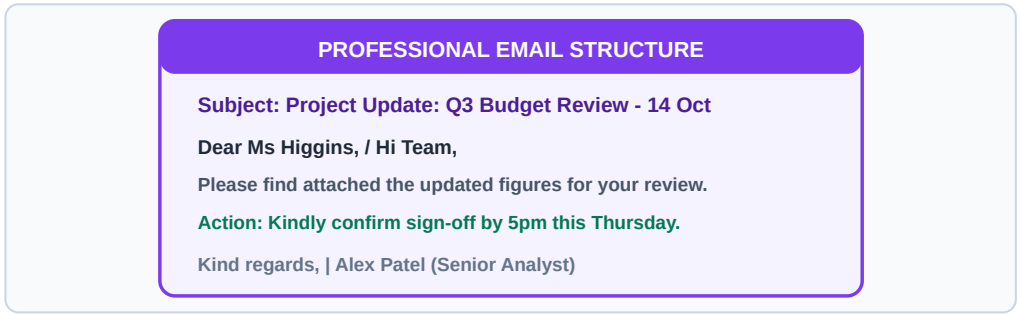
⚠️ Watch Out: Always keep your original .docx or .xlsx master file in case you need to edit later.

25. Professional Email Structure

COMMUNICATING ONLINE

Clear, Actionable Email Hierarchy

A professional email includes: 1. A clear, specific subject line. 2. A polite formal salutation. 3. Concise body paragraphs stating the main purpose. 4. An unambiguous call-to-action or deadline. 5. A professional sign-off and signature block.



⋆ Golden Rule: Always write a descriptive subject line before composing the email body!

⚠️ Watch Out: Avoid vague or blank subject lines like 'urgent' or 'hi' which trigger spam filters.

Challenge: Collaboration: Track Changes

CREATING CONTENT

🎯 QUICK EXAM CHALLENGE — DFS 2.13

What tool allows multiple team members to review a Word document and propose text edits without overwriting the original text?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Track Changes.

Challenge: Presentations: Speaker Notes

CREATING CONTENT

🎯 QUICK EXAM CHALLENGE — DFS 2.12

Where should a presenter put detailed talking points that they do not want the audience to see on the main screen?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

In the Speaker Notes pane (using Presenter View).

Challenge: Professional Email Structure

COMMUNICATING ONLINE

🎯 QUICK EXAM CHALLENGE — DFS 3.1

Why is 'Q3 Budget Review - Approval Needed by Thursday' far more effective than just 'Budget' as an email subject line?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

It specifies the exact topic, the action required (approval), and the deadline, allowing the recipient to prioritize immediately.

Challenge: Exporting & Converting to PDF

CREATING CONTENT

🎯 QUICK EXAM CHALLENGE — DFS 2.14

Give two reasons why job applications require CVs to be submitted as PDF files.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

1. Formatting, fonts, and layout remain fixed across all devices. 2. The recipient cannot easily edit or corrupt the text.

26. Email Fields: To, Cc & Bcc

COMMUNICATING ONLINE

Direct Action vs Courtesy vs Privacy

To: Direct recipients expected to read and take action. Cc (Carbon Copy): People kept informed for visibility; no direct action expected. Bcc (Blind Carbon Copy): Hidden recipients; preserves privacy and complies with UK GDPR when emailing groups.

To:	Action Owner (e.g. John - needs to complete task)
Cc:	Informed only (e.g. Manager - keeps in the loop)
Bcc:	Hidden / Blind (protects privacy & UK GDPR) Recipients cannot see each other's email addresses

⋆ Golden Rule: Use Bcc for group emails to keep recipient email addresses private!

⚠️ Watch Out: Putting client emails in 'To' or 'Cc' is a direct GDPR violation.

27. Email Attachments & Cloud Links

COMMUNICATING ONLINE

Direct File vs Cloud Share Link

Email Attachments: Suitable for small individual files under 15-20 MB. Cloud Links (OneDrive, Google Drive): Best for large files, videos, or collaborative live editing. Cloud links prevent mailbox quota exhaustion and allow permission revoking.

DIRECT ATTACHMENT	CLOUD SHARE LINK
• File size limit (~20 MB)	• Unlimited file size
• Static snapshot copy	• Real-time live collaboration
• Cannot revoke access	• Permissions can be revoked
Best for: Final invoices, receipts	Best for: Large decks, team docs

⋆ Golden Rule: Use cloud links instead of heavy attachments to avoid bouncing emails!

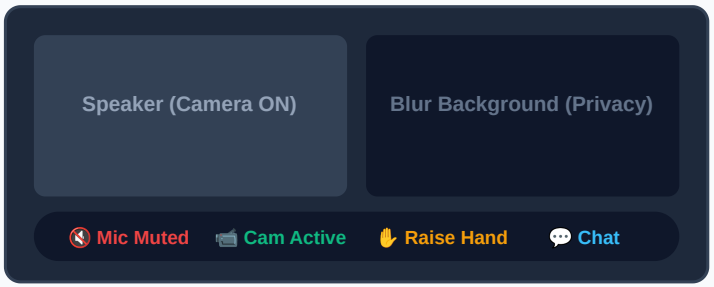
⚠️ Watch Out: Check sharing permissions before sending a cloud link so recipients can actually open it.

28. Video Calling Setup & Etiquette

COMMUNICATING ONLINE

Mute, Backgrounds & Participation

Key video meeting protocols: 1. Keep microphone MUTED when not speaking to eliminate background noise. 2. Enable Background Blur or a neutral virtual background for privacy. 3. Use 'Raise Hand' feature to speak without interrupting. 4. Maintain eye contact near camera.



⋆ Golden Rule: Mute your microphone when not speaking to eliminate background noise!

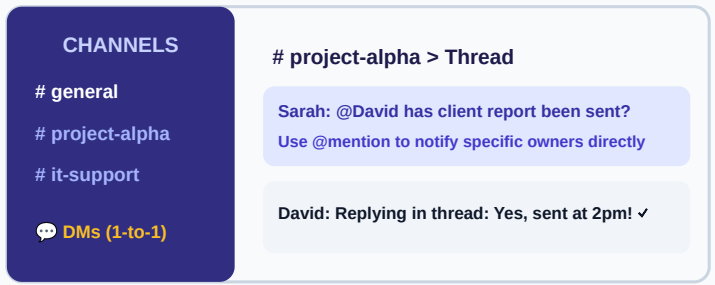
⚠️ Watch Out: Place light in FRONT of you — backlighting turns you into a shadow.

29. Workplace Messaging (Teams/Slack)

COMMUNICATING ONLINE

Channels vs Direct Messages (DMs)

Organise chat communication efficiently: 1. Use Channels for team-wide project topics and public discussions. 2. Use Direct Messages (DMs) for private, 1-to-1 conversations. 3. Use thread replies to keep specific conversations clustered. 4. Use @mentions sparingly.



⋆ Golden Rule: Use @mentions only when an immediate action or response is required!

⚠️ Watch Out: Avoid @everyone for routine updates — it creates notification fatigue.

Challenge: Email Attachments & Cloud Links

COMMUNICATING ONLINE

🎯 QUICK EXAM CHALLENGE — DFS 3.3

Why might an email containing a 65 MB video attachment fail to reach the recipient?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Most email providers enforce attachment size limits between 20 MB and 25 MB, causing larger files to be rejected.

Challenge: Email Fields: To, Cc & Bcc

COMMUNICATING ONLINE

🎯 QUICK EXAM CHALLENGE — DFS 3.2

A community organiser is emailing 120 residents about a neighbourhood meeting. Should they use To, Cc, or Bcc?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Bcc (Blind Carbon Copy), to prevent exposing residents' private email addresses to one another in violation of UK GDPR.

Challenge: Workplace Messaging (Teams/Slack)

COMMUNICATING ONLINE

🎯 QUICK EXAM CHALLENGE — DFS 3.5

When should you discuss a work issue in a public project channel rather than sending a private direct message?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

When the information, decision, or answer is relevant to the entire team and should be searchable in future.

Challenge: Video Calling Setup & Etiquette

COMMUNICATING ONLINE

🎯 QUICK EXAM CHALLENGE — DFS 3.4

What two features in Microsoft Teams help protect your home privacy and prevent talking over colleagues?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Background Blur (protects privacy) and the 'Raise Hand' button (prevents speaking over others).

30. Online Calendar Invites & Scheduling

COMMUNICATING ONLINE

Creating Invites, Times & Agendas

To schedule effective meetings: 1. Add clear Title and meeting purpose. 2. Specify Date, Start time, and Duration. 3. Include meeting link (Teams/Meet) or physical room. 4. Use Scheduling Assistant to check attendees' free/busy availability. 5. Include an agenda.



⋆ Golden Rule: Always include the video link and agenda in the calendar description!

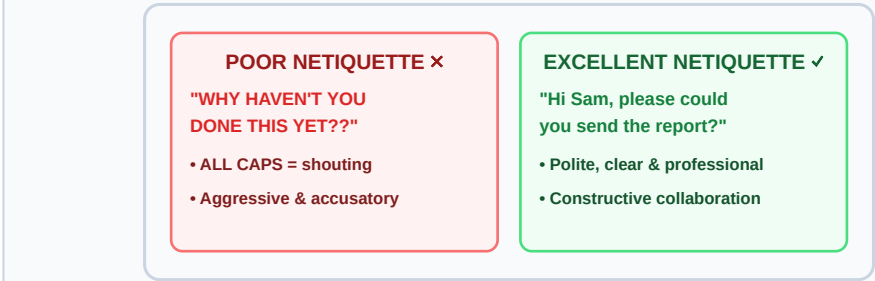
⚠️ Watch Out: Check colleague availability and working hours before sending invites.

31. Digital Netiquette & Tone

COMMUNICATING ONLINE

Written Nuance in Online Spaces

Digital messages lack body language and vocal inflections. Follow core netiquette: 1. NEVER WRITE IN ALL CAPS (read as shouting). 2. Avoid ambiguous sarcasm or irony. 3. Be polite, constructive, and concise. 4. Proofread messages before hitting send.



⋆ Golden Rule: Re-read your message through the recipient's eyes before clicking send!

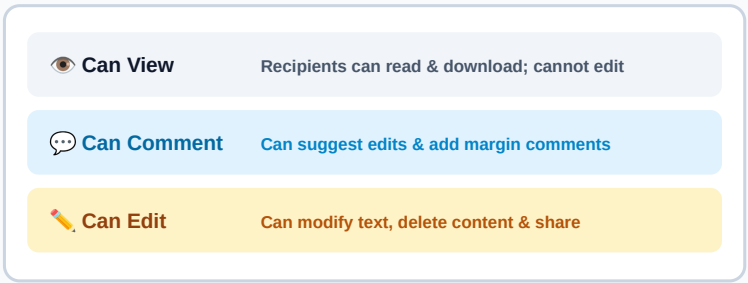
⚠️ Watch Out: Words typed in CAPITAL LETTERS are universally understood online as shouting.

32. Document Sharing Permissions

COMMUNICATING ONLINE

Viewer vs Commenter vs Editor Roles

Protect cloud files by assigning the minimum required permissions: 1. Can View (Read-Only; cannot modify or delete content). 2. Can Comment (can read and leave margin feedback notes without altering text). 3. Can Edit (Full read/write permissions). Restrict links to specific people.



⋆ Golden Rule: Set permissions to 'Can View' unless collaborative editing is needed!

⚠️ Watch Out: Never set links to 'Anyone with the link can edit' for company files.

33. Secure Online Purchasing & HTTPS

TRANSACTING ONLINE

HTTPS, SSL Encryption & Merchant Checks

Safe online shopping requirements: 1. Confirm the address begins with 'https://' with a padlock icon (data encrypted in transit). 2. Inspect domain spelling carefully to avoid fake clones. 3. Check independent reviews on Trustpilot. 4. Ensure clear contact details and postal address exist.



⋆ Golden Rule: Look for 'https://' and the padlock icon before entering card data!

⚠️ Watch Out: Avoid online shopping on public, open Wi-Fi networks without a VPN.

Challenge: Digital Netiquette & Tone

COMMUNICATING ONLINE

🎯 QUICK EXAM CHALLENGE — DFS 3.7

Why is typing workplace communications in capital letters considered unacceptable digital practice?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Typing in all capital letters is universally perceived as shouting, aggressive, or hostile.

Challenge: Online Calendar Invites & Scheduling

COMMUNICATING ONLINE

🎯 QUICK EXAM CHALLENGE — DFS 3.6

What calendar feature lets you view whether your colleagues are in other meetings before scheduling an appointment?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

The Scheduling Assistant (or 'Find a Time' view).

Challenge: Secure Online Purchasing & HTTPS

TRANSACTING ONLINE

🎯 QUICK EXAM CHALLENGE — DFS 4.1

What does the 's' in 'https://' indicate about the communication between your browser and the website?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

It stands for 'Secure' (SSL/TLS encryption), meaning data transferred between your browser and the site is encrypted.

Challenge: Document Sharing Permissions

COMMUNICATING ONLINE

🎯 QUICK EXAM CHALLENGE — DFS 3.8

Which sharing permission should you give a client to let them review a proposal and suggest changes without risking text deletion?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

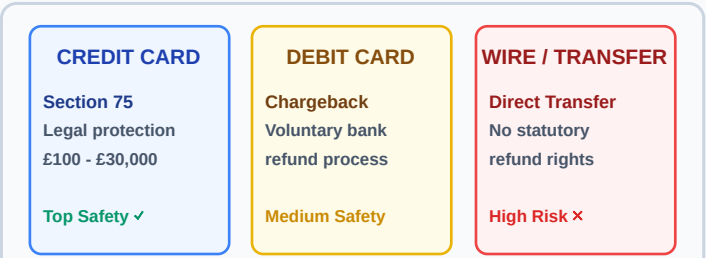
'Can Comment' (Commenter permission).

34. Payment Methods & Section 75

TRANSACTING ONLINE

Credit vs Debit vs Digital Wallets

Payment safeguards: 1. Credit Cards: UK Section 75 Consumer Credit Act legally protects single purchases over £100 up to £30,000 if goods fail to arrive or seller collapses. 2. Debit Cards: Protected by voluntary Chargeback scheme. 3. Digital Wallets (Apple Pay/PayPal): Tokenises card details.



★ Golden Rule: Use a credit card for £100+ purchases for Section 75 legal refund cover!

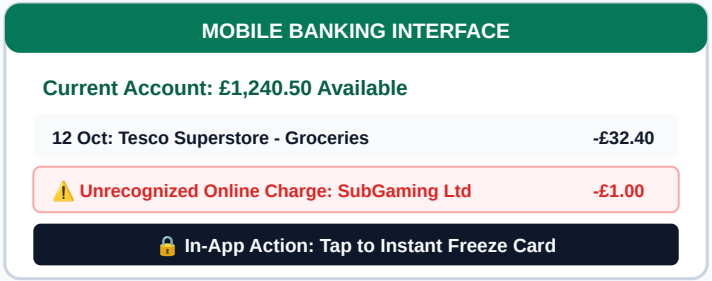
⚠ Watch Out: Direct bank transfers offer almost no buyer protection if scammed.

35. Online Banking & Statements

TRANSACTING ONLINE

Reconciling Transactions & Card Freezes

Banking safely online: 1. Regularly review transaction history to spot unauthorized micro-charges. 2. Verify all active Direct Debits and standing orders. 3. Immediately 'Freeze' lost cards inside mobile banking app. 4. Never access banking on public shared PCs.



★ Golden Rule: Review online banking weekly to spot unauthorised charges promptly!

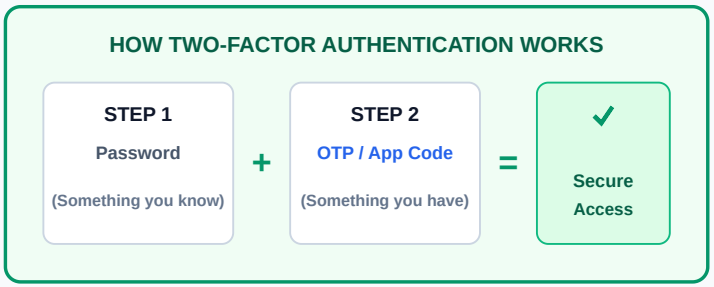
⚠ Watch Out: Never share one-time passcodes (OTPs) with anyone — banks never ask.

36. Two-Factor Authentication (2FA)

TRANSACTING ONLINE

Something You Know + Something You Have

2FA adds an essential second security barrier: 1. Factor 1: Password (something you know). 2. Factor 2: One-Time Passcode (OTP) via authenticator app, SMS, or biometric fingerprint (something you have/are). If hackers steal your password, they still cannot gain access.



★ Golden Rule: Enable 2FA everywhere — authenticator apps are much safer than SMS!

⚠ Watch Out: SMS codes are vulnerable to SIM-swap fraud; prefer app-based 2FA.

37. Accessing Public Services Online

TRANSACTING ONLINE

GOV.UK, NHS App & Council Portals

Official UK public digital services: 1. GOV.UK: Apply for passports, check tax codes, Universal Credit, and renew driving licences. 2. NHS App: Order repeat prescriptions, view GP records, book blood tests. 3. Local Council: Pay Council Tax, register to vote, apply for parking permits.



★ Golden Rule: Access public services directly via GOV.UK or NHS.UK official domains!

⚠ Watch Out: Beware of copycat websites charging fees for free public services.

Challenge: Online Banking & Statements

TRANSACTING ONLINE

🎯 QUICK EXAM CHALLENGE — DFS 4.3

What in-app mobile banking tool allows you to immediately block all transactions if you misplace your card?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

The 'Freeze' or 'Lock Card' toggle switch.

Challenge: Payment Methods & Section 75

TRANSACTING ONLINE

🎯 QUICK EXAM CHALLENGE — DFS 4.2

Why is paying with a credit card safer than paying via direct bank transfer when purchasing a £400 laptop online?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Under Section 75 of the Consumer Credit Act, the credit card provider is jointly liable with the retailer if goods are faulty or never delivered.

Challenge: Accessing Public Services Online

TRANSACTING ONLINE

🎯 QUICK EXAM CHALLENGE — DFS 4.5

How can you verify that a website offering driving licence renewals is the legitimate government service?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Check that the URL ends in '.gov.uk' (the official UK government domain).

Challenge: Two-Factor Authentication (2FA)

TRANSACTING ONLINE

🎯 QUICK EXAM CHALLENGE — DFS 4.4

Why does two-factor authentication (2FA) protect your account even if a cybercriminal steals your password?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

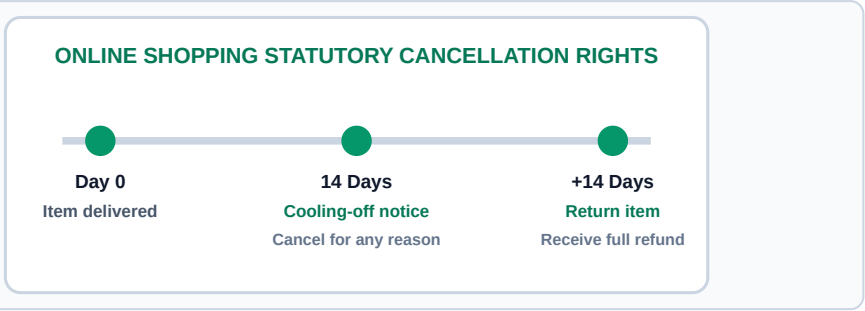
Because login is blocked without the second factor (the unique code sent to your physical device or generated by your authenticator app).

38. Online Consumer Rights & Returns

TRANSACTING ONLINE

14-Day Cooling-Off Period & Refunds

UK Consumer Contracts Regulations 2013 give statutory rights for online distance shopping: 1. 14 days from delivery date to cancel an order for any reason. 2. A further 14 days to return the item. 3. Full refund including standard delivery cost. Exemptions: Perishables, custom-made goods, unsealed hygiene items.



★ Golden Rule: Online orders give a statutory 14-day cooling-off cancellation right!

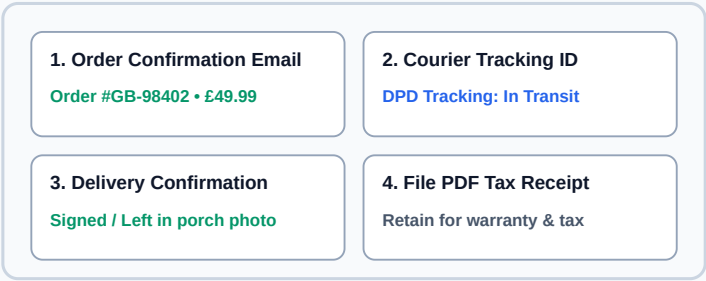
⚠ Watch Out: Perishable and custom-made items are exempt from statutory returns.

39. Digital Receipts & Order Tracking

TRANSACTING ONLINE

Saving Proof of Purchase & Tracking

Keep accurate records for online transactions: 1. Save confirmation emails and PDF receipts in a dedicated 'Purchases' email folder. 2. Record the Order ID, vendor name, and date. 3. Use parcel tracking numbers to track courier delivery progress. 4. Inspect goods promptly on arrival.



★ Golden Rule: Save order confirmations into a dedicated receipts folder or cloud PDF!

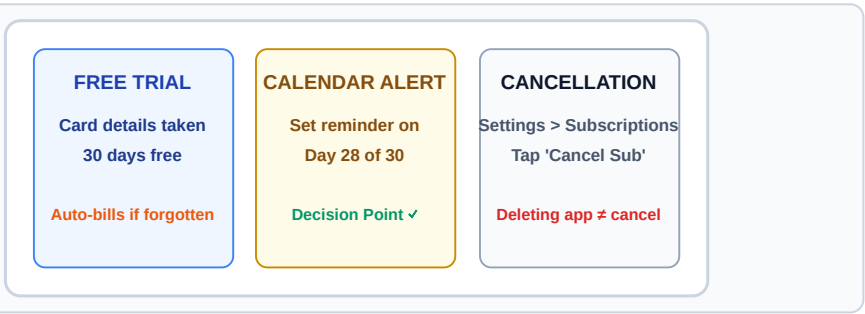
⚠ Watch Out: Retailers can close accounts — always save an offline copy of receipts.

40. Subscriptions & Auto-Renewals

TRANSACTING ONLINE

Free Trial Traps & Direct Debit Guarantee

Managing online subscriptions: 1. 'Free trials' automatically roll into paid monthly recurring subscriptions unless cancelled prior to renewal. 2. Set a calendar alert 2 days before the trial finishes. 3. Manage subscriptions via Apple App Store / Google Play subscriptions. 4. Direct Debit Guarantee covers unnotified debits.



★ Golden Rule: Set a calendar alert to review or cancel free trials before renewal!

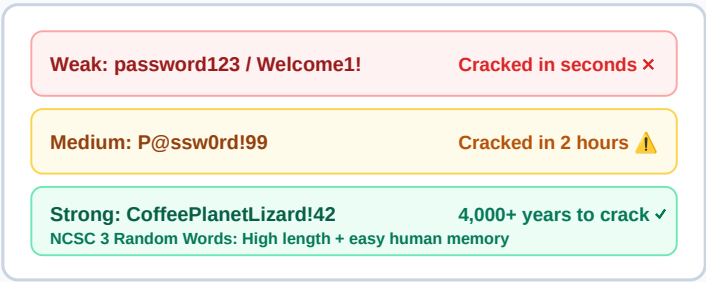
⚠ Watch Out: Deleting an app does NOT cancel an active recurring paid subscription.

41. Strong Passwords & Passphrases

SAFE & RESPONSIBLE

Length Trumps Complexity: Three Random Words

UK NCSC recommendation: Use 'Three Random Words' to build long, uncrackable passphrases that are easy to remember (e.g. 'CoffeePlanetLizard!'). Minimum 12-16 characters. Combine uppercase, lowercase, numbers, and symbols. NEVER reuse passwords across accounts.



★ Golden Rule: Combine three random words into a 14+ character secure passphrase!

⚠ Watch Out: Never reuse your primary email password on any other website.

Challenge: Digital Receipts & Order Tracking

TRANSACTING ONLINE

🎯 QUICK EXAM CHALLENGE — DFS 4.7

What key piece of information in an order confirmation email is required when contacting customer support about a missing parcel?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

The Order Reference Number (Order ID).

Challenge: Online Consumer Rights & Returns

TRANSACTING ONLINE

🎯 QUICK EXAM CHALLENGE — DFS 4.6

A customer purchases a pair of standard trainers online. How many days do they legally have after delivery to cancel the purchase?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

14 calendar days from the day of delivery under the Consumer Contracts Regulations.

Challenge: Strong Passwords & Passphrases

SAFE & RESPONSIBLE

🎯 QUICK EXAM CHALLENGE — DFS 5.1

Why is 'RocketTeapotBlanket!' a stronger password than 'P@ssw0rd1' despite being easier to remember?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Because length is the primary defence against brute-force attacks; 19 characters takes centuries to crack, while 9 characters takes minutes.

Challenge: Subscriptions & Auto-Renewals

TRANSACTING ONLINE

🎯 QUICK EXAM CHALLENGE — DFS 4.8

If you delete a paid fitness application from your smartphone, will your recurring monthly payments stop automatically?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

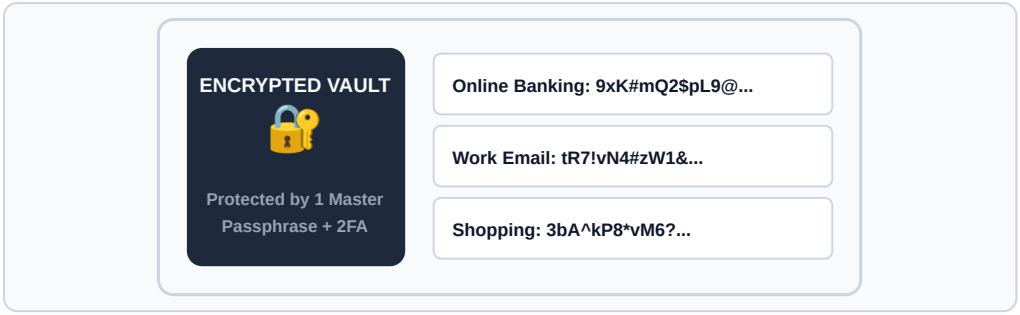
No. You must explicitly cancel the subscription inside your device's App Store / Google Play account settings or provider website.

42. Password Managers & Safe Storage

SAFE & RESPONSIBLE

Encrypted Vaults & Unique Logins

Password managers (Bitwarden, 1Password, Apple Keychain) store all your passwords inside a military-grade encrypted digital vault. Benefits: 1. You only need to remember ONE strong Master Password. 2. Auto-generates random 20-character passwords. 3. Autofills logins on trusted sites.



★ Golden Rule: Use a password manager so every account has a unique, strong password!

⚠ Watch Out: Never write passwords on sticky notes or keep them in unencrypted files.

43. Spotting Phishing & Smishing

SAFE & RESPONSIBLE

Urgency, Sender Spoofing & Malicious Links

Spotting fraudulent messages: 1. Artificial urgency ('Account suspended in 24 hrs!'). 2. Fake sender email domain (e.g. service@hmrc-tax-refunds.com instead of gov.uk). 3. Generic salutations ('Dear Customer'). 4. Suspicious links (hover cursor to view true destination URL).



★ Golden Rule: Hover over links to verify the actual destination URL before clicking!

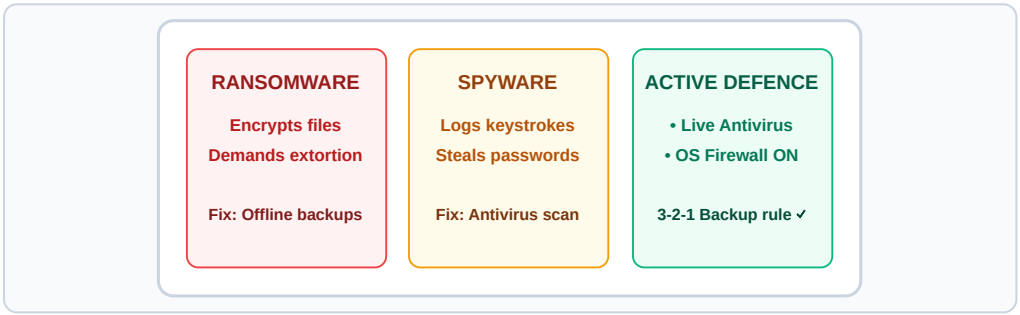
⚠ Watch Out: HMRC and banks never text or email links asking for security details.

44. Malware, Viruses & Ransomware

SAFE & RESPONSIBLE

Malicious Software Types & Defence

Key malware categories: 1. Virus/Trojan: Malicious code hidden inside innocent-looking files. 2. Spyware/Keylogger: Silently records passwords and browsing habits. 3. Ransomware: Encrypts your files and demands money for decryption. Defence: Active antivirus and regular offline backups.



★ Golden Rule: Maintain offline backups of critical data to safeguard from ransomware!

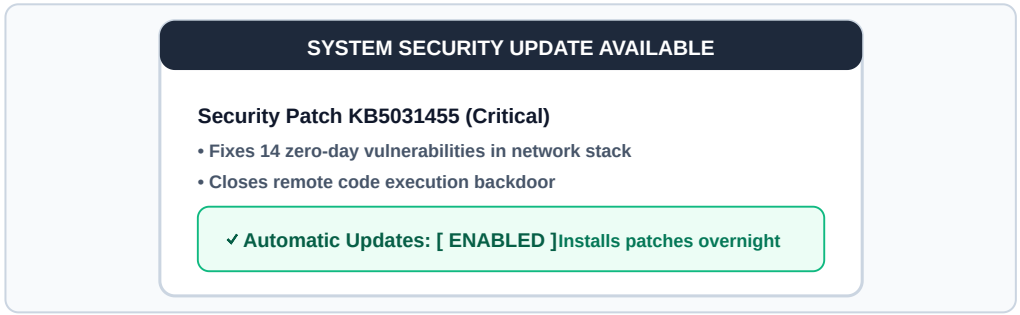
⚠ Watch Out: Never enable macros on unexpected email attachments from strangers.

45. Software Updates & Security Patches

SAFE & RESPONSIBLE

Zero-Day Flaws & Automatic Updates

Software and operating system updates provide critical security patches that close vulnerabilities before cybercriminals exploit them. Always enable automatic updates for: Windows/macOS, smartphones (iOS/Android), web browsers (Chrome/Edge), and antivirus software.



★ Golden Rule: Enable automatic updates so operating systems and apps patch flaws fast!

⚠ Watch Out: Postponing security updates leaves systems vulnerable to known exploits.

Challenge: Spotting Phishing & Smishing

SAFE & RESPONSIBLE

🎯 QUICK EXAM CHALLENGE — DFS 5.3

What should you do before clicking on a link in an unexpected email claiming your account has been compromised?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Hover your mouse over the link to verify the true destination URL, or visit the official website directly via your browser.

Challenge: Password Managers & Safe Storage

SAFE & RESPONSIBLE

🎯 QUICK EXAM CHALLENGE — DFS 5.2

What is the primary security advantage of using a password manager over using the same password for 10 different websites?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

If one website suffers a security breach, your other accounts remain completely secure because each site has a unique password.

Challenge: Software Updates & Security Patches

SAFE & RESPONSIBLE

🎯 QUICK EXAM CHALLENGE — DFS 5.5

Why is installing operating system updates vital even if you do not care about new visual features?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Because updates contain security patches that fix discovered vulnerabilities, preventing hackers and malware from exploiting them.

Challenge: Malware, Viruses & Ransomware

SAFE & RESPONSIBLE

🎯 QUICK EXAM CHALLENGE — DFS 5.4

What is ransomware, and what proactive step best ensures you never have to pay a ransom to recover files?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

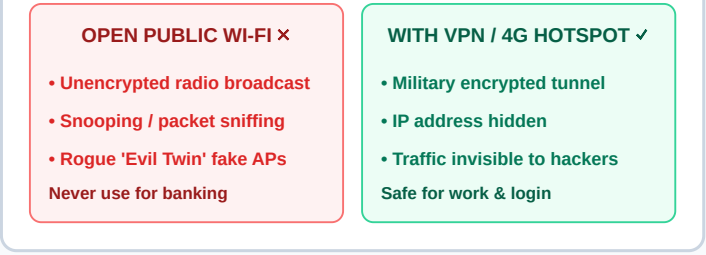
Ransomware is malicious software that encrypts your files and demands payment. Maintaining regular offline backups ensures you can restore files without paying.

46. Public Wi-Fi Risks & VPNs

SAFE & RESPONSIBLE

Unencrypted Networks & Encrypted Tunnels

Public Wi-Fi in cafes, hotels, and airports carries risks: 1. Man-in-the-Middle attacks (eavesdropping unencrypted packets). 2. Fake rogue hotspots ('Evil Twins'). Safety rules: Avoid banking on public Wi-Fi; use mobile 4G/5G tethering; use a reputable Virtual Private Network (VPN).



★ Golden Rule: Use mobile data hotspot or a VPN when connecting on public Wi-Fi!

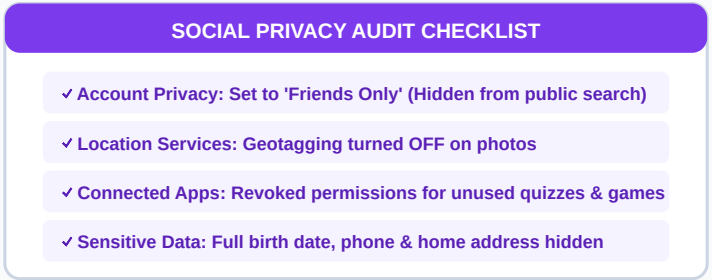
⚠️ Watch Out: Rogue open Wi-Fi hotspots can intercept unencrypted network passwords.

47. Social Media Privacy Settings

SAFE & RESPONSIBLE

Auditing Visibility, Location Tags & Permissions

Safeguarding social media profiles: 1. Set account visibility to 'Private' or 'Friends Only' (never Public). 2. Disable location tagging on posts and photos. 3. Audit third-party apps connected to your profile. 4. Never post identifying details (birth year, mother's maiden name, holiday dates).



★ Golden Rule: Keep profiles private and avoid posting live location updates when away!

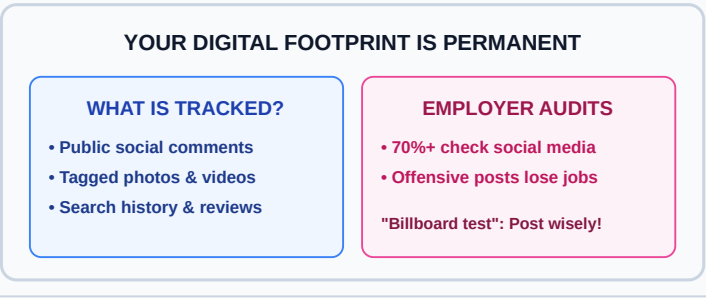
⚠️ Watch Out: Never post photos of boarding passes — barcodes contain sensitive data.

48. Digital Footprint & Reputation

SAFE & RESPONSIBLE

Permanent Traces & Employer Background Checks

Your digital footprint is the permanent trail of data left by your online activities: comments, posts, tagged photos, forum reviews. Facts: 1. Deleted posts can be archived or screenshotted. 2. 70%+ of UK employers check applicants' public online profiles. 3. Follow the 'billboard test'.



★ Golden Rule: Only post content you would be comfortable showing to an employer!

⚠️ Watch Out: Deleting a post does not remove cached versions or viewer screenshots.

49. Digital Wellbeing & Ergonomics

SAFE & RESPONSIBLE

The 20-20-20 Rule, Posture & Screen Limits

Health habits for computer users: 1. 20-20-20 Rule: Every 20 minutes, look at an object 20 feet away for 20 seconds to prevent digital eye strain. 2. Ergonomics: Top of screen at eye level, 90° elbow angle, feet flat. 3. Night light / Blue light filter to safeguard sleep cycles.



★ Golden Rule: Follow the 20-20-20 rule to prevent digital eye strain during desk work!

⚠️ Watch Out: Blue light in bed disrupts natural melatonin and degrades sleep quality.

Challenge: Social Media Privacy Settings

SAFE & RESPONSIBLE

🎯 QUICK EXAM CHALLENGE — DFS 5.7

Why is announcing on public social media that your family is departing for a 2-week holiday a significant security risk?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

It alerts potential burglars that your home will be unoccupied during that exact timeframe.

Challenge: Public Wi-Fi Risks & VPNs

SAFE & RESPONSIBLE

🎯 QUICK EXAM CHALLENGE — DFS 5.6

What does a Virtual Private Network (VPN) do to protect your data when using public Wi-Fi?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

It creates an encrypted tunnel between your device and the internet, preventing anyone on the public network from viewing your data.

Challenge: Digital Wellbeing & Ergonomics

SAFE & RESPONSIBLE

🎯 QUICK EXAM CHALLENGE — DFS 5.9

What does the 20-20-20 rule recommend to help prevent screen-related eye strain?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Every 20 minutes, take a 20-second break to look at an object at least 20 feet away.

Challenge: Digital Footprint & Reputation

SAFE & RESPONSIBLE

🎯 QUICK EXAM CHALLENGE — DFS 5.8

What is a 'digital footprint', and how can it impact a person's career prospects?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

It is the permanent trail of online activity; employers routinely research candidates online, and inappropriate posts can result in rejected applications.

50. Copyright & Creative Commons

SAFE & RESPONSIBLE

Intellectual Property, CC Licences & Attribution

Digital media ownership rules: 1. Copyright (©): Automatic legal ownership; you cannot copy or distribute without the author's permission. 2. Creative Commons (CC): Licences allowing free reuse under specific conditions (e.g. CC-BY requires attribution). 3. Public Domain: Free for anyone.



⋆ Golden Rule: Verify image license terms before using web images in published work!

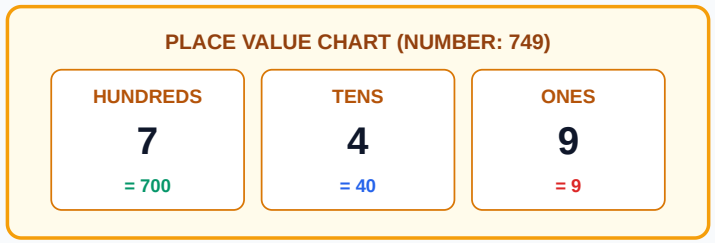
⚠️ Watch Out: Google Image results are NOT automatically free or cleared for use.

1. Place Value up to 1,000

NUMBER SYSTEM

Hundreds • Tens • Ones (H-T-O)

Every digit in a 3-digit number has a value based on its column: Hundreds (×100), Tens (×10), and Ones (×1). Zero (0) holds an empty column in place.



⋆ Golden Rule: Read 3-digit numbers from left to right: 407 is Four hundred and seven!

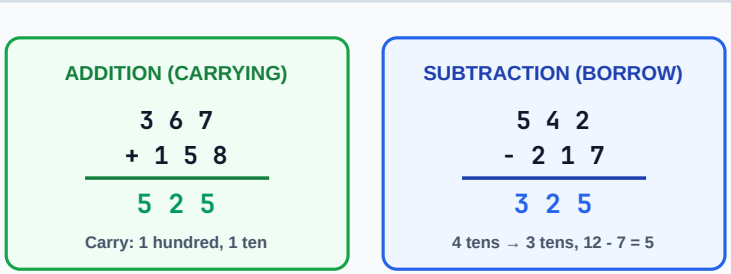
⚠️ Watch Out: Never drop the middle zero: 'Six hundred and five' is 605, NOT 65!

2. Column Addition & Subtraction

NUMBER SYSTEM

Align Right • Start at Ones • Regroup

Always line up Hundreds, Tens, and Ones on the right. Start at the Ones column. In addition, carry 10 to the next column; in subtraction, borrow from the left if the top digit is smaller.



⋆ Golden Rule: Always start adding or subtracting from the ONES column on the far right!

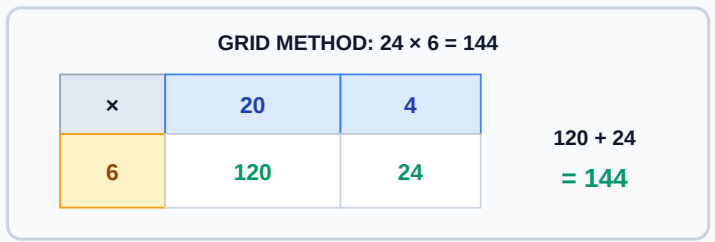
⚠️ Watch Out: In 342 - 158, do NOT do 8 - 2 = 6! You must borrow 1 ten from the 4 so 12 - 8 = 4.

3. Multiplying 2-Digit Numbers

NUMBER SYSTEM

Partition into Tens & Ones (Grid Method)

To multiply a 2-digit number by a single digit (e.g. 24 × 6), split 24 into 20 and 4. Multiply each part by 6 (20×6 = 120, 4×6 = 24), then add the answers together.



⋆ Golden Rule: When multiplying by 10, move digits one place left and place a 0 at the end!

⚠️ Watch Out: Don't forget the zero when multiplying the Tens part: 30 × 4 = 120, NOT 12!

Challenge: Place Value up to 1,000

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE E3.N1

What is the value of the digit 8 in the number 842, and how do you write 'seven hundred and nine' in digits?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Value of 8 = 800 (8 hundreds). 'Seven hundred and nine' = 709.

Challenge: Copyright & Creative Commons

SAFE & RESPONSIBLE

🎯 QUICK EXAM CHALLENGE — DFS 5.10

What does a Creative Commons 'Attribution' (CC-BY) license require you to do when using someone else's digital graphic?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

You must credit the original creator, provide a link to the license, and indicate if changes were made.

Challenge: Multiplying 2-Digit Numbers

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE E3.N3

A crate holds 36 bottles of water. How many bottles are in 4 crates?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Partition 36 into 30 + 6. (30 × 4 = 120) + (6 × 4 = 24) = 144 bottles.

Challenge: Column Addition & Subtraction

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE E3.N2

Calculate: (a) 468 + 275 (b) 604 - 238

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

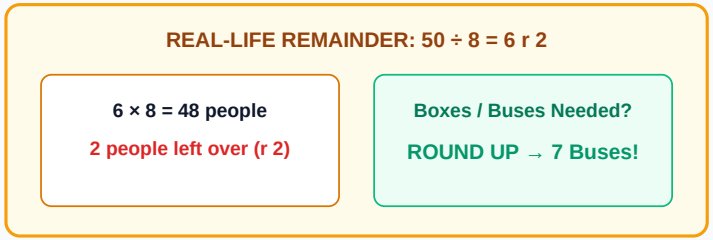
(a) 468 + 275 = 743 (carry 1 ten and 1 hundred). (b) 604 - 238 = 366.

4. Division & Interpreting Remainders

NUMBER SYSTEM

Bus Stop Division • Round Up or Down in Context

Divide from left to right. Any leftover amount is the remainder (r). In real-life problems, decide whether to round UP (e.g. minibuses/boxes needed) or round DOWN (what you can afford).



★ Golden Rule: Round UP when booking seats or buying tins so nothing is left short!

⚠ Watch Out: If asked for 'full packs', ignore remainder and round DOWN.

5. Rounding to Nearest 10 & 100

NUMBER SYSTEM

5 or More Rounds Up • 4 or Less Stays Same

To round to the nearest 10, look at the Ones digit. To round to the nearest 100, look at the Tens digit. If that neighbour digit is 5, 6, 7, 8, or 9, round UP; if 0–4, stay the same.



★ Golden Rule: Use rounded numbers to ESTIMATE and check if your calculator answer is sensible!

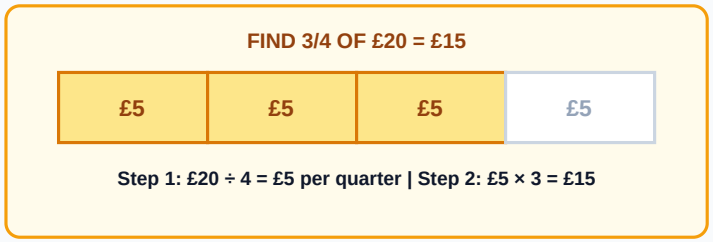
⚠ Watch Out: When rounding 674 to the nearest 100, look at the Tens digit (7), NOT the Ones digit (4)!

6. Fractions: Thirds, Quarters, Fifths & Tenths

NUMBER SYSTEM

Divide by the Bottom • Multiply by the Top

A fraction splits a whole into equal parts. To find a fraction of an amount (e.g. 3/4 of £20): Step 1: Divide the amount by the bottom number (20 ÷ 4 = 5). Step 2: Multiply by the top number (5 × 3 = £15).



★ Golden Rule: Divide by the bottom (denominator), then times by the top (numerator)!

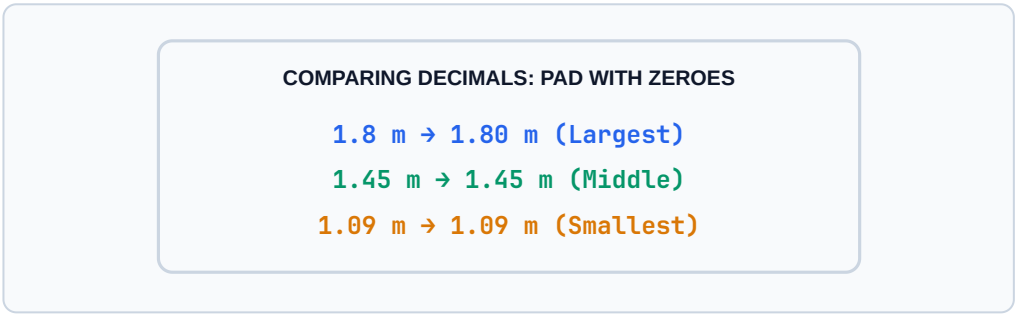
⚠ Watch Out: 1/3 is BIGGER than 1/5 — a smaller bottom number means larger slices!

7. Decimals up to 2 Decimal Places

NUMBER SYSTEM

Line Up the Decimal Points Vertically

Decimals show tenths (0.1) and hundredths (0.01), just like pounds and pence (£3.45). When adding or subtracting decimals, always line up the decimal points in a straight vertical column.



★ Golden Rule: Add a trailing zero (0.5 = 0.50) so both numbers have 2 decimals!

⚠ Watch Out: 0.4 is larger than 0.25 because 0.40 has 4 tenths while 0.25 has 2.

Challenge: Rounding to Nearest 10 & 100

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE E3.N5

Round 463: (a) to the nearest 10, and (b) to the nearest 100.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

(a) Nearest 10: Ones digit is 3 (< 5) → 460. (b) Nearest 100: Tens digit is 6 (≥ 5) → 500.

Challenge: Division & Interpreting Remainders

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE E3.N4

50 care home residents are going on a trip. Each minibus holds 8 people. How many minibuses are needed?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

50 ÷ 8 = 6 remainder 2. Because 2 people cannot be left behind, round UP to 7 minibuses.

Challenge: Decimals up to 2 Decimal Places

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE E3.N8

Put these lengths in order from shortest to longest: 1.45 m, 1.8 m, 1.09 m.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Write with 2 decimal places: 1.45 m, 1.80 m, 1.09 m. Order: 1.09 m, 1.45 m, 1.8 m.

Challenge: Fractions: Thirds, Quarters, Fifths & Tenths

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE E3.N7

A jacket costs £45. It is reduced by 1/5 in a sale. How much money is taken off?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

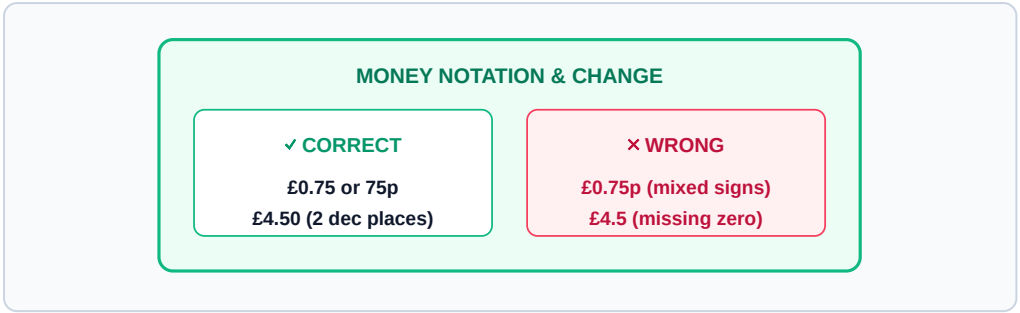
1/5 of £45 = 45 ÷ 5 = £9 taken off.

8. Money: Pounds, Pence & Change

MEASURES & MONEY

£1 = 100p • Never Mix £ and p Signs

Convert pence to pounds by dividing by 100 (e.g. 245p = £2.45). Write money with either a £ sign and 2 decimal places (£0.65) OR a p sign (65p), never both (£0.65p is wrong).



⋆ Golden Rule: Always write two digits after the dot for pounds: £4.5 on a calculator must be written as £4.50!

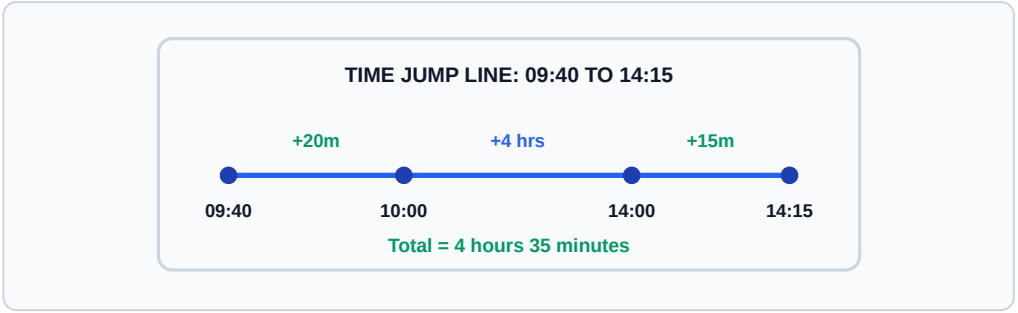
⚠ Watch Out: Before adding 85p and £2.40, convert 85p into £0.85 so both are in pounds!

9. Time: Roman Numerals, 12-Hour & 24-Hour

MEASURES & MONEY

Add 12 to PM Hours • 1 Hour = 60 Minutes

On clock faces: IV = 4, VI = 6, IX = 9, XII = 12. For 24-hour time, add 12 to afternoon (pm) hours from 1:00 pm onwards (e.g. 3:15 pm → 15:15). Count time intervals across the top of the hour (:00).



⋆ Golden Rule: There are 60 minutes in 1 hour — never use column subtraction as if 1 hour had 100 minutes!

⚠ Watch Out: 12:30 pm stays 12:30 in 24-hour time; 12:30 am (midnight) becomes 00:30.

10. Metric Units: Length, Weight & Capacity

MEASURES & MONEY

1 m = 100 cm • 1 kg = 1,000 g • 1 L = 1,000 ml

Length: 1 cm = 10 mm, 1 m = 100 cm, 1 km = 1,000 m. Weight: 1 kg = 1,000 g. Capacity: 1 Litre = 1,000 ml. Always convert to the SAME unit before adding or comparing.



⋆ Golden Rule: 'Kilo' (km, kg) and 'Milli' (ml, mm in a m) mean 1,000; 'Centi' (cm) means 100!

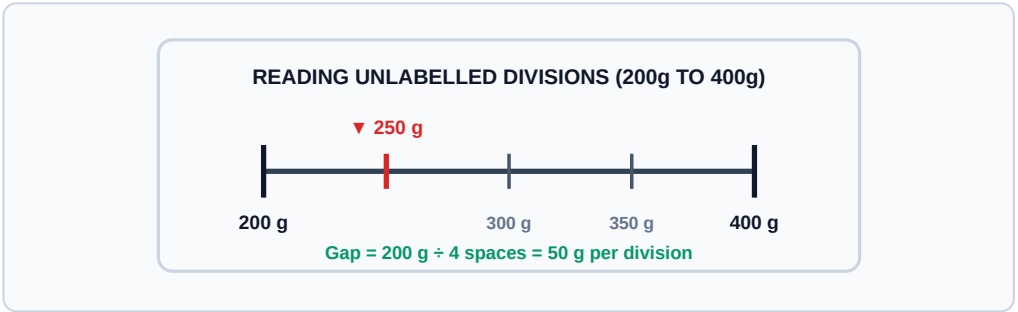
⚠ Watch Out: 1 metre is 100 cm (NOT 1,000 cm!). Half a kilogram (0.5 kg) is 500 g.

11. Reading Scales & Dials

MEASURES & MONEY

Find the Gap Divided by the Number of Steps

To read an unlabelled tick mark on a scale, thermometer, or weighing dial: (1) Subtract two neighbouring numbered marks to find the gap, then (2) Divide by the number of spaces between them.



⋆ Golden Rule: Count the SPACES (jumps) between numbered marks, not the lines!

⚠ Watch Out: If there are 5 spaces between 0 g and 100 g, each mark is 100 ÷ 5 = 20 g (not 10 g!).

Challenge: Time: Roman Numerals, 12-Hour & 24-Hour

MEASURES & MONEY

🎯 QUICK EXAM CHALLENGE — DfE E3.M12

A shift starts at 9:40 am and finishes at 2:15 pm (14:15). How long is the shift in hours and minutes?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

9:40 am to 10:00 am = 20 mins. 10:00 am to 2:00 pm = 4 hours. 2:00 pm to 2:15 pm = 15 mins. Total = 4 hours 35 minutes.

Challenge: Money: Pounds, Pence & Change

MEASURES & MONEY

🎯 QUICK EXAM CHALLENGE — DfE E3.M10

You buy a sandwich for £3.25 and a tea for 90p. How much change do you get from a £10 note?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

90p = £0.90. Total cost = £3.25 + £0.90 = £4.15. Change = £10.00 - £4.15 = £5.85.

Challenge: Reading Scales & Dials

MEASURES & MONEY

🎯 QUICK EXAM CHALLENGE — DfE E3.M18

A kitchen scale has 4 equal spaces between 200 g and 400 g. The pointer is 1 mark past 200 g. What is the weight?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Gap = 400 - 200 = 200 g. Step size = 200 ÷ 4 = 50 g per mark. Weight = 200 + 50 = 250 g.

Challenge: Metric Units: Length, Weight & Capacity

MEASURES & MONEY

🎯 QUICK EXAM CHALLENGE — DfE E3.M14

A jug holds 2 Litres of juice. You pour out three 250 ml glasses. How many millilitres (ml) are left in the jug?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

2 Litres = 2,000 ml. Poured out: 3 × 250 ml = 750 ml. Left: 2,000 - 750 = 1,250 ml (or 1.25 L).

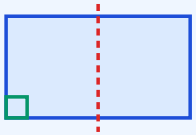
12. 2D & 3D Shapes, Right Angles & Symmetry

SHAPE & SPACE

Faces, Edges & Vertices • 90° Right Angles

A right angle (90°) is a square corner like a book edge. A line of symmetry folds a 2D shape into two matching halves. 3D shapes have flat/curved Faces, Edges, and Vertices (corners).

RIGHT ANGLE & SYMMETRY



4 Right Angles (90° corners)

3D SHAPE PROPERTIES

- **Cube/Cuboid: 6 Faces,** 12 Edges, 8 Vertices
- **Cylinder: 3 Faces (2 flat,** 1 curved), 2 Edges, 0 Vertices

★ **Golden Rule: A cuboid (box) has 6 rectangular faces, 12 edges, and 8 vertices (corners)!**

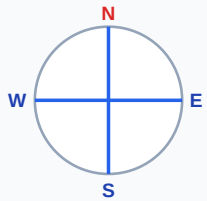
⚠️ **Watch Out: A rectangle has 2 lines of symmetry (vertical & horizontal), NOT 4 — folding diagonally does not match!**

13. Compass Directions & Turns

SHAPE & SPACE

Never Eat Shredded Wheat (N • E • S • W)

The 8 compass points go clockwise: North (N), North-East (NE), East (E), South-East (SE), South (S), South-West (SW), West (W), North-West (NW). A quarter turn is 1 right angle (90°).



8-POINT COMPASS & TURNS

- Quarter turn = 1 right angle (N → E)
- Half turn = Opposite way (N → S)
- Diagonal points: NE, SE, SW, NW

★ **Golden Rule: Clockwise follows clock hands; Anti-clockwise turns the opposite way!**

⚠️ **Watch Out: Check the facing direction before calculating quarter or half turns.**

14. Bar Charts, Pictograms & Tally Tables

HANDLING DATA

Always Check the Pictogram Key First!

In a pictogram, one symbol often stands for 2, 4, or 10 items — always read the Key first! In a bar chart, read across from the top of the bar to the vertical frequency axis.

PICTOGRAM KEY RULE: 1 CIRCLE (●) = 4 ITEMS

Monday: ●●●◗ 4 + 4 + 4 + 2 = 14

Full symbol = 4 | Half symbol = 2 | Quarter symbol = 1

★ **Golden Rule: If 1 circle = 4 items, then half a circle = 2 items and a quarter circle = 1 item!**

⚠️ **Watch Out: Never count pictogram symbols as '1 each' without checking the Key box!**

15. Main Purpose of a Text

READING BASICS

To Inform • To Instruct • To Persuade

Every text has a main purpose: To Inform (gives facts, dates, or news), To Instruct (step-by-step recipes or safety rules using command verbs), or To Persuade (adverts encouraging you to buy or join).

INFORM	INSTRUCT	PERSUADE
Gives facts, times & details	Numbered steps & command verbs	Encourages you to buy, join, or act
Notice / Timetable	Recipe / Manual	Poster / Advert

★ **Golden Rule: When naming purpose, state both the verb AND topic (e.g. To warn about)!**

⚠️ **Watch Out: Never just write 'To inform' — write 'To inform staff about fire rules'.**

Challenge: Compass Directions & Turns

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE E3.M20

You are facing North. You turn a quarter-turn (90°) anti-clockwise. Which direction are you facing now?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Facing West (1 quarter-turn anti-clockwise from North).

Challenge: 2D & 3D Shapes, Right Angles & Symmetry

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE E3.M19

How many faces, edges, and vertices does a cube have, and how many right angles are inside a rectangle?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Cube: 6 faces, 12 edges, 8 vertices. A rectangle has 4 right angles (90° corners).

Challenge: Main Purpose of a Text

READING BASICS

🎯 QUICK EXAM CHALLENGE — DfE E3.R8

A leaflet titled '5 Easy Steps to Reset Your Work Password' uses numbered steps. What is its main purpose?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

To instruct (or explain to) staff how to reset their work password step by step.

Challenge: Bar Charts, Pictograms & Tally Tables

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE E3.HD21

In a pictogram, Key: 1 star = 4 books. Monday shows 3 full stars and a half star. How many books were borrowed on Monday?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

3 full stars = 3 × 4 = 12 books. Half star = 4 ÷ 2 = 2 books. Total = 12 + 2 = 14 books.

16. Organizational Features (Headings & Bullets)

READING BASICS

Headings, Subheadings, Bullet Points & Boxes

Writers use organizational features to help readers find information fast: Headings show the main topic, Subheadings divide sections, Bullet points list key facts, and Text boxes highlight warnings or prices.

LAYOUT FEATURES THAT HELP THE READER

- **Main Heading:** Grabs attention & states overall topic
- **Subheadings:** Split text into clear, easy-to-scan sections
- **Bullet Points / Numbered Lists:** Make key steps stand out
- **Bold Text & Captions:** Highlight important words & photos

★ **Golden Rule: Use subheadings like signposts to jump straight to the needed section!**

⚠️ **Watch Out: Name the specific feature (e.g. Subheading, Table, Bullet points).**

17. Using a Dictionary & Alphabetical Order

READING BASICS

Compare the 1st, 2nd, then 3rd Letter

To put words in alphabetical (A–Z) order or look them up in a dictionary/index: compare the first letter. If the first letters are the same, compare the second letter (or third letter).

ALPHABETICAL ORDER: CHECK THE NEXT LETTER

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

Compare: Paper • Postcode • Printer

Since A → O → R: Paper, then Postcode, then Printer!

★ **Golden Rule: Check the 3rd and 4th letters when words start with the same letters!**

⚠️ **Watch Out: 'Staff' comes before 'Store' because 'a' precedes 'o' in the alphabet.**

18. Writing Emails & Short Notes

EVERYDAY WRITING

To • Subject Line • Greeting • Paragraphs • Sign-Off

Every Entry 3 email needs: (1) A clear Subject Line, (2) A polite greeting ('Dear Mr Patel,' or 'Hi Sarah,'), (3) Short paragraphs answering every bullet point in the task, and (4) A polite sign-off ('Kind regards,').

✉️ **EMAIL CHECKLIST (ENTRY LEVEL 3)**

Subject: Shift Swap Request – Tuesday 14 May

Dear Manager,

I am writing to ask if I can swap my Tuesday morning shift...

Kind regards, Alex Carter

★ **Golden Rule: Tick off each task bullet point to ensure full content marks are won!**

⚠️ **Watch Out: Never leave the email Subject line blank — it costs an easy mark.**

19. Paragraphs & Sequencing Ideas

EVERYDAY WRITING

Start a New Paragraph for Each New Topic

Break your writing into at least 3 clear paragraphs: (1) Opening (why you are writing), (2) Middle (details and examples), and (3) Ending (what happens next). Use time linkers like First, Next, Then, Finally.

3-PARAGRAPH STRUCTURE + SEQUENCING WORDS

1. OPENING	2. DETAILS	3. ENDING
Why you are writing	First, Next, Also, Because	Finally, action & thank you

★ **Golden Rule: Leave a clear blank line between paragraphs to show structure clearly!**

⚠️ **Watch Out: Writing one giant block of text without breaks loses layout marks.**

Challenge: Using a Dictionary & Alphabetical Order

READING BASICS

🎯 QUICK EXAM CHALLENGE — DfE E3.R10

Put these four words in alphabetical order: Printer, Paper, Postcode, Folder.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

1. Folder (F), 2. Paper (Pa), 3. Postcode (Po), 4. Printer (Pr).

Challenge: Organizational Features (Headings & Bullets)

READING BASICS

🎯 QUICK EXAM CHALLENGE — DfE E3.R9

Why does a safety poster use bullet points instead of one long paragraph?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Bullet points break up the text so key safety rules are quick and easy to read at a glance.

Challenge: Paragraphs & Sequencing Ideas

EVERYDAY WRITING

🎯 QUICK EXAM CHALLENGE — DfE E3.W19

Give three sequencing words you can use to order instructions for a new colleague.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

First (or Firstly), Next (or Then / After that), and Finally.

Challenge: Writing Emails & Short Notes

EVERYDAY WRITING

🎯 QUICK EXAM CHALLENGE — DfE E3.W18

Write a clear Subject Line and Greeting for an email to your manager, Mrs Singh, requesting Friday off.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Subject: Annual Leave Request – Friday 18 October | Greeting: Dear Mrs Singh,

20. Sentence Punctuation & Capital Letters

SPAG FOUNDATIONS

M.I.N.T.S. Capitals • Full Stops • Question Marks

Start every sentence with a Capital Letter and end with a Full Stop (.), Question Mark (?), or Exclamation Mark (!). Always capitalise Months/Days, the word 'I', Names of people, and Places (Oxford, Tuesday).

M.I.N.T.S. CAPITAL LETTER CHECKLIST

- **M — Months & Days:** Friday, November
- **I — The word 'I':** I work at the clinic.
- **N — Names & Places:** Abdul, Oxford, High Street
- **S — Start of Every Sentence (+ end with . ? !)**

★ **Golden Rule: Every question must end with a question mark (?), not a full stop!**

⚠️ **Watch Out: Don't join two complete sentences with a comma — use a full stop.**

21. Commas in Lists & Apostrophes

SPAG FOUNDATIONS

Commas Separate List Items • Apostrophes for Missing Letters

Use commas to separate items in a list ('We need pens, paper, folders and tape'). Use an apostrophe (') to show where letters are missing in contractions (don't = do not, can't = cannot, I'm = I am).

COMMAS IN A LIST

"Bring a pen, a notepad, your ID badge and water."

Separates 3 or more items

APOSTROPHE (OMISSION)

- do not → don't
- it is → it's
- you are → you're

★ **Golden Rule: Put the apostrophe in the exact spot where the missing letter was taken out!**

⚠️ **Watch Out: Never put an apostrophe in simple plural words: 'Three coffees and two cakes' has NO apostrophe!**

22. Subject-Verb Agreement (Was/Were & Is/Are)

SPAG FOUNDATIONS

Singular Takes 'Is/Was' • Plural Takes 'Are/Were'

Match the verb to the subject: One person/thing (I, he, she, it, the bus) takes 'is' or 'was'. More than one (we, they, the buses) and 'you' ALWAYS take 'are' or 'were' ('We were happy', NEVER 'We was').

SINGULAR (ONE)

- I was / He was / She was
- The train is / was late
- There is one seat left

PLURAL (2+ OR 'YOU')

- We were / They were
- You were (NEVER 'you was')
- There are many options

★ **Golden Rule: Always write 'We were', 'They were', and 'You were' (never 'We was')!**

⚠️ **Watch Out: 'There is' is for one item; 'There are' is for multiple items.**

23. Plurals (-s, -es, -ies) & Common Spellings

SPAG FOUNDATIONS

Ch/Sh/S/X Add -es • Consonant + Y Changes to -ies

Most nouns add -s for plurals. Words ending in -ch, -sh, -s, -x, or -z add -es (box → boxes, bus → buses). Words ending in a consonant + y drop the y and add -ies (baby → babies, city → cities).

THE 3 PLURAL SPELLING RULES

1. Ends in -s, -sh, -ch, -x → Add -ES: box → boxes, watch → watches
2. Consonant + Y → Drop Y, add -IES: city → cities, lorry → lorries
3. Vowel + Y → Keep Y, add -S: day → days, trolley → trolleys

★ **Golden Rule: Consonant + y becomes -ies (party → parties); vowel + y adds -s!**

⚠️ **Watch Out: Irregular plurals change completely: child → children, person → people.**

Challenge: Commas in Lists & Apostrophes

SPAG FOUNDATIONS

🎯 QUICK EXAM CHALLENGE — DfE E3.W14

Add commas and an apostrophe to fix this sentence: 'Weve packed sandwiches apples crisps and water.'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'We've packed sandwiches, apples, crisps and water.'

Challenge: Sentence Punctuation & Capital Letters

SPAG FOUNDATIONS

🎯 QUICK EXAM CHALLENGE — DfE E3.W13

Fix the 4 errors in this sentence: 'on monday i am going to london, when does the train leave.'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'On Monday I am going to London. When does the train leave?' (Capitals for On, Monday, I, London + Full Stop + Question Mark).

Challenge: Plurals (-s, -es, -ies) & Common Spellings

SPAG FOUNDATIONS

🎯 QUICK EXAM CHALLENGE — DfE E3.W17

Write the plural of these three words: (a) box (b) family (c) key

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

(a) boxes (add -es) (b) families (consonant + y → -ies) (c) keys (vowel + y → add -s).

Challenge: Subject-Verb Agreement (Was/Were & Is/Are)

SPAG FOUNDATIONS

🎯 QUICK EXAM CHALLENGE — DfE E3.W15

Choose the correct verbs: 'We (was / were) waiting at the stop because the buses (was / were) delayed.'

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

'We were waiting at the stop because the buses were delayed.'

24. SLC E3: Listening, Questions & Turns

SPEAKING & LISTENING

Listen Actively • Ask Relevant Questions • Take Turns

In your Entry 3 Speaking & Listening discussion: (1) Listen carefully without interrupting, (2) State your views clearly with a reason ('I think... because...'), and (3) Ask at least two relevant questions to others.

ENTRY 3 SPEAKING & LISTENING SUCCESS

- ✓ Speak Clearly: "I think we should... because..."
- ✓ Ask Questions: "What do you think is the best day for...?"
- ✓ Respect Turns: Nod, keep eye contact, and don't interrupt

⋆ Golden Rule: Always add 'because...' to explain your reasons during discussions!

⚠️ Watch Out: Don't stay silent after speaking — invite others to share views.

DfE E3.SLC1 • Entry Level 3 | Turn over for Practice Challenge ➔ | A6 Pocket Card

25. Device Settings, Wi-Fi & Accessibility

USING DEVICES

Wi-Fi Padlock • Display Zoom • Volume & Contrast

Use your device Settings app to connect to secure Wi-Fi (look for the padlock icon), adjust screen brightness/text size for readability, and install system updates to fix security bugs.

ESSENTIAL DEVICE SETTINGS (ENTRY 3)

- 🔒 Secure Wi-Fi: Look for the padlock icon & enter password
- 🔍 Accessibility: Increase Text Size, Zoom, or High Contrast
- 🔄 Software Updates: Install updates to patch security risks

⋆ Golden Rule: Only connect to Wi-Fi networks that show a padlock icon and require a password!

⚠️ Watch Out: Never log into online banking on an open, unlocked public Wi-Fi network!

DFSQ E3.1 • Entry Level 3 | Turn over for Practice Challenge ➔ | A6 Pocket Card

26. Files, Folders & Keyboard Shortcuts

USING DEVICES

Clear File Names • Ctrl+C (Copy) • Ctrl+V (Paste)

Save files into named folders with clear, descriptive filenames (e.g. 'CV_Alex_Carter.pdf', not 'doc1'). Essential shortcuts: Ctrl+C = Copy, Ctrl+X = Cut, Ctrl+V = Paste, Ctrl+Z = Undo, Ctrl+S = Save.

THE 4 GOLDEN KEYBOARD SHORTCUTS

Ctrl + C = COPY	Ctrl + V = PASTE
Ctrl + Z = UNDO	Ctrl + S = SAVE

⋆ Golden Rule: Press Ctrl + Z immediately if you make a mistake to undo your action!

⚠️ Watch Out: Know file extensions: .docx = Word, .pdf = Print, .png = Image.

DFSQ E3.2 • Entry Level 3 | Turn over for Practice Challenge ➔ | A6 Pocket Card

27. Word Processing & Formatting Documents

CREATING & EDITING

Bold (B) • Centre Align • Bullet Lists • Spellcheck

In a poster or notice: make the main title Larger (18–24pt), Bold (B), and Centre-aligned. Keep body text Left-aligned (11–12pt) so it is easy to read, and use Bullet Points for lists.

[B] Bold | [I] Italic | [U] Underline | [•] Bullets

STAFF FIRE SAFETY NOTICE (Bold, 18pt, Centred)

- Leave the building calmly via the nearest green exit sign.
- Report to Assembly Point B in the main car park.

(Body text: Left-aligned, 12pt, clean bullet points)

⋆ Golden Rule: Right-click red wavy underlines to correct spellings before saving!

⚠️ Watch Out: Keep body text left-aligned — never centre-align long paragraphs.

DFSQ E3.6 • Entry Level 3 | Turn over for Practice Challenge ➔ | A6 Pocket Card

Challenge: Device Settings, Wi-Fi & Accessibility

USING DEVICES

🎯 QUICK EXAM CHALLENGE — DFSQ E3.1

A colleague finds the text on their tablet screen too small to read. Which device setting should they adjust?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Open Settings → Display / Accessibility → Increase Font Size (or Screen Zoom).

Challenge: SLC E3: Listening, Questions & Turns

SPEAKING & LISTENING

🎯 QUICK EXAM CHALLENGE — DfE E3.SLC1

Give one polite question you could ask a classmate during a discussion about local recycling.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

"How easy do you find it to use the recycling bins on your street?" (or "What could the council do to help people recycle more?").

Challenge: Word Processing & Formatting Documents

CREATING & EDITING

🎯 QUICK EXAM CHALLENGE — DFSQ E3.6

How should you format the main heading of a workplace safety notice so it stands out clearly?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Make the heading Bold, increase the font size (e.g. 18–22pt), and Centre-align it at the top of the page.

Challenge: Files, Folders & Keyboard Shortcuts

USING DEVICES

🎯 QUICK EXAM CHALLENGE — DFSQ E3.2

Which keyboard shortcut copies selected text, and which shortcut pastes it into a new place?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Ctrl + C copies the selected text; Ctrl + V pastes it.

28. Simple Spreadsheets & Totals (=SUM)

CREATING & EDITING

Columns (A, B, C) • Rows (1, 2, 3) • Always Start with =

Each box in a spreadsheet is a cell named by its Column letter and Row number (e.g. B4). Every formula MUST start with an equals sign (=): use =B2+B3 or =SUM(B2:B5) to add numbers.

SPREADSHEET FORMULA SYMBOLS

```
=SUM(B2:B5) → Adds all cells from B2 down to B5
=B2-C2 → Subtracts C2 from B2
=B2*C2 → Multiplies B2 by C2 (* means ×)
=B2/C2 → Divides B2 by C2 (/ means ÷)
```

★ Golden Rule: Always begin spreadsheet formulas with '=' (e.g. =SUM(A1:A5))!

⚠️ Watch Out: Use '*' for multiply and '/' for divide — never type the letter 'x'.

29. Email Attachments, Reply vs Reply All

COMMUNICATING ONLINE

Paperclip = Attach File • Check Recipient Before Sending

Click the Paperclip icon to attach a file (like a PDF or photo) to an email. Use 'Reply' to answer ONLY the sender; use 'Reply All' only when everyone on the thread needs to see your answer.

↩️ REPLY (ONE PERSON)

Sends your message ONLY to the person who emailed

✓ Safe for private info

📎 ATTACHMENTS

Click Paperclip icon (📎) to attach a PDF or photo

✓ Check file name & size

★ Golden Rule: Always attach the file before sending if writing 'please find attached'!

⚠️ Watch Out: Never click 'Reply All' when sending sensitive personal information.

30. Online Safety: Passwords & Scams

BEING SAFE & SECURE

3 Random Words + Symbols • Check HTTPS Padlock

Create strong passwords using three random words plus numbers/symbols (e.g. 'Green#kettle42!'). When shopping or banking online, check the browser address bar starts with 'https://' and has a padlock icon.

3 GOLDEN RULES FOR STAYING SAFE ONLINE

1. Strong Password: 3 random words + numbers/symbols + 2FA code
2. Safe Websites: Check for https:// and the Padlock icon in URL bar
3. Spot Phishing: Ignore urgent panic links & fake sender addresses

★ Golden Rule: Never click links in unexpected security alerts or parcel texts!

⚠️ Watch Out: Never use obvious passwords like 'password123' or your date of birth.

Challenge: Email Attachments, Reply vs Reply All

COMMUNICATING ONLINE

🎯 QUICK EXAM CHALLENGE — DFSQ E3.10

You receive a company-wide email from HR asking you to send your private bank details. Should you click 'Reply' or 'Reply All'?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Click 'Reply' (so only HR receives your private details, NOT the entire company).

Challenge: Simple Spreadsheets & Totals (=SUM)

CREATING & EDITING

🎯 QUICK EXAM CHALLENGE — DFSQ E3.8

Write the spreadsheet formula to add the costs in cells B2, B3, B4, and B5.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

=SUM(B2:B5) (or =B2+B3+B4+B5).

Challenge: Online Safety: Passwords & Scams

BEING SAFE & SECURE

🎯 QUICK EXAM CHALLENGE — DFSQ E3.12

State two signs that a text message claiming 'Your parcel is held—click here to pay £1.99' is a phishing scam.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

1. Urgent demand for payment via an unfamiliar web link. 2. Sent from an unknown mobile number with a generic greeting.